

BAI ENDSEM

05/13/2026 22:59:18

BAI (april 2023).pdf BAI(2024 nov).pdf BAI(april 2024).pdf BAI(nov 2023).pdf i...
Yes. I analyzed all 4 PYQs of SPPU BAI (Business Analytics & Intelligence) and...
extract exact question with marks which appeared in exam unitwise from high...
UNIT 3 → (Q1 & Q2) 🔥 MOST REPEATED QUESTIONS 1) Balanced Scorecard...
i want to preparr for exam and score outoff Generate a **sppu university exa...
Your prompt is already very strong. I improved it further so ChatGPT generate...
unit 3 1) Balanced Scorecard Repeated Questions "What is a Balanced...
Q. What is a Balanced Scorecard? Discuss the four perspectives of a balanced...
2."How is a balanced scorecard used in Business?" [5]
Q. How is a Balanced Scorecard used in Business? [5 Marks] 1 Simple...
3) Dashboard Design / Dashboard Repeated Questions "Discuss the common...
Q. Discuss the common pitfalls in Dashboard Design. [6 Marks] 1 Simple...
4."What is the role of the Performance dashboards?" [5] Apr 2024 "Discuss wit...
Q. What is the role of the Performance Dashboards? [5 Marks] 1 Simple...
"What is data visualization? Explain the need for data visualization in Business...
Q. What is Data Visualization? Explain the need for Data Visualization in Busine...
"List and briefly explain different types of charts used for visualization." [8]
Q. List and briefly explain different types of charts used for visualization. [8...
"Explain with neat diagrams various forms of charts and graphs used in visual...
Q. Explain with neat diagrams various forms of charts and graphs used in Visu...
4) BPM (Business Performance Management) Repeated Questions "Write a not...
Q. Write a note on Business Performance Management (BPM). [6 Marks] 1...
"List and explain the four processes that define a closed-loop BPM cycle." [8]
Q. List and explain the four processes that define a Closed-Loop BPM Cycle. [8...
5) KPI "What is KPI? Explain the types of KPIs." [5] Nov 2023 "List and justify an...
Q. What is KPI? Explain the types of KPIs. [5 Marks] 1 Simple Introduction KPI...
⚡ MEDIUM REPEATED 5) Power BI / Tableau as BI Tool Asked: Tableau → Apr...

Q. Discuss the use of Tableau as a Business Intelligence (BI) Tool. [5 Marks]  1...

7) Business Reports Asked: Nov 2024 Apr 2024 Important: Types of reports...

Q. List and briefly explain different types of Business Reports. [5 Marks]  1...

UNIT 4 → (Q3 & Q4)  MOST REPEATED QUESTIONS 1) What-if Analysis...

Q. Write a note on: i) Multiple Goals ii) Sensitivity Analysis iii) What-If Analysis...

Bread Seller What-if Analysis Question [8]

Q. Consider a scenario that a bread seller charges Rs. 30 per loaf of bread, and...

Bookstore What-if Analysis Question [9] Nov 2023 Income & Expense What-if...

Q. Consider a scenario that you own a bookstore and have 100 books in storag...

2) Descriptive / Predictive / Prescriptive Analytics Repeated Questions "Write a...

Q. Write a short note on: i) Descriptive Analytics ii) Predictive Analytics iii)...

"What is descriptive analytics? Explain the importance of descriptive analytics...

Q. What is Descriptive Analytics? Explain the importance of Descriptive Analytic...

3) Certainty, Uncertainty, Risk "Write short note on : i) Certainty ii) Uncertainty i...

Q. Write short note on: i) Certainty ii) Uncertainty iii) Risk [9 Marks]  1 Simple...

4) Mathematical Models "Discuss structure of mathematical model for decision...

Q. Discuss structure of Mathematical Model for Decision Support. [8 Marks]  1...

"Discuss the various Components of Decision Support Mathematical Models." [...

Q. Discuss the various Components of Decision Support Mathematical Models....

6) Decision Trees / Decision Tables Asked: Apr 2023

Q. Explain Decision Trees and Decision Tables with suitable examples. [8 Marks...

UNIT 5 → (Q5 & Q6)  MOST REPEATED QUESTIONS 1) BI in Finance "Enlist th...

Q. Enlist the advantages of using Business Intelligence in Finance. [4 Marks] ...

"Explain the role of BI in finance with respect to i) Monitor Financial...

Q. Explain the role of BI in Finance with respect to: i) Monitor Financial...

2) Fraud Detection "Explain application of Business Intelligence in Fraud...

Q. Explain application of Business Intelligence in Fraud Detection with a suitab...

"With the help of a real time case study explain the applications of Business...

Q. With the help of a real time case study explain the applications of Business...

3) CRM using BI "Discuss the real time Applications of BI in Customer...

Q. Discuss the real time Applications of BI in Customer Relationship...

"Enlist the advantages of using Business Intelligence in Customer Relationship...

Q. Enlist the advantages of using Business Intelligence in Customer Relationsh...

4) ERP and BI "Write a short note on: ERP and Business Intelligence." [6]

Q. Write a short note on: ERP and Business Intelligence. [6 Marks] **1** Simple...

⚡ MEDIUM REPEATED 5) Banking Applications of BI Asked: Apr 2023 Nov 2024...

Q. Explain the applications of Business Intelligence (BI) in Banking Sector. [8...

8) Types of Business Analytics / BA vs BI Asked: Apr 2024 Nov 2023

Q. Explain the Types of Business Analytics. [8 Marks] **1** Simple Introduction...

UNIT 6 → (Q7 & Q8) 🔥 MOST REPEATED QUESTIONS 1) Issues of Legality,...

Q. Write a short note on: Issues of Legality, Privacy and Ethics. [6 Marks] **1**...

2) Cloud Computing and BI "Write a short note on: Cloud Computing and...

Q. Write a short note on: Cloud Computing and Business Intelligence. [6 Marks...

3) BI Search and Text Analytics "Explain BI Search and Text Analytics with the...

Q. Explain BI Search and Text Analytics with the help of diagram. [6 Marks] **1**...

4) Analytics Applications for Consumers "Discuss the Analytical Applications for...

Q. Discuss the Analytical Applications for Consumers with respect to Online...

⚡ MEDIUM REPEATED 5) BI Strategy / Critical Success Factors Asked: Nov 202...

Q. Explain Business Intelligence (BI) Strategy and Critical Success Factors. [8...

6) Predicting Future using Data Analysis Asked: Apr 2024 Nov 2023 7) Rich...

Q. Explain Predicting Future using Data Analysis. [6 Marks] **1** Simple...



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BAI(2024 nov).pdf



BAI(april 2024).pdf



BAI(nov 2023).pdf

i am a final yera enginerring student studied in sppu brach
IT
here i attched pyq can u extract most repeater to less
repeated question in unit 3 means Q.1,2 and uit 4 means
Q.3,4, unit 5 means Q.5,6 and unit 6 means Q.7,8

5/11/2026, 2:01:33 PM



Yes. I analyzed all 4 PYQs of SPPU BAI (Business Analytics & Intelligence) and extracted the **most repeated** → **less repeated questions** unit-wise.

Sources used:

April 2023 BAI (april 2023).pdf

Nov 2023 BAI(nov 2023).pdf

April 2024 BAI(april 2024).pdf

Nov 2024 BAI(2024 nov).pdf

UNIT 3 → (Q1 & Q2)

MOST REPEATED QUESTIONS

1) Balanced Scorecard / Four Perspectives

Asked in:

- Apr 2023
- Nov 2024
- Apr 2024

Important Topics:

- Definition
 - 4 Perspectives
 - Financial
 - Customer
 - Internal Process
 - Learning & Growth
 - Diagram
 - Example
-

2) Dashboard Design / Characteristics / Pitfalls

Asked in:

- Apr 2023
- Nov 2023
- Apr 2024

Important:

- Dashboard characteristics
- Dashboard design
- Pitfalls in dashboard design
- Best practices

3) Data Visualization

Asked in:

- Nov 2024
- Apr 2024
- Nov 2023

Important:

- Definition
 - Need
 - Importance
 - Types of charts
-

4) BPM (Business Performance Management)

Asked in:

- Apr 2023
- Nov 2024
- Apr 2024

Important:

- Closed loop BPM cycle
 - BPM definition
 - Processes
-

MEDIUM REPEATED

5) Power BI / Tableau as BI Tool

Asked:

- Tableau → Apr 2023
- Power BI → Apr 2023 OR, Nov 2023

Important:

- Features
 - Advantages
 - Use in BI
-

6) KPI (Key Performance Indicators)

Asked:

- Nov 2023
- Nov 2024

Important:

- Definition
 - Types
 - Real-time examples
-

7) Business Reports

Asked:

- Nov 2024
- Apr 2024

Important:

- Types of reports
 - Components of Business Reporting System
-

UNIT 4 → (Q3 & Q4)

MOST REPEATED QUESTIONS

1) What-if Analysis

Asked in ALL papers:

- Apr 2023
- Nov 2023
- Apr 2024
- Nov 2024

VERY IMPORTANT 

Practice:

- Bread seller problem
 - Bookstore problem
 - Income-expense problem
-

2) Descriptive / Predictive / Prescriptive Analytics

Asked in:

- Nov 2024
- Apr 2024
- Nov 2023

Important:

- Definitions
 - Differences
 - Examples
-

3) Certainty, Uncertainty, Risk

Asked in:

- Apr 2023
- Nov 2024

Short note type but repeated.

4) Mathematical Model / Decision Support Model

Asked in:

- Nov 2024
- Apr 2024
- Nov 2023

Important:

- Structure of mathematical model
 - Components
 - Decision support
-

MEDIUM REPEATED

5) Mathematical Programming Optimization

Asked:

- Apr 2023
 - Nov 2023
-

6) Decision Trees / Decision Tables

Asked:

- Apr 2023
 - Nov 2023
-

7) Predictive Analytics Importance

Asked:

- Apr 2023
 - Apr 2024
-

UNIT 5 → (Q5 & Q6)

MOST REPEATED QUESTIONS

1) BI in Finance

Asked in:

- Apr 2023
- Nov 2024
- Nov 2023

VERY IMPORTANT 

Topics:

- Financial reporting
 - Forecasting
 - Budgeting
 - Monitoring performance
-

2) Fraud Detection using BI

Asked in:

- Nov 2024
- Apr 2024
- Nov 2023

Important:

- Real-time case study

- Banking fraud
 - Detection methods
-

3) ERP and Business Intelligence

Asked in:

- Apr 2023
- Nov 2024

Short note repeated.

4) Customer Relationship Management (CRM) using BI

Asked in:

- Apr 2023
 - Nov 2024
 - Apr 2024
-

 **MEDIUM REPEATED**

5) Banking Applications of BI

Asked:

- Apr 2023
- Nov 2024
- Nov 2023

6) Visual Analytics improving Revenue

Asked:

- Nov 2024
 - Nov 2023
-

7) Logistics & Production

Asked:

- Apr 2024
 - Nov 2023
-

8) Types of Business Analytics / BA vs BI

Asked:

- Apr 2024
 - Nov 2023
-

UNIT 6 → (Q7 & Q8)

MOST REPEATED QUESTIONS

1) Issues of Legality, Privacy & Ethics

Asked in:

- Apr 2023
- Nov 2024
- Apr 2024

VERY IMPORTANT ★★★★★

2) Cloud Computing and BI

Asked in:

- Nov 2024
 - Apr 2024
 - Nov 2023
-

3) BI Search and Text Analytics

Asked in:

- Apr 2023
- Nov 2024
- Nov 2023

Diagram-based important question.

4) Analytics Applications for Consumers

Asked in:

- Apr 2023
 - Nov 2024
 - Apr 2024
-

MEDIUM REPEATED

5) BI Strategy / Critical Success Factors

Asked:

- Nov 2024
 - Apr 2024
 - Nov 2023
-

6) Predicting Future using Data Analysis

Asked:

- Apr 2024
 - Nov 2023
-

7) Rich Reports

Asked:

- Apr 2023
 - Nov 2023
-



FINAL EXAM PRIORITY LIST (MOST IMPORTANT)

UNIT 3

1. Balanced Scorecard
 2. Dashboard Design + Pitfalls
 3. Data Visualization
 4. BPM Cycle
 5. KPI
-

UNIT 4

1. What-if Analysis ★★★★★
 2. Descriptive/Predictive/Prescriptive Analytics
 3. Mathematical Models
 4. Certainty-Uncertainty-Risk
-

UNIT 5

1. BI in Finance ★★★★★
2. Fraud Detection
3. CRM using BI
4. Banking Applications
5. ERP + BI



UNIT 6

1. Legality, Privacy & Ethics ★★★★★
2. Cloud Computing + BI
3. BI Search & Text Analytics
4. Consumer Analytics
5. BI Strategy Factors

If you want, next I can also make:

-  **Unit-wise IMP questions only**
-  **Most expected 2026 questions**
-  **Simple exam answers**
-  **Easy diagrams**
-  **2-page revision notes**
-  **70 marks preparation strategy**

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extract exact question with marks which appeared in exam unitwise from higher repeated to lower repeated i want exact question like above but exact questions

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UNIT 3 → (Q1 & Q2)



MOST REPEATED QUESTIONS

1) Balanced Scorecard

Repeated Questions

1. **"What is a Balanced Scorecard? Discuss the four perspectives of a balanced scorecard."** [6]

Apr 2023 BAI(april 2023).pdf

2. **"List and explain the four perspectives of Balanced Scorecards."** [8]

Nov 2024 BAI(2024 nov).pdf

3. **"How is a balanced scorecard used in Business?"** [5]

Apr 2024 BAI(april 2024).pdf

2) Dashboard Design / Dashboard

Repeated Questions

1. **"Discuss the common pitfalls in dashboard design."** [6]

Apr 2023 BAI(april 2023).pdf

2. **"Explain with example the best practices used in dashboard design." [5]**
Nov 2023 BAI(nov 2023).pdf
 3. **"Write a note on Dashboard design. Also explain the common characteristics of Dashboard." [5]**
Nov 2023 BAI(nov 2023).pdf
 4. **"What is the role of the Performance dashboards?" [5]**
Apr 2024 BAI(april 2024).pdf
 5. **"Discuss with appropriate examples the various types of dashboard." [8]**
Apr 2023 BAI (april 2023).pdf
-

3) Data Visualization

Repeated Questions

1. **"What is data visualization? Explain the need for data visualization in Business intelligence." [5]**
Nov 2024 BAI(2024 nov).pdf
2. **"What is data visualization? Explain the need for data visualization in Business intelligence." [5]**
Apr 2024 BAI(april 2024).pdf
3. **"Discuss the importance of data visualization and explain the types of data visualization." [8]**
Nov 2023 BAI(nov 2023).pdf
4. **"List and briefly explain different types of charts used for visualization." [8]**
Apr 2024 BAI(april 2024).pdf
5. **"Explain with neat diagrams various forms of charts and graphs used in visual analytics." [8]**
Nov 2023 BAI(nov 2023).pdf

4) BPM (Business Performance Management)

Repeated Questions

1. **"Write a note on Business Performance Management."** [6]
Apr 2023 BAI(april 2023).pdf
2. **"List and explain the four processes that define a closed-loop BPM cycle."** [8]
Nov 2024 BAI(2024 nov).pdf
3. **"List and explain the four processes that define a closed-loop BPM cycle."** [8]
Apr 2024 BAI(april 2024).pdf

5) KPI

1. **"What is KPI? Explain the types of KPIs."** [5]
Nov 2023 BAI(nov 2023).pdf
2. **"List and justify any five KPIs for a real time business scenario."** [5]
Nov 2024 BAI(2024 nov).pdf

UNIT 4 → (Q3 & Q4)

MOST REPEATED QUESTIONS

1) What-if Analysis

Repeated Questions

1. **"Write a note on : i) Multiple Goals ii) Sensitivity Analysis iii) What-If Analysis and Goal Seeking" [9]**
Apr 2023 BAI(april 2023).pdf
 2. **Bread Seller What-if Analysis Question [8]**
Nov 2024 BAI(2024 nov).pdf
 3. **Bread Seller What-if Analysis Question [8]**
Apr 2024 BAI(april 2024).pdf
 4. **Bookstore What-if Analysis Question [9]**
Nov 2023 BAI(nov 2023).pdf
 5. **Income & Expense What-if Analysis Question [9]**
Apr 2024 BAI(april 2024).pdf
-

2) Descriptive / Predictive / Prescriptive Analytics

Repeated Questions

1. **"Write a short note on: i) Descriptive analytics ii) Predictive analytics iii) Prescriptive analytics" [9]**
Nov 2024 BAI(2024 nov).pdf
2. **"Write a short note on: i) Descriptive analytics ii) Predictive analytics iii) Prescriptive analytics" [9]**
Apr 2024 BAI(april 2024).pdf
3. **"What is descriptive analytics? Explain the importance of descriptive analytics with the help of an example." [5]**
Nov 2023 BAI(nov 2023).pdf
4. **"What is predictive analytics? Explain the importance of predictive analytics with the help of an example." [5]**
Apr 2023 BAI(april 2023).pdf

3) Certainty, Uncertainty, Risk

1. **"Write short note on : i) Certainty ii) Uncertainty iii) Risk" [9]**

Apr 2023 BAI (april 2023).pdf

2. **"Write short note on: i) Certainty ii) Uncertainty iii) Risk" [9]**

Nov 2024 BAI(2024 nov).pdf

4) Mathematical Models

1. **"Discuss structure of mathematical model for decision support." [8]**

Nov 2024 BAI(2024 nov).pdf

2. **"Discuss structure of mathematical model for decision support." [8]**

Apr 2024 BAI(april 2024).pdf

3. **"Discuss the various Components of Decision Support Mathematical Models." [5]**

Nov 2023 BAI(nov 2023).pdf

4. **"Discuss Mathematical Programming Optimization with an example." [5]**

Apr 2023 BAI (april 2023).pdf

5. **"Write a note on Mathematical Programming Optimization with an example." [7]**

Nov 2023 BAI(nov 2023).pdf

UNIT 5 → (Q5 & Q6)

MOST REPEATED QUESTIONS

1) BI in Finance

1. **"Enlist the advantages of using Business Intelligence in Finance." [4]**
Apr 2023 BAI (april 2023).pdf
 2. **"Explain role of Business Intelligence (BI) in finance with the help of an example." [6]**
Nov 2024 BAI(2024 nov).pdf
 3. **"Explain the role of BI in finance with respect to i) Monitor Financial Performance ii) Budgeting and Forecasting iii) Financial Reporting" [7]**
Nov 2023 BAI(nov 2023).pdf
-

2) Fraud Detection

1. **"Explain application of Business Intelligence in Fraud Detection with a suitable example." [6]**
Nov 2024 BAI(2024 nov).pdf
 2. **"Explain application of Business Intelligence in Fraud Detection with a suitable example." [6]**
Apr 2024 BAI(april 2024).pdf
 3. **"With the help of a real time case study explain the applications of Business Intelligence in Fraud Detection." [7]**
Nov 2023 BAI(nov 2023).pdf
-

3) CRM using BI

1. **"Discuss the real time Applications of BI in Customer Relationship Management with appropriate case study." [6]**
Apr 2023 BAI (april 2023).pdf

2. **"Enlist the advantages of using Business Intelligence in Customer Relationship Management."** [6]

Nov 2024 BAI(2024 nov).pdf

3. **"Enlist the advantages of using Business Intelligence in Customer Relationship Management."** [6]

Apr 2024 BAI(april 2024).pdf

4) ERP and BI

1. **"Write a short note on: ERP and Business Intelligence."** [6]

Apr 2023 BAI (april 2023).pdf

2. **"Write a short note on: ERP and Business Intelligence."** [6]

Nov 2024 BAI(2024 nov).pdf

UNIT 6 → (Q7 & Q8)

MOST REPEATED QUESTIONS

1) Issues of Legality, Privacy and Ethics

1. **"Write a short note on: Issues of legality, Privacy and Ethics."** [6]

Apr 2023 BAI (april 2023).pdf

2. **"Write a short note on: Issues of Legality, Privacy, and Ethics."** [6]

Nov 2024 BAI(2024 nov).pdf

3. **"Write a short note on: Issues of Legality, Privacy and Ethics."** [6]

Apr 2024 BAI(april 2024).pdf

2) Cloud Computing and BI

1. **"Write a short note on: Cloud Computing and Business Intelligence."** [6]
Nov 2024 BAI(2024 nov).pdf
 2. **"Write a short note on: Cloud Computing and Business Intelligence."** [6]
Apr 2024 BAI(april 2024).pdf
 3. **"Write a short note on: Cloud Computing and Business Intelligence."** [6]
Nov 2023 BAI(nov 2023).pdf
-

3) BI Search and Text Analytics

1. **"Explain BI Search and Text Analytics with the help of diagram."** [6]
Apr 2023 BAI (april 2023).pdf
 2. **"Explain BI Search and Text Analytics with the help of diagram."** [5]
Nov 2024 BAI(2024 nov).pdf
 3. **"Explain BI Search and Text Analytics with the help of diagram."** [5]
Nov 2023 BAI(nov 2023).pdf
-

4) Analytics Applications for Consumers

1. **"Discuss the Analytical Applications for Consumers with respect to online shopping websites."** [5]
Apr 2023 BAI (april 2023).pdf
2. **"Write a short note on Analytics Applications for Consumers."** [6]
Nov 2024 BAI(2024 nov).pdf

3. **"Write a short note on Analytics Applications for Consumers."** [5]

Apr 2024 BAI(april 2024).pdf

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i want to preparr for exam and score outoff

Generate a **sppu** university exam-ready answer for Business Analytics and Intelligence (BAI).

Goal: The answer should be written so that a student can **understand, remember, and reproduce it in the exam after reading only once.**

Requirements:

- Use **very simple English** (avoid heavy technical words).
- Explain the concept in **clear logical steps** so it is easy to remember.
- Do **not** give textbook definitions only. Explain the idea clearly.
- Include **simple real-life examples** from business, hospitals, e-commerce, logistics, education, etc.
- After giving the example, **explain the example clearly** so the concept becomes obvious.
- Use **short paragraphs + bullet points** so the answer looks clean in exam.
- Include a **simple diagram whenever possible**.
- After the diagram, **explain what the diagram means in simple words**.
- Highlight **important keywords** a student should write in exam.
- Add **memory tricks** if possible to help quick revision.

Answer Structure:

1 Simple Introduction (2–3 lines)

Explain what the concept means in easy language.

2 Concept Explanation (Step-by-Step)

Explain the idea in simple logical steps.

3 Simple Real-Life Example

Give a relatable example (business, hospital, online store,

etc.).

4 **Explain the Example**

Clearly explain how the example relates to the concept.

5 **Simple Diagram (if applicable)**

Draw a basic exam-style diagram using text.

6 **Diagram Explanation**

Explain each part of the diagram briefly.

7 **Key Points / Features**

Write important points students should remember for exams.

8 **Memory Trick (Optional but helpful)**

Give an easy way to remember the concept.

9 **Short Conclusion (2 lines)**

Summarize the idea clearly.

Subject: ****Business Analytics and Intelligence (BAI)****

Question: [Paste your BAI question here]

Marks: [5 / 8 / 10]

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Your prompt is already very strong.

I improved it further so ChatGPT generates **maximum-scoring SPPU exam answers** exactly in university-writing style.



FINAL IMPROVED MASTER PROMPT FOR SPPU BAI

Copy this prompt and paste it whenever you want any BAI answer.

Generate a **SPPU university end-sem exam-ready answer for Business Analytics and Intelligence (BAI)** that helps a student **score maximum marks in theory exams**.

The answer should be written in a way that a student can:

- understand the concept quickly,
 - remember it easily,
 - reproduce it in the exam without confusion,
 - and write a high-scoring university answer even with basic knowledge.
-

IMPORTANT INSTRUCTIONS

Language Style

- Use **very simple English**.
 - Avoid difficult technical jargon.
 - Explain concepts like teaching a beginner student.
 - Use short and easy sentences.
-

WRITING STYLE (VERY IMPORTANT)

The answer must look exactly like a **proper SPPU university paper answer**.

Use:

- Proper headings
- Point-wise explanation

- Numbering
- Short paragraphs
- Bullet points
- Clean formatting

The answer should look neat and easy to write in exams.

CONTENT REQUIREMENTS

1 Simple Introduction (2–3 lines)

- Explain the concept in easy words.
 - Do not use difficult textbook definitions only.
 - Make the meaning very clear.
-

2 Main Concept Explanation (Step-by-Step)

- Explain the topic in logical steps.
 - Use simple language.
 - Each point should be easy to remember.
 - Important keywords should be highlighted in **bold**.
-

3 Real-Life Example

Give one simple and relatable example from:

- Business
 - Hospital
 - Banking
 - E-commerce
 - Education
 - Logistics
 - Retail
 - Social Media
 - Online Shopping
etc.
-

4 Explain the Example Clearly

After giving the example:

- Explain HOW the example relates to the concept.
 - Make the concept obvious to the student.
-

5 Simple Exam Diagram (Very Important)

Whenever possible:

- Draw a neat and simple text diagram.
- Diagram should look easy to reproduce in exam.
- Avoid complex diagrams.

Example format:

Data Collection



Data Analysis



Decision Making



Better Business

6 Diagram Explanation

Explain each block of the diagram in 1–2 simple lines.

7 Key Points / Features / Advantages

Write important exam points such as:

- Features
- Advantages
- Applications
- Importance
- Benefits
- Limitations (if needed)

Use bullets for better presentation.

8 Memory Trick / Easy Shortcut (IMPORTANT)

If possible:

- Give a small memory trick,
 - acronym,
 - shortcut sentence,
 - or easy logic to remember points quickly during revision.
-

9 Exam Writing Tips

Mention:

- which keywords are important,
 - what should be underlined,
 - which diagram can fetch extra marks.
-

10 Short Conclusion (2 lines)

Summarize the concept in simple words.

EXTRA RULES FOR HIGH SCORING ANSWERS

- Write the answer according to **SPPU paper-writing style**.
- The answer should feel like a **topper's answer sheet**.
- Add proper spacing between points.

- Keep explanations simple but meaningful.
 - Make the answer easy to revise one day before exam.
 - If the question is for:
 - **5 marks** → medium answer
 - **8 marks** → detailed answer with diagram + example
 - **10 marks** → full detailed university answer
-

SUBJECT

Business Analytics and Intelligence (BAI)

QUESTION

[Paste Question Here]

MARKS

[5 / 8 / 10]

OPTIONAL EXTRA INSTRUCTION (VERY USEFUL)

At the end also provide:

- ✓ "Last Minute Revision Points"
- ✓ "Most Important Keywords"

- ✔ "Possible Viva Questions with Answers"
- ✔ "Expected Follow-up Questions in SPPU Exam"

This helps in theory + viva preparation together.

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unit 3

1) Balanced Scorecard

Repeated Questions

"What is a Balanced Scorecard? Discuss the four perspectives of a balanced scorecard." [10]

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Q. What is a Balanced Scorecard? Discuss the four perspectives of a balanced scorecard. [10 Marks]

1 Simple Introduction

A **Balanced Scorecard (BSC)** is a performance management tool used by organizations to measure and improve business performance.

It helps a company check whether it is achieving its goals not only financially, but also in customer satisfaction, internal work process, and employee growth.

👉 It is called "**Balanced**" because it balances both:

- Financial performance
- Non-financial performance

2 Concept Explanation (Step-by-Step)



Earlier, companies focused only on:

- Profit
- Sales
- Revenue

But this was not enough because:

- Happy customers are important
- Employee learning is important
- Good internal processes are important

So, the **Balanced Scorecard** was introduced.

It measures performance from **4 important perspectives**:

1. Financial Perspective
2. Customer Perspective
3. Internal Business Process Perspective
4. Learning and Growth Perspective

These four perspectives together give a complete picture of business performance.

3 Four Perspectives of Balanced Scorecard

A) Financial Perspective

Meaning:

It checks whether the company is financially successful.

Focus:

- Profit
- Revenue
- Cost reduction
- Return on investment

Example:

A company checks whether yearly profit increased by 15%.

Important Keywords:

- Profitability
 - Revenue Growth
 - Cost Control
 - ROI
-

B) Customer Perspective

Meaning:

It checks whether customers are happy with products and services.

Focus:

- Customer satisfaction
- Customer retention
- Product quality

- Service quality

Example:

An online shopping company checks customer ratings and feedback.

Important Keywords:

- Customer Satisfaction
 - Loyalty
 - Market Share
 - Customer Service
-

C) Internal Business Process Perspective

Meaning:

It checks how efficiently internal business operations are working.

Focus:

- Production speed
- Error reduction
- Quality improvement
- Process efficiency

Example:

A hospital reduces patient waiting time using better management.

Important Keywords:

- Efficiency
 - Quality
 - Productivity
 - Process Improvement
-

D) Learning and Growth Perspective

Meaning:

It focuses on employee improvement and future growth.

Focus:

- Employee training
- Skill development
- Innovation
- Technology improvement

Example:

An IT company gives AI training to employees.

Important Keywords:

- Employee Development
 - Innovation
 - Learning
 - Growth
-

4 Simple Real-Life Example

Example: Amazon / Online Shopping Company

Amazon uses Balanced Scorecard to improve overall business performance.

Perspective	Example
Financial	Increase yearly profit
Customer	Fast delivery & good service
Internal Process	Improve warehouse management
Learning & Growth	Employee technology training

5 Explain the Example Clearly

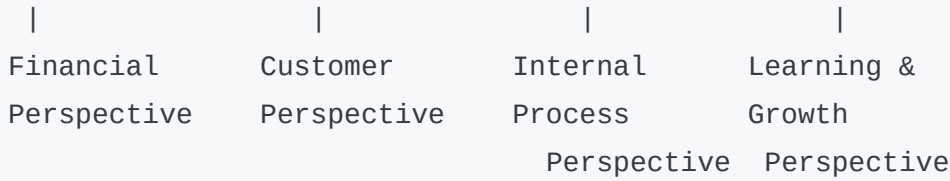
- Amazon checks profits → Financial Perspective
- Amazon improves customer service → Customer Perspective
- Amazon improves delivery system → Internal Process Perspective
- Amazon trains employees → Learning & Growth Perspective

Thus, Balanced Scorecard helps Amazon improve in all areas, not only money.

6 Simple Diagram

Balanced Scorecard

|



7 Diagram Explanation

Financial Perspective

Measures profit and revenue performance.

Customer Perspective

Measures customer happiness and satisfaction.

Internal Process Perspective

Measures efficiency of business operations.

Learning & Growth Perspective

Measures employee learning and future development.

8 Key Points / Features

Features of Balanced Scorecard

- Measures complete business performance
- Balances financial and non-financial factors

- Improves decision making
 - Helps achieve business goals
 - Improves customer satisfaction
 - Supports long-term growth
 - Increases employee performance
-

9 Advantages

- Better business planning
 - Improved communication
 - Higher customer satisfaction
 - Better employee performance
 - Better monitoring of business activities
-

10 Memory Trick

Memory Trick for 4 Perspectives:

F C I L

👉 **"Friendly Customers Improve Learning"**

- **F** → Financial
- **C** → Customer
- **I** → Internal Process
- **L** → Learning & Growth

Easy to remember in exam.

1 1 Exam Writing Tips

✓ Draw the diagram properly

✓ Underline keywords like:

- Financial Perspective
- Customer Satisfaction
- Internal Process
- Learning & Growth

✓ Write one example for each perspective

✓ Use headings and bullet points for neat presentation

1 2 Short Conclusion

Balanced Scorecard is an important tool used to measure overall organizational performance. It helps businesses improve financial performance, customer satisfaction, internal efficiency, and employee growth together.

Last Minute Revision Points

- Balanced Scorecard = Performance Measurement Tool
- 4 Perspectives:

1. Financial
 2. Customer
 3. Internal Process
 4. Learning & Growth
- Used for overall business improvement
 - Balances financial + non-financial performance
-

Most Important Keywords

- Balanced Scorecard
 - Performance Management
 - Financial Perspective
 - Customer Satisfaction
 - Internal Business Process
 - Learning & Growth
 - KPI
 - Business Performance
-

Possible Viva Questions with Answers

Q1) What is Balanced Scorecard?

Balanced Scorecard is a tool used to measure organizational performance from different perspectives.

Q2) Why is it called Balanced?

Because it balances financial and non-financial performance measures.

Q3) Name the four perspectives.

- Financial
 - Customer
 - Internal Process
 - Learning & Growth
-

Q4) Which perspective focuses on employees?

Learning and Growth Perspective.

Q5) Which perspective focuses on customers?

Customer Perspective.

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2. "How is a balanced scorecard used in Business?" [5]

5/13/2026, 7:55:44 PM



Q. How is a Balanced Scorecard used in Business? [5 Marks]

1 Simple Introduction

A **Balanced Scorecard (BSC)** is a tool used by businesses to measure and improve overall performance.

It helps organizations check whether they are achieving their business goals in different areas like profit, customers, internal work, and employee growth.

2 How Balanced Scorecard is Used in Business

Businesses use Balanced Scorecard in the following ways:

1) Measuring Financial Performance

Companies use BSC to track:

- Profit
- Revenue
- Sales growth
- Cost reduction

Example:

A retail company checks whether yearly profit has increased.

2) Improving Customer Satisfaction

Businesses use BSC to understand:

- Customer feedback
- Customer loyalty
- Service quality

Example:

Amazon tracks delivery speed and customer ratings.

3) Improving Internal Business Processes

Balanced Scorecard helps improve:

- Work efficiency
- Production quality
- Error reduction

Example:

A hospital reduces patient waiting time.

4) Employee Learning and Growth

Businesses use BSC to improve:

- Employee skills
- Training
- Innovation
- Technology knowledge

Example:

An IT company gives AI training to employees.

5) Better Decision Making

Managers use Balanced Scorecard reports to:

- Identify problems
 - Set goals
 - Improve business performance
-

3 Simple Real-Life Example

Example: Online Shopping Company

An e-commerce company uses Balanced Scorecard:

Area	Usage
Financial	Increase profit
Customer	Fast delivery
Internal Process	Better warehouse management
Learning & Growth	Employee training

4 Explain the Example

- Profit checking helps financial growth.
- Fast delivery improves customer satisfaction.
- Better warehouse system improves internal processes.
- Employee training improves future business growth.

Thus, Balanced Scorecard helps improve the complete business.

5 Simple Diagram



6 Diagram Explanation

Finance

Checks profit and revenue.

Customer

Checks customer happiness.

Internal Process

Checks efficiency of business operations.

Learning & Growth

Checks employee improvement and innovation.

7 Key Points / Features

- Measures overall business performance
- Improves customer satisfaction
- Helps in decision making
- Improves employee performance
- Supports long-term business growth
- Balances financial and non-financial goals

8 Memory Trick

Memory Trick → F C I L

👉 **F** → Financial

👉 **C** → Customer

👉 **I** → Internal Process

👉 **L** → Learning & Growth

Sentence:

“Friendly Customers Improve Learning”

9 Exam Writing Tips

✅ Draw simple diagram

✅ Underline keywords:

- Customer Satisfaction
- Financial Performance
- Internal Process
- Learning & Growth

✅ Write one example for better marks

10 Short Conclusion

Balanced Scorecard is an important business tool used to measure and improve organizational performance from multiple perspectives. It helps businesses achieve both short-term and long-term goals effectively.

★ Last Minute Revision Points

- BSC = Performance Measurement Tool
 - Used in:
 - Finance
 - Customer Service
 - Internal Processes
 - Employee Growth
 - Helps in better business decisions
-

★ Most Important Keywords

- Balanced Scorecard
 - Financial Performance
 - Customer Satisfaction
 - Internal Process
 - Learning & Growth
 - Business Goals
 - Decision Making
-

★ Viva Questions with Answers

Q1) What is Balanced Scorecard?

Balanced Scorecard is a tool used to measure business performance.

Q2) Why is Balanced Scorecard important?

It helps improve overall business performance.

Q3) Which four areas are measured in Balanced Scorecard?

- Financial
- Customer
- Internal Process
- Learning & Growth

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3) Dashboard Design / Dashboard

Repeated Questions

"Discuss the common pitfalls in dashboard design." [6]

Apr 2023

"Explain with example the best practices used in dashboard design." [5]

Nov 2023

"Write a note on Dashboard design. Also explain the common characteristics of Dashboard." [5]

5/13/2026, 7:59:36 PM



Q. Discuss the common pitfalls in Dashboard Design. [6 Marks]

1 Simple Introduction

A **dashboard** is a visual screen that shows important business information like sales, profit, customer data, and performance in one place.

If a dashboard is not designed properly, users may get confused and make wrong business decisions. These mistakes are called **dashboard design pitfalls**.

2 Common Pitfalls in Dashboard Design

1) Too Much Information

Meaning:

Adding too many charts, tables, and data on one screen.

Problem:

Users become confused and cannot identify important information.

Example:

A sales dashboard showing 50 graphs together.

2) Poor Choice of Charts

Meaning:

Using the wrong chart type for data visualization.

Problem:

Data becomes difficult to understand.

Example:

Using a pie chart for large complex data instead of a bar chart.

3) Too Many Colors

Meaning:

Using excessive bright colors and animations.

Problem:

Dashboard looks confusing and unprofessional.

Example:

Red, green, blue, yellow colors everywhere on one screen.

4) Lack of Important KPIs

Meaning:

Dashboard does not show important Key Performance Indicators (KPIs).

Problem:

Managers cannot monitor business performance properly.

Example:

A hospital dashboard without patient waiting time KPI.

5) Slow Dashboard Performance

Meaning:

Dashboard takes too much time to load.

Problem:

Users waste time and productivity decreases.

Example:

Large reports causing dashboard delay.

6) No Clear Goal

Meaning:

Dashboard is created without understanding user requirements.

Problem:

Irrelevant information is displayed.

Example:

Finance manager dashboard showing unnecessary HR data.

3 Simple Real-Life Example

Example: E-Commerce Dashboard

An online shopping company created a dashboard with:

- Too many graphs
- Bright colors
- Unnecessary information

Managers could not quickly identify:

- Best-selling products
- Customer complaints

- Daily sales

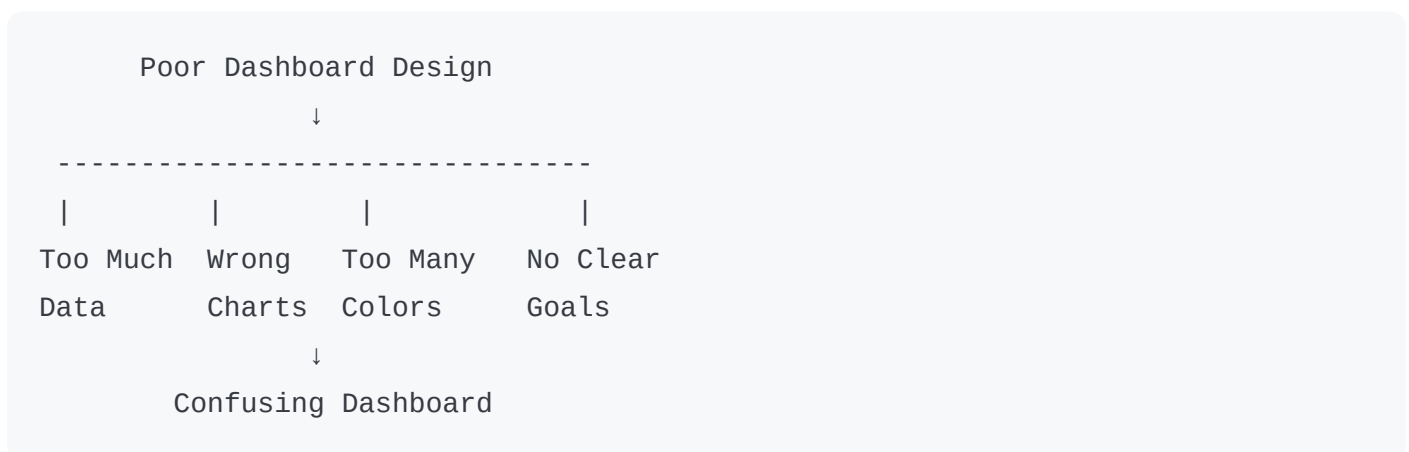
As a result, decision-making became difficult.

4 Explain the Example

- Too much data created confusion.
- Wrong chart selection reduced understanding.
- Missing important KPIs affected monitoring.
- Dashboard became less useful for managers.

Thus, good dashboard design is very important.

5 Simple Diagram



6 Diagram Explanation

Too Much Data

Creates information overload.

Wrong Charts

Makes analysis difficult.

Too Many Colors

Reduces readability.

No Clear Goals

Dashboard becomes less useful.

All these problems lead to a confusing dashboard.

Key Points / Features

Common Dashboard Design Pitfalls

- Information overload
 - Wrong visualization techniques
 - Excessive colors and graphics
 - Missing KPIs
 - Slow performance
 - Lack of clarity
 - Poor layout design
-

8 Memory Trick

Memory Trick → D C C K G

- 👉 **D** → Too much Data
- 👉 **C** → Wrong Charts
- 👉 **C** → Too many Colors
- 👉 **K** → Missing KPIs
- 👉 **G** → No Goal

Sentence:

“Data Charts Colors Kill Goals”

9 Exam Writing Tips

- ✅ Draw diagram neatly
 - ✅ Underline keywords:
 - KPI
 - Information Overload
 - Visualization
 - Dashboard Performance
 - ✅ Write one real-life example for extra marks
-

10 Short Conclusion

Dashboard design should be simple, clear, and user-friendly. Avoiding common pitfalls helps organizations make faster and better business decisions.

★ Last Minute Revision Points

- Dashboard = Visual business report
 - Common pitfalls:
 - Too much information
 - Wrong charts
 - Too many colors
 - Missing KPIs
 - Slow performance
 - No clear goals
-

★ Most Important Keywords

- Dashboard Design
 - KPI
 - Information Overload
 - Visualization
 - User-Friendly
 - Business Decision
 - Dashboard Performance
-

★ Viva Questions with Answers

Q1) What is a dashboard?

A dashboard is a visual screen showing important business information.

Q2) What is the biggest dashboard design mistake?

Showing too much unnecessary information.

Q3) Why are KPIs important in dashboards?

KPIs help measure business performance.

Q4) Why should dashboards use limited colors?

Too many colors create confusion.

Q. Explain with example the best practices used in Dashboard Design. [5 Marks]

1 Simple Introduction

Dashboard design means arranging business information in a clear and simple visual format.

Good dashboard design helps managers understand data quickly and make better decisions.

2 Best Practices in Dashboard Design

1) Keep Dashboard Simple

Only important information should be shown.

Example:

Sales dashboard showing only:

- Sales
 - Profit
 - Orders
-

2) Use Proper Charts

Choose correct charts for correct data.

Example:

- Bar chart → comparison

- Pie chart → percentage
-

3) Use Limited Colors

Use simple and meaningful colors.

Example:

Green = Profit

Red = Loss

4) Highlight Important KPIs

Important business metrics should be clearly visible.

Example:

Daily sales at top of dashboard.

5) Arrange Information Properly

Most important data should appear first.

3 Simple Real-Life Example

Example: Hospital Dashboard

A hospital dashboard shows:

- Patient count
- Waiting time
- Available beds
- Emergency cases

All information is shown clearly using proper charts and limited colors.

4 Diagram

Good Dashboard Design



| Simple | Clear | KPI | Charts |



Better Decision Making

5 Key Points

- Simple layout
- Correct visualization
- Limited colors
- Proper KPI display
- Easy understanding

6 Memory Trick

"S C K C"

- **S** → Simple
 - **C** → Clear
 - **K** → KPI
 - **C** → Correct Charts
-

7 Conclusion

Good dashboard design improves understanding, saves time, and supports better business decisions.

Q. Write a note on Dashboard Design. Also explain the common characteristics of Dashboard. [5 Marks]

1 Simple Introduction

Dashboard design is the process of presenting important business information visually in an organized and easy-to-understand format.

Dashboards help managers monitor business performance quickly.

2 Common Characteristics of Dashboard

1) Visual Presentation

Uses:

- Charts
 - Graphs
 - Tables
 - KPIs
-

2) Real-Time Information

Shows updated business data continuously.

Example:

Live sales dashboard.

3) User Friendly

Dashboard should be simple and easy to understand.

4) Interactive Nature

Users can filter and analyze data easily.

5) Quick Decision Support

Helps managers take fast decisions.

3 Simple Diagram

Dashboard



| Visual | Real-Time | Simple |



Better Business Decisions

4 Key Points

- Easy visualization
 - Real-time monitoring
 - Interactive design
 - Supports decision-making
 - User-friendly interface
-

5 Memory Trick

“V R U I”

- **V** → Visual
 - **R** → Real-time
 - **U** → User-friendly
 - **I** → Interactive
-

6 Conclusion

A well-designed dashboard helps organizations monitor performance, analyze data, and make quick business decisions effectively.

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4. "What is the role of the Performance dashboards?" [5]
Apr 2024
"Discuss with appropriate examples the various types of dashboard." [8]

5/13/2026, 8:07:31 PM



Q. What is the role of the Performance Dashboards? [5 Marks]

1 Simple Introduction

A **Performance Dashboard** is a visual tool that displays important business data like sales, profit, customer feedback, and employee performance in one screen.

It helps managers monitor business activities and make quick decisions.

2 Role of Performance Dashboards

1) Monitoring Business Performance

Dashboards help organizations track:

- Sales
- Profit

- Productivity
- Customer satisfaction

Example:

A retail company checks daily sales performance.

2) Quick Decision Making

Managers can quickly identify problems and take action.

Example:

A hospital dashboard shows increase in patient waiting time, so management adds more staff.

3) Real-Time Information

Dashboards show updated business data continuously.

Example:

An online shopping company monitors live order status.

4) Identifying Problems Easily

Performance dashboards help find:

- Low sales

- Customer complaints
 - Delayed delivery
 - Employee performance issues
-

5) Improving Communication

All departments can view the same business information.

Example:

Finance, HR, and Sales teams use the same dashboard.

3 Simple Real-Life Example

Example: E-Commerce Company

An online shopping company uses a dashboard to monitor:

- Daily orders
- Product returns
- Delivery status
- Customer ratings

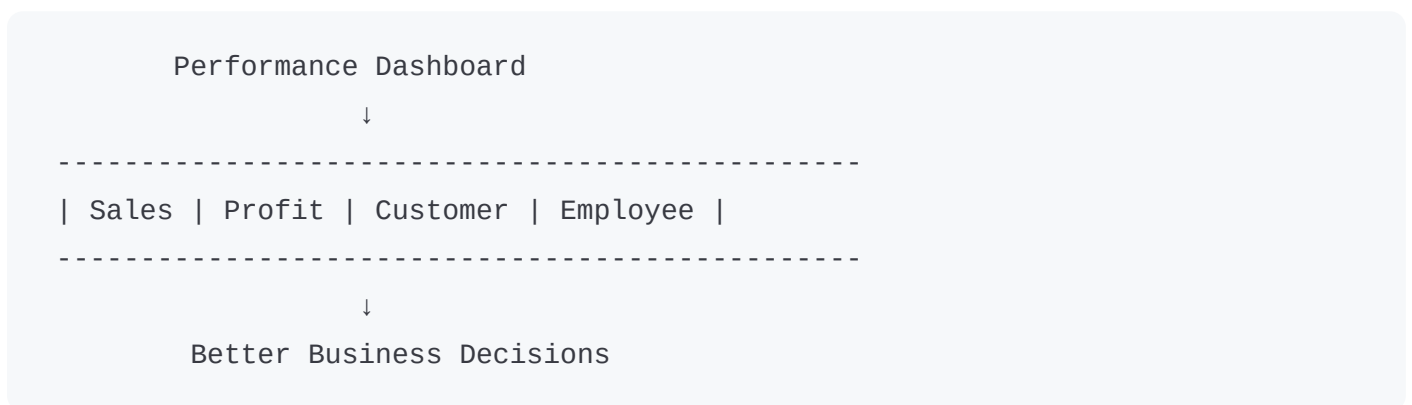
Managers use this information to improve service quality and profit.

4 Explain the Example

- Dashboard shows real-time sales data.
- Customer complaints can be identified quickly.
- Delivery delays can be reduced.
- Managers can make faster decisions.

Thus, dashboards improve overall business performance.

5 Simple Diagram



6 Diagram Explanation

Sales

Shows product sales performance.

Profit

Displays financial growth.

Customer

Shows customer satisfaction data.

Employee

Measures employee productivity.

All together help in better business management.

7 Key Points / Features

Roles of Performance Dashboard

- Monitors business activities
 - Supports quick decision making
 - Shows real-time information
 - Improves communication
 - Identifies problems quickly
 - Improves business performance
-

8 Memory Trick

“M Q R I”

- **M** → Monitoring
- **Q** → Quick Decisions
- **R** → Real-Time Data
- **I** → Identify Problems

Sentence:

9 Exam Writing Tips

✓ Draw neat dashboard diagram

✓ Underline keywords:

- Real-Time Information
- Monitoring
- KPI
- Decision Making

✓ Write one real-life example for extra marks

10 Short Conclusion

Performance dashboards help organizations monitor important business information visually and support faster and better decision making.

★ Last Minute Revision Points

- Dashboard = Visual business monitoring tool
- Main roles:
 - Monitoring
 - Real-time updates
 - Quick decisions

- Problem identification
-

★ Viva Questions with Answers

Q1) What is a performance dashboard?

A performance dashboard is a visual tool used to monitor business performance.

Q2) Why are dashboards important?

They help managers make quick and better decisions.

Q3) What type of information is shown on dashboards?

Sales, profit, customer data, KPIs, employee performance.

Q. Discuss with appropriate examples the various types of Dashboard. [8 Marks]

1 Simple Introduction

A dashboard is a visual display of important business information shown using charts, graphs, and KPIs.

Different dashboards are used for different business purposes like monitoring performance, analysis, and decision making.

2 Types of Dashboard

Mainly dashboards are of three types:

1. Strategic Dashboard
 2. Operational Dashboard
 3. Analytical Dashboard
-

3 Strategic Dashboard

Meaning

Strategic dashboards are used by top-level management for long-term planning and business goals.

Focus

- Business growth
- Profit

- Market performance
- Company targets

Example

CEO dashboard showing yearly profit and market growth.

Operational Dashboard

Meaning

Operational dashboards monitor daily business activities.

Focus

- Daily sales
- Employee attendance
- Production status
- Delivery tracking

Example

A hospital dashboard showing daily patient count.

Analytical Dashboard

Meaning

Analytical dashboards are used for detailed data analysis and business insights.

Focus

- Trends
- Comparisons
- Predictions
- Data analysis

Example

An e-commerce company analyzing customer buying patterns.

6 Simple Real-Life Example

Example: Amazon

Dashboard Type	Usage
Strategic	Annual business growth
Operational	Daily orders and delivery
Analytical	Customer purchase analysis

7 Explain the Example

- Strategic dashboard helps Amazon plan future growth.
- Operational dashboard manages daily business operations.
- Analytical dashboard helps understand customer behavior.

Thus, different dashboards serve different business purposes.

8 Simple Diagram

Types of Dashboard



| Strategic | Operational | Analytical |

9 Diagram Explanation

Strategic Dashboard

Used for long-term goals and planning.

Operational Dashboard

Used for daily operations monitoring.

Analytical Dashboard

Used for detailed analysis and decision making.

10 Key Points / Features

Strategic Dashboard

- Long-term planning
- Used by top management

Operational Dashboard

- Real-time monitoring
- Daily business activities

Analytical Dashboard

- Data analysis
 - Trend identification
-

11 Memory Trick

“S O A”

- **S** → Strategic
- **O** → Operational
- **A** → Analytical

Sentence:

“Smart Organizations Analyze”

1 2 Exam Writing Tips

✓ Draw dashboard types diagram

✓ Underline:

- Strategic
- Operational
- Analytical

✓ Write one example for each type

1 3 Short Conclusion

Different types of dashboards help organizations monitor, analyze, and improve business performance effectively according to business needs.

★ Last Minute Revision Points

- 3 Types of Dashboard:
 1. Strategic
 2. Operational
 3. Analytical
- Strategic → Long-term planning
- Operational → Daily monitoring

★ Viva Questions with Answers

Q1) Name the types of dashboards.

- Strategic
 - Operational
 - Analytical
-

Q2) Which dashboard is used for daily monitoring?

Operational dashboard.

Q3) Which dashboard is used by top management?

Strategic dashboard.

Q4) Which dashboard is used for data analysis?

Analytical dashboard.

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"What is data visualization? Explain the need for data visualization in Business intelligence." [5]

Apr 2024

"Discuss the importance of data visualization and explain the types of data visualization." [8]

5/13/2026, 9:33:19 PM



Q. What is Data Visualization? Explain the need for Data Visualization in Business Intelligence. [5 Marks]

1 Simple Introduction

Data Visualization means representing data in visual form using:

- Charts
- Graphs
- Maps
- Dashboards

It helps people understand large amounts of data quickly and easily.

In **Business Intelligence (BI)**, data visualization helps managers analyze business data and make better decisions.

2 Need for Data Visualization in Business Intelligence

1) Easy Understanding of Data

Large data in table format is difficult to understand.

Visualization converts complex data into simple graphs and charts.

Example:

A sales graph is easier to understand than a long Excel table.

2) Faster Decision Making

Managers can quickly identify trends and problems.

Example:

A company quickly identifies low sales in one region using a bar chart.

3) Identifying Trends and Patterns

Visualization helps find:

- Growth trends
- Customer behavior
- Seasonal changes

Example:

An online shopping company identifies festival season sales increase.

4) Better Comparison

Charts help compare products, departments, or yearly performance easily.

Example:

Comparing yearly profits using a line chart.

5) Real-Time Monitoring

Dashboards show live business information.

Example:

A hospital monitors live patient data using dashboards.

3 Simple Real-Life Example

Example: E-Commerce Company

An online shopping company uses:

- Bar charts for sales
- Pie charts for product categories

- Dashboards for live orders

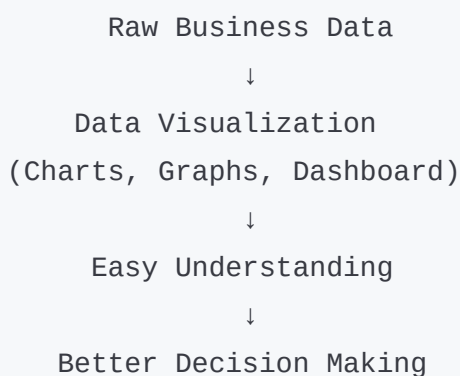
Managers can easily understand customer buying behavior and improve sales.

4 Explain the Example

- Charts make sales data easy to understand.
- Dashboards help monitor orders in real time.
- Visualization helps managers take quick business decisions.

Thus, data visualization improves business intelligence.

5 Simple Diagram



6 Diagram Explanation

Raw Business Data

Large amount of business information.

Data Visualization

Converts data into visual format.

Easy Understanding

Users understand information quickly.

Better Decision Making

Managers take faster and smarter decisions.

7 Key Points / Features

Need for Data Visualization

- Easy understanding of complex data
 - Quick decision making
 - Trend identification
 - Better comparison
 - Real-time monitoring
 - Improved business analysis
-

8 Memory Trick

“U D T C R”

- **U** → Understanding
- **D** → Decision Making
- **T** → Trends
- **C** → Comparison
- **R** → Real-Time Monitoring

Sentence:

“Understanding Data Through Clear Reports”

9 Exam Writing Tips

✓ Draw diagram neatly

✓ Underline keywords:

- Charts
- Graphs
- Dashboards
- Decision Making
- Trends

✓ Give one real-life example

10 Short Conclusion

Data visualization helps organizations understand business data easily and supports better, faster, and smarter business decisions.

★ Last Minute Revision Points

- Data Visualization = Visual representation of data
- Uses:
 - Charts
 - Graphs
 - Dashboards
- Need:
 - Easy understanding
 - Quick decisions
 - Trend analysis

★ Viva Questions with Answers

Q1) What is data visualization?

Data visualization means representing data visually using charts and graphs.

Q2) Why is data visualization important?

It helps understand complex data easily.

Q3) Name some visualization tools.

- Charts
 - Graphs
 - Dashboards
 - Maps
-

Q. Discuss the importance of Data Visualization and explain the types of Data Visualization. [8 Marks]

1 Simple Introduction

Data Visualization is the process of presenting data visually using graphs, charts, dashboards, and maps.

It helps businesses understand data quickly and make better decisions.

2 Importance of Data Visualization

1) Simplifies Complex Data

Large data becomes easy to understand.

Example:

Sales reports shown as graphs instead of large tables.

2) Helps in Quick Decision Making

Managers can quickly identify business problems.

Example:

Low sales identified using dashboards.

3) Identifies Trends and Patterns

Visualization helps identify:

- Customer behavior
 - Market trends
 - Business growth
-

4) Improves Communication

Visual reports are easier to explain in meetings.

5) Supports Business Intelligence

Visualization improves data analysis and decision-making process.

3 Types of Data Visualization

A) Bar Chart

Use:

Used for comparison.

Example:

Comparing sales of different products.

B) Pie Chart

Use:

Shows percentage distribution.

Example:

Market share of companies.

C) Line Chart

Use:

Shows trends over time.

Example:

Monthly sales growth.

D) Histogram

Use:

Shows frequency distribution of data.

Example:

Student marks distribution.

E) Scatter Plot

Use:

Shows relationship between two variables.

Example:

Advertising cost vs sales.

F) Dashboard

Use:

Shows multiple business metrics together.

Example:

Hospital performance dashboard.

4 Simple Real-Life Example

Example: Hospital Dashboard

A hospital uses:

- Bar chart → Number of patients
 - Pie chart → Disease categories
 - Line chart → Monthly patient growth
 - Dashboard → Overall hospital performance
-

5 Explain the Example

Visualization helps hospital management:

- Monitor patient growth
 - Analyze diseases
 - Improve hospital services
 - Take faster decisions
-

6 Simple Diagram

Data Visualization



| Bar | Pie | Line | Histogram | Dashboard |

7 Diagram Explanation

Bar Chart

Used for comparison.

Pie Chart

Used for percentage representation.

Line Chart

Used for trend analysis.

Histogram

Used for frequency analysis.

Dashboard

Used for overall business monitoring.

8 Key Points / Features

Importance of Data Visualization

- Easy understanding
 - Better analysis
 - Faster decisions
 - Trend identification
 - Improved communication
-

9 Memory Trick

“B P L H D”

- **B** → Bar Chart
- **P** → Pie Chart
- **L** → Line Chart
- **H** → Histogram
- **D** → Dashboard

Sentence:

“Big People Love Helpful Dashboards”

10 Exam Writing Tips

✓ Draw charts diagram

✓ Underline:

- Trend Analysis
- Comparison
- Dashboard
- Visualization

✓ Write one example for each chart type

1 1 Short Conclusion

Data visualization plays an important role in Business Intelligence by simplifying data analysis and helping organizations make better business decisions quickly.

★ Last Minute Revision Points

Types:

- Bar Chart
- Pie Chart
- Line Chart
- Histogram
- Dashboard

Importance:

- Quick understanding
 - Better decisions
 - Trend analysis
 - Business monitoring
-

Viva Questions with Answers

Q1) What is data visualization?

It is the visual representation of data.

Q2) Which chart is used for comparison?

Bar chart.

Q3) Which chart is used for trends?

Line chart.

Q4) Why are dashboards important?

They show important business information in one screen.

5/13/2026, 9:33:21 PM



"List and briefly explain different types of charts used for visualization." [8]

5/13/2026, 9:39:14 PM



Q. List and briefly explain different types of charts used for visualization. [8 Marks]

1 Simple Introduction

Charts are graphical representations of data used in **Data Visualization**.

They help users understand business information quickly and easily by converting complex data into visual form.

Different charts are used for different purposes like:

- Comparison
- Trend analysis
- Percentage analysis
- Relationship analysis

2 Types of Charts used for Visualization

1) Bar Chart

Meaning

A bar chart uses rectangular bars to compare data values.

Use

Used for comparison between categories.

Example

Comparing sales of different products.

Example:

Product	Sales
Mobile	500
Laptop	300

Bar chart easily shows which product has higher sales.

2) Pie Chart

Meaning

A pie chart shows data in percentage form using slices of a circle.

Use

Used to show proportion or percentage distribution.

Example

Market share of companies.

Example:

- Samsung → 40%
 - Apple → 30%
 - Vivo → 20%
 - Others → 10%
-

3) Line Chart

Meaning

A line chart connects data points using lines.

Use

Used to show trends over time.

Example

Monthly sales growth of a company.

4) Histogram

Meaning

A histogram shows frequency distribution of continuous data.

Use

Used to analyze data distribution.

Example

Distribution of student marks in a class.

5) Scatter Plot

Meaning

Scatter plot shows relationship between two variables using dots.

Use

Used to identify correlation between variables.

Example

Advertising cost vs sales increase.

6) Area Chart

Meaning

An area chart is similar to a line chart but the area below the line is filled with color.

Use

Used to show quantity changes over time.

Example

Website traffic growth.

7) Bubble Chart

Meaning

Bubble chart displays data using bubbles of different sizes.

Use

Used to compare three dimensions of data.

Example

Sales, profit, and market share comparison.

8) Dashboard Chart

Meaning

Dashboard combines multiple charts and KPIs in one screen.

Use

Used for overall business monitoring.

Example

Hospital dashboard showing:

- Patient count
 - Bed availability
 - Emergency cases
-

3 Simple Real-Life Example

Example: Online Shopping Company

An e-commerce company uses:

- Bar chart → Product sales comparison
- Pie chart → Customer categories
- Line chart → Monthly revenue growth
- Dashboard → Overall performance monitoring

4 Explain the Example

- Bar chart helps compare product sales.
- Pie chart shows customer distribution.
- Line chart identifies growth trends.
- Dashboard helps managers monitor all activities together.

Thus, different charts help businesses analyze data effectively.

5 Simple Diagram

Types of Charts



| Bar | Pie | Line | Histogram | Scatter |



Better Data Visualization

6 Diagram Explanation

Bar Chart

Used for comparison.

Pie Chart

Used for percentage distribution.

Line Chart

Used for trend analysis.

Histogram

Used for frequency analysis.

Scatter Plot

Used for relationship analysis.

All charts help users understand data easily.

7 Key Points / Features

Advantages of Charts

- Easy understanding of data
- Quick comparison
- Better decision making
- Trend identification

- Attractive visual presentation
 - Simplifies complex data
-

8 Memory Trick

“B P L H S”

- **B** → Bar Chart
- **P** → Pie Chart
- **L** → Line Chart
- **H** → Histogram
- **S** → Scatter Plot

Sentence:

“Big People Like Helpful Statistics”

9 Exam Writing Tips

✓ Draw small chart diagrams if possible

✓ Underline keywords:

- Comparison
- Trend Analysis
- Percentage Distribution
- Correlation

✓ Write one example for each chart type

Short Conclusion

Different types of charts are used in data visualization to present business data clearly and effectively. They help organizations analyze information and make better business decisions.

Last Minute Revision Points

Important Charts:

- Bar Chart
- Pie Chart
- Line Chart
- Histogram
- Scatter Plot
- Dashboard

Uses:

- Comparison
 - Trends
 - Percentage
 - Relationship Analysis
-

Viva Questions with Answers

Q1) Which chart is used for comparison?

Bar chart.

Q2) Which chart is used for trend analysis?

Line chart.

Q3) Which chart shows percentage distribution?

Pie chart.

Q4) What is a dashboard?

Dashboard is a visual screen showing multiple charts and KPIs together.

5/13/2026, 9:39:16 PM



"Explain with neat diagrams various forms of charts and graphs used in visual analytics." [8]

5/13/2026, 9:39:30 PM



Q. Explain with neat diagrams various forms of charts and graphs used in Visual Analytics. [8 Marks]

1 Simple Introduction

Visual Analytics means analyzing data using visual tools like:

- Charts
- Graphs
- Dashboards
- Maps

These visual forms help businesses understand complex data easily and make better decisions.

Different charts are used for different purposes such as:

- Comparison
- Trend analysis
- Percentage analysis
- Relationship analysis

2 Various Forms of Charts and Graphs used in Visual Analytics

1) Bar Chart

Meaning

A bar chart represents data using rectangular bars.

Use

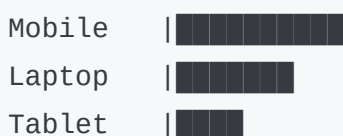
Used for comparing different categories.

Example

Comparing sales of products.

Diagram

Sales Comparison



Explanation

- Longer bar = higher value
- Easy comparison between products

2) Pie Chart

Meaning

Pie chart shows percentage distribution using slices of a circle.

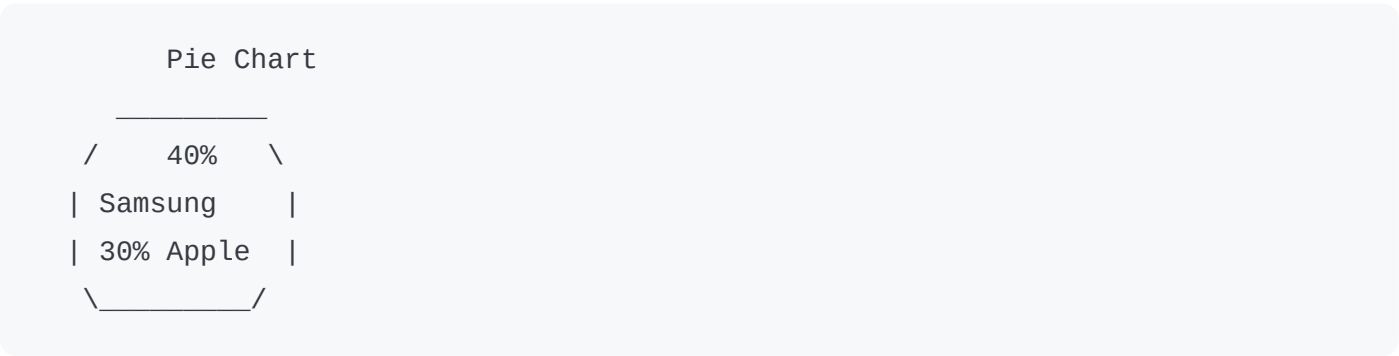
Use

Used to represent proportions.

Example

Market share of companies.

Diagram



Explanation

- Each slice represents percentage share.
 - Total circle = 100%
-

3) Line Chart

Meaning

Line chart connects data points with lines.

Use

Used to show trends over time.

Example

Monthly sales growth.

Diagram

Sales Trend

Value	Point
100	*
80	*
60	*

40 | *

Jan Feb Mar Apr

Explanation

- Shows increase or decrease over time.
- Useful for trend analysis.

4) Histogram

Meaning

Histogram shows frequency distribution of continuous data.

Use

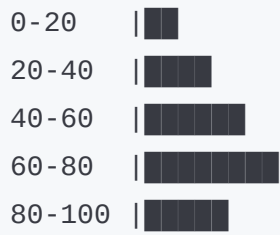
Used for statistical analysis.

Example

Student marks distribution.

Diagram

Marks Distribution



Explanation

- Shows how data is distributed.
- Higher bar = more frequency.

5) Scatter Plot

Meaning

Scatter plot shows relationship between two variables using dots.

Use

Used to identify correlation.

Example

Advertising cost vs sales.

Diagram



Explanation

- Dots show relationship between variables.
- Helps identify positive or negative correlation.

6) Area Chart

Meaning

Area chart is similar to line chart with filled area.

Use

Used to show quantity changes over time.

Example

Website traffic analysis.

Diagram

Traffic Growth



Explanation

- Filled area improves visibility.
 - Useful for growth analysis.
-

7) Dashboard

Meaning

Dashboard combines multiple charts and KPIs together.

Use

Used for overall business monitoring.

Example

Hospital management dashboard.

Diagram

```
-----  
| Sales | Profit | Users |  
-----  
| Charts | Graphs | KPIs |  
-----
```

Explanation

- Displays multiple business metrics.
 - Helps in quick decision making.
-

3 Simple Real-Life Example

Example: E-Commerce Company

An online shopping company uses:

- Bar chart → Product sales
- Pie chart → Customer categories
- Line chart → Revenue growth

- Dashboard → Overall business performance
-

4 Explain the Example

- Bar chart compares product sales.
- Pie chart shows customer percentage.
- Line chart identifies growth trend.
- Dashboard monitors all activities together.

Thus, visual analytics helps businesses understand data quickly and make smarter decisions.

5 Summary Diagram

Visual Analytics



| Bar | Pie | Line | Histogram | Scatter |



Better Business Decisions

6 Key Points / Features

Advantages of Charts and Graphs

- Easy understanding of data
 - Attractive presentation
 - Better comparison
 - Trend identification
 - Quick decision making
 - Simplifies complex data
-

7 Memory Trick

“B P L H S”

- **B** → Bar Chart
- **P** → Pie Chart
- **L** → Line Chart
- **H** → Histogram
- **S** → Scatter Plot

Sentence:

“Big People Love Helpful Statistics”

8 Exam Writing Tips

- ✓ Draw neat small diagrams
- ✓ Underline keywords:
 - Comparison

- Trend Analysis
- Correlation
- Visualization

✅ Write one use and example for every chart

Short Conclusion

Charts and graphs are important tools in visual analytics that help organizations understand data clearly and make better business decisions quickly and effectively.

Last Minute Revision Points

Important Charts

- Bar Chart
- Pie Chart
- Line Chart
- Histogram
- Scatter Plot
- Dashboard

Main Uses

- Comparison
- Trend analysis

- Percentage representation
 - Correlation analysis
-

★ Viva Questions with Answers

Q1) Which chart is used for comparison?

Bar chart.

Q2) Which chart is used for trend analysis?

Line chart.

Q3) Which chart shows percentage distribution?

Pie chart.

Q4) What is visual analytics?

Visual analytics means analyzing data using charts and graphs.

Q5) What is the use of dashboards?

Dashboards help monitor overall business performance.

5/13/2026, 9:39:32 PM



4) BPM (Business Performance Management)

Repeated Questions

"Write a note on Business Performance Management." [6]

5/13/2026, 10:01:29 PM



Q. Write a note on Business Performance Management (BPM). [6 Marks]

1 Simple Introduction

Business Performance Management (BPM) is a process used by organizations to monitor, measure, and improve business performance.

It helps companies achieve their goals by analyzing business data and improving decision making.

2 Concept Explanation (Step-by-Step)

Business Performance Management works in the following steps:

1) Setting Business Goals

Organizations first define business objectives.

Example:

- Increase sales
 - Reduce costs
 - Improve customer satisfaction
-

2) Collecting Business Data

Business data is collected from:

- Sales systems
 - Customer records
 - Financial reports
 - Employee reports
-

3) Monitoring Performance

Managers compare actual performance with business targets using:

- KPIs
 - Dashboards
 - Reports
-

4) Analyzing Performance

Organizations identify:

- Problems
 - Weak areas
 - Improvement opportunities
-

5) Taking Corrective Actions

Based on analysis, organizations take actions to improve performance.

Example:

Increasing staff if customer waiting time is high.

3 Simple Real-Life Example

Example: Hospital Management

A hospital uses BPM to:

- Monitor patient waiting time
- Track doctor performance
- Improve patient satisfaction
- Reduce operational costs

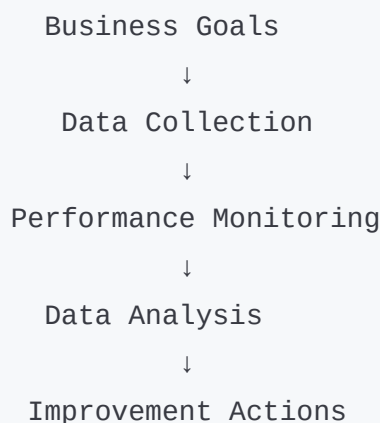
Managers analyze dashboard reports and improve hospital services.

4 Explain the Example

- Data is collected from hospital systems.
- Dashboard shows patient delays.
- Management identifies problems quickly.
- Correct actions improve hospital performance.

Thus, BPM helps improve organizational efficiency and service quality.

5 Simple Diagram



6 Diagram Explanation

Business Goals

Organization defines targets.

Data Collection

Business data is gathered.

Performance Monitoring

KPIs and dashboards monitor performance.

Data Analysis

Problems and opportunities are identified.

Improvement Actions

Corrective actions improve business performance.

7 Key Points / Features

Features of BPM

- Monitors business performance
 - Uses KPIs and dashboards
 - Supports decision making
 - Improves business efficiency
 - Helps achieve organizational goals
 - Supports continuous improvement
-

8 Advantages of BPM

- Better decision making

- Increased productivity
 - Improved customer satisfaction
 - Better resource management
 - Faster problem identification
-

9 Memory Trick

“G M A I”

- **G** → Goals
- **M** → Monitoring
- **A** → Analysis
- **I** → Improvement

Sentence:

“Good Managers Always Improve”

10 Exam Writing Tips

- ✓ Draw BPM cycle diagram
- ✓ Underline keywords:

- KPI
- Performance Monitoring
- Data Analysis
- Improvement

✓ Write one real-life example

1 1 Short Conclusion

Business Performance Management helps organizations monitor and improve business activities using data analysis, dashboards, and performance measurement techniques.

★ Last Minute Revision Points

- BPM = Monitor + Measure + Improve
 - Uses:
 - KPIs
 - Dashboards
 - Reports
 - Helps in:
 - Better decisions
 - Improved performance
 - Goal achievement
-

★ Most Important Keywords

- Business Performance Management
- KPI

- Dashboard
 - Monitoring
 - Data Analysis
 - Decision Making
 - Continuous Improvement
-

★ Viva Questions with Answers

Q1) What is BPM?

BPM is a process used to monitor and improve business performance.

Q2) What tools are used in BPM?

- KPIs
 - Dashboards
 - Reports
-

Q3) Why is BPM important?

It improves business efficiency and decision making.

Q4) What is the main goal of BPM?

To achieve business objectives and improve performance.

5/13/2026, 10:01:29 PM



"List and explain the four processes that define a closed-loop BPM cycle." [8]

5/13/2026, 10:01:53 PM



Q. List and explain the four processes that define a Closed-Loop BPM Cycle. [8 Marks]

1 Simple Introduction

A **Closed-Loop BPM (Business Performance Management) Cycle** is a continuous process used by organizations to:

- Plan business goals
- Monitor performance
- Analyze results
- Improve business activities

It is called "**Closed-Loop**" because the process repeats continuously for continuous business improvement.

2 Four Processes of Closed-Loop BPM Cycle

The Closed-Loop BPM Cycle mainly consists of four processes:

1. Strategize
 2. Plan
 3. Monitor and Analyze
 4. Take Corrective Action
-

3 Process 1: Strategize

Meaning

Organizations define:

- Business goals
- Mission
- Vision
- Strategies

Example

A company sets a goal to increase sales by 20%.

Importance

- Gives business direction
 - Helps achieve organizational objectives
-

4 Process 2: Plan

Meaning

Organizations prepare plans and allocate resources to achieve goals.

Activities

- Budget planning
- Resource allocation
- Setting targets

Example

A hospital plans additional staff for better patient service.

Importance

- Proper planning improves efficiency
 - Helps manage resources effectively
-

5 Process 3: Monitor and Analyze

Meaning

Business performance is monitored using:

- KPIs

- Dashboards
- Reports

Data is analyzed to identify problems and trends.

Example

An online shopping company monitors daily sales using dashboards.

Importance

- Helps identify weak areas
 - Supports better decision making
-

Process 4: Take Corrective Action

Meaning

Organizations take improvement actions based on analysis results.

Example

If sales decrease, the company may:

- Improve marketing
 - Reduce prices
 - Improve customer service
-

Importance

- Improves business performance
- Solves business problems quickly

7 Simple Real-Life Example

Example: E-Commerce Company

An online shopping company follows Closed-Loop BPM:

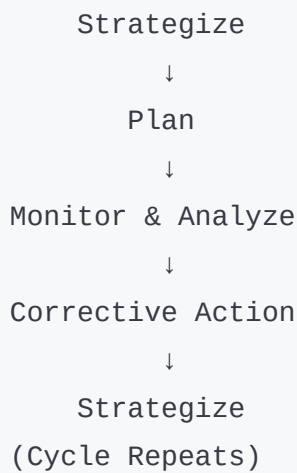
Process	Example
Strategize	Increase online sales
Plan	Launch discount offers
Monitor & Analyze	Check sales dashboard
Corrective Action	Improve delivery service

8 Explain the Example

- Company first sets sales goals.
- Then plans marketing campaigns.
- Dashboard monitors daily sales performance.
- If sales are low, corrective actions are taken.

Thus, the BPM cycle continuously improves business performance.

9 Simple Diagram



10 Diagram Explanation

Strategize

Set business goals and strategies.

Plan

Prepare action plans and budgets.

Monitor & Analyze

Track performance using dashboards and KPIs.

Corrective Action

Take actions to improve performance.

The cycle repeats continuously for continuous improvement.

11 Key Points / Features

Features of Closed-Loop BPM

- Continuous improvement process
- Supports strategic planning
- Uses KPIs and dashboards
- Helps in performance monitoring
- Improves decision making
- Aligns business goals with actions

12 Advantages

- Better business control
- Faster problem solving
- Improved productivity
- Better resource utilization
- Increased organizational efficiency

13 Memory Trick

"S P M C"

- **S** → Strategize
- **P** → Plan
- **M** → Monitor & Analyze
- **C** → Corrective Action

Sentence:

“Smart People Monitor Carefully”

14 Exam Writing Tips

✓ Draw BPM cycle diagram neatly

✓ Underline keywords:

- Strategize
- Plan
- Monitor
- Corrective Action

✓ Write one example for each process

15 Short Conclusion

The Closed-Loop BPM Cycle helps organizations continuously monitor, analyze, and improve business performance for achieving organizational goals effectively.

Last Minute Revision Points

Four Processes:

1. Strategize
2. Plan
3. Monitor & Analyze
4. Corrective Action

Main Goal:

Continuous business improvement.

★ Most Important Keywords

- Closed-Loop BPM
 - Strategize
 - Planning
 - Monitoring
 - KPI
 - Dashboard
 - Corrective Action
 - Continuous Improvement
-

★ Viva Questions with Answers

Q1) Why is BPM called closed-loop?

Because the process continuously repeats for improvement.

Q2) What is the first process in BPM cycle?

Strategize.

Q3) Which process uses KPIs and dashboards?

Monitor and Analyze.

Q4) What happens in corrective action?

Organizations improve performance by solving problems.

5/13/2026, 10:01:54 PM



5) KPI

"What is KPI? Explain the types of KPIs." [5]

Nov 2023

"List and justify any five KPIs for a real time business scenario." [5]

5/13/2026, 10:03:20 PM



Q. What is KPI? Explain the types of KPIs. [5 Marks]

1 Simple Introduction

KPI (Key Performance Indicator) is a measurable value used to check how effectively an organization is achieving its business goals.

KPIs help businesses monitor performance, identify problems, and improve decision making.

2 Concept Explanation

- KPI measures business success.
- It shows whether business goals are achieved or not.
- Organizations use KPIs to monitor:
 - Sales
 - Profit

- Customer satisfaction
 - Employee performance
-

3 Types of KPIs

1) Financial KPI

Meaning

Measures financial performance of a business.

Examples

- Profit
- Revenue
- Return on Investment (ROI)

Example

A company checks monthly profit growth.

2) Customer KPI

Meaning

Measures customer satisfaction and customer behavior.

Examples

- Customer satisfaction score
- Customer retention
- Customer complaints

Example

Amazon monitors customer ratings.

3) Operational KPI

Meaning

Measures efficiency of daily business operations.

Examples

- Production speed
- Delivery time
- Error rate

Example

A logistics company tracks delivery time.

4) Employee KPI

Meaning

Measures employee performance and productivity.

Examples

- Attendance
- Task completion
- Training performance

Example

An IT company tracks employee productivity.

Simple Real-Life Example

Example: Hospital

A hospital uses KPIs like:

- Patient waiting time
- Doctor performance
- Patient satisfaction
- Number of emergency cases handled

These KPIs help improve hospital services.

5 Explain the Example

- Patient waiting time measures operational efficiency.
- Patient feedback measures customer satisfaction.
- Doctor performance measures employee productivity.

Thus, KPIs help organizations improve overall performance.

6 Simple Diagram

KPI



| Financial | Customer | Operational | Employee |

7 Diagram Explanation

Financial KPI

Measures profit and revenue.

Customer KPI

Measures customer satisfaction.

Operational KPI

Measures business efficiency.

Employee KPI

Measures employee productivity.

8 Key Points / Features

Features of KPI

- Measures business performance
 - Helps achieve business goals
 - Supports decision making
 - Identifies weak areas
 - Improves efficiency
 - Supports business monitoring
-

9 Memory Trick

“F C O E”

- **F** → Financial
- **C** → Customer
- **O** → Operational
- **E** → Employee

Sentence:

10 Exam Writing Tips

✓ Draw KPI types diagram

✓ Underline keywords:

- Financial KPI
- Customer KPI
- Operational KPI
- Employee KPI

✓ Write one example for each KPI

1 1 Short Conclusion

KPIs are important business metrics used to measure organizational performance and help businesses achieve their goals effectively.

Last Minute Revision Points

Types of KPI

1. Financial KPI
2. Customer KPI
3. Operational KPI

4. Employee KPI

Main Purpose

- Performance measurement
 - Decision making
 - Goal achievement
-

★ Most Important Keywords

- KPI
 - Performance Measurement
 - Financial KPI
 - Customer Satisfaction
 - Operational Efficiency
 - Employee Productivity
-

★ Viva Questions with Answers

Q1) What is KPI?

KPI is a measurable value used to measure business performance.

Q2) Why are KPIs important?

They help organizations achieve business goals.

Q3) Give examples of KPIs.

- Profit
 - Customer satisfaction
 - Delivery time
 - Employee productivity
-

Q. List and justify any five KPIs for a real time business scenario. [5 Marks]

1 Simple Introduction

KPIs (Key Performance Indicators) are measurable values used by organizations to evaluate business performance.

They help managers monitor important business activities and improve decision making.

2 Five KPIs for Real-Time Business Scenario

Example: E-Commerce Company

1) Sales Revenue

Meaning

Measures total income from sales.

Justification

Helps company understand business growth.

2) Customer Satisfaction Score

Meaning

Measures customer happiness.

Justification

Satisfied customers increase business success.

3) Delivery Time

Meaning

Measures how quickly products are delivered.

Justification

Fast delivery improves customer experience.

4) Website Traffic

Meaning

Measures number of visitors on website.

Justification

More visitors can increase sales opportunities.

5) Product Return Rate

Meaning

Measures how many products customers return.

Justification

High return rate may indicate poor product quality.

3 Simple Real-Life Example

Example: Amazon

Amazon monitors:

- Daily sales
- Delivery speed
- Customer ratings
- Product returns
- Website traffic

These KPIs help Amazon improve customer satisfaction and profit.

4 Explain the Example

- Sales KPI measures revenue growth.
- Delivery KPI improves service quality.
- Return rate helps identify product problems.
- Website traffic measures business popularity.

Thus, KPIs help businesses improve performance.

5 Simple Diagram

Business KPIs



| Sales | Customers | Delivery | Traffic |

6 Key Points

- KPIs measure business performance
- Help in monitoring activities
- Improve business decisions
- Support business growth

7 Memory Trick

“S C D T R”

- **S** → Sales
- **C** → Customer Satisfaction
- **D** → Delivery Time
- **T** → Traffic
- **R** → Return Rate

Sentence:

Conclusion

KPIs help organizations measure business success, monitor performance, and improve decision making in real-time business environments.

Viva Questions with Answers

Q1) What is a KPI?

A KPI is a measurable value used to monitor business performance.

Q2) Give one financial KPI.

Sales revenue.

Q3) Which KPI measures customer happiness?

Customer Satisfaction Score.

5/13/2026, 10:03:20 PM



⚡ MEDIUM REPEATED

5) Power BI / Tableau as BI Tool

Asked:

Tableau → Apr 2023

Power BI → Apr 2023 OR, Nov 2023

Important:

Features

Advantages

Use in BI

5/13/2026, 10:04:25 PM



Q. Discuss the use of Tableau as a Business Intelligence (BI) Tool. [5 Marks]

1 Simple Introduction

Tableau is a popular **Business Intelligence (BI)** and **Data Visualization tool** used to analyze and present business data visually.

It helps organizations convert raw data into:

- Charts
- Graphs
- Dashboards

- Reports

for better decision making.

2 Uses of Tableau as a BI Tool

1) Data Visualization

Tableau converts complex business data into simple visual formats.

Example

Sales data shown using bar charts and line graphs.

2) Dashboard Creation

Tableau creates interactive dashboards showing:

- Sales
- Profit
- Customer data
- KPIs

Example

Retail company sales dashboard.

3) Data Analysis

Businesses use Tableau to:

- Identify trends
- Compare performance
- Analyze customer behavior

Example

E-commerce company analyzing monthly sales growth.

4) Real-Time Reporting

Tableau provides live business reports.

Example

Hospital dashboard showing live patient information.

5) Better Decision Making

Managers can quickly understand data and make better business decisions.

3 Features of Tableau

- User-friendly interface

- Interactive dashboards
 - Drag-and-drop functionality
 - Real-time analytics
 - Fast data visualization
 - Supports multiple data sources
-

4 Advantages of Tableau

- Easy to use
 - Attractive visual reports
 - Faster data analysis
 - Improves decision making
 - Supports business intelligence
 - Handles large data easily
-

5 Simple Real-Life Example

Example: Online Shopping Company

An e-commerce company uses Tableau to:

- Monitor daily sales
- Analyze customer buying patterns
- Track product performance
- Create business dashboards

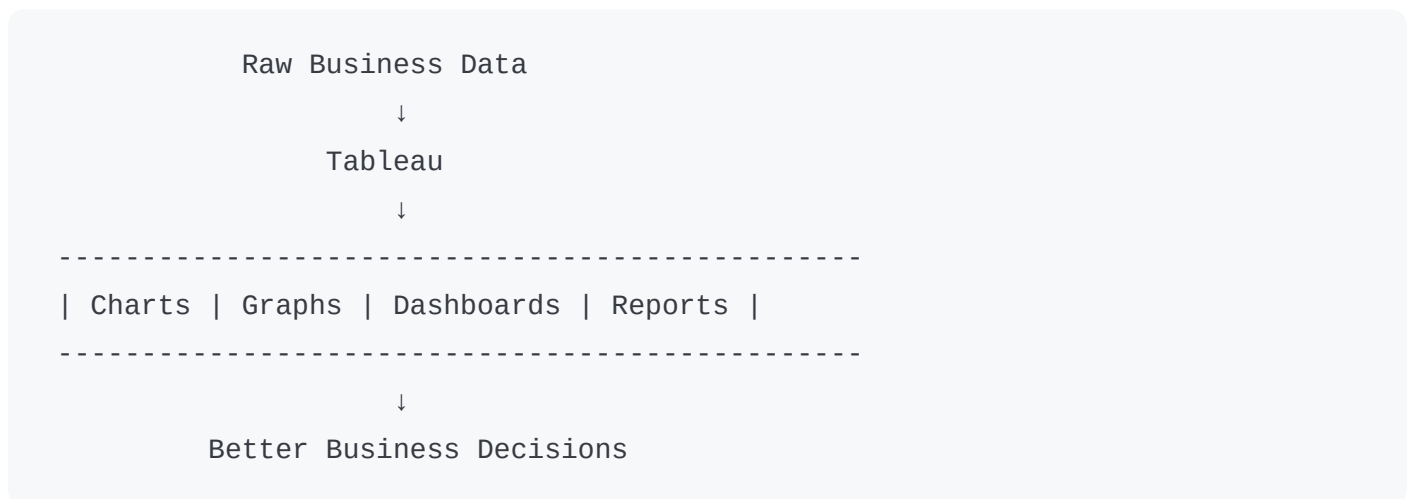
Managers use these dashboards to improve sales and customer satisfaction.

6 Explain the Example

- Tableau converts raw sales data into graphs.
- Dashboards help managers monitor business performance.
- Data analysis identifies best-selling products.

Thus, Tableau improves business intelligence and decision making.

7 Simple Diagram



8 Diagram Explanation

Raw Business Data

Data collected from business systems.

Tableau

Processes and visualizes data.

Charts and Dashboards

Display information visually.

Better Decisions

Managers understand data easily.

9 Key Points / Features

Important Features

- Interactive dashboards
 - Data visualization
 - Real-time reporting
 - Business analysis
 - User-friendly interface
-

10 Memory Trick

“D D A R”

- **D** → Data Visualization
- **D** → Dashboards

- **A** → Analysis
- **R** → Reporting

Sentence:

“Data Dashboards Analyze Reports”

1 1 Exam Writing Tips

✓ Draw Tableau workflow diagram

✓ Underline keywords:

- Dashboard
- Visualization
- BI Tool
- Real-Time Reporting

✓ Write one business example

1 2 Short Conclusion

Tableau is a powerful BI tool that helps organizations visualize data, analyze business performance, and make better decisions quickly and effectively.

Last Minute Revision Points

Tableau Uses

- Visualization
- Dashboard creation
- Reporting
- Trend analysis

Advantages

- Easy to use
 - Fast analysis
 - Better decisions
-

Viva Questions with Answers

Q1) What is Tableau?

Tableau is a Business Intelligence and Data Visualization tool.

Q2) What is the main use of Tableau?

To create charts, dashboards, and reports.

Q3) Why is Tableau important?

It helps organizations analyze data visually.

Q. Discuss the use of Power BI as a Business Intelligence (BI) Tool. [5 Marks]

1 Simple Introduction

Power BI is a Business Intelligence (BI) tool developed by Microsoft.

It is used to:

- Analyze business data
- Create reports
- Build dashboards
- Visualize information

Power BI helps organizations make smart business decisions.

2 Uses of Power BI as a BI Tool

1) Data Visualization

Power BI converts raw data into:

- Charts
- Graphs
- Maps
- Dashboards

Example

Company sales represented using graphs.

2) Interactive Dashboards

Power BI creates dashboards for:

- Sales monitoring
- KPI tracking
- Financial analysis

Example

Retail store dashboard showing daily sales.

3) Data Analysis

Businesses use Power BI to:

- Analyze trends
- Compare performance
- Identify business problems

Example

Bank analyzing customer transaction patterns.

4) Real-Time Monitoring

Power BI provides live business reports.

Example

Hospital dashboard showing patient data in real time.

5) Better Decision Making

Managers understand data easily and make quick decisions.

3 Features of Power BI

- Interactive dashboards
 - Real-time analytics
 - Cloud support
 - Data sharing
 - Easy report generation
 - User-friendly interface
-

4 Advantages of Power BI

- Easy to learn
 - Fast reporting
 - Better visualization
 - Supports large data
 - Improves business analysis
 - Low cost compared to other BI tools
-

5 Simple Real-Life Example

Example: Banking Sector

A bank uses Power BI to:

- Detect fraud
- Monitor customer transactions
- Analyze loan performance
- Track branch performance

Managers use reports and dashboards for decision making.

6 Explain the Example

- Power BI visualizes banking data.
- Dashboards show fraud alerts and transaction trends.

- Managers identify risks quickly.

Thus, Power BI improves business intelligence and operational efficiency.

7 Simple Diagram



8 Diagram Explanation

Business Data

Raw information collected from organization.

Power BI

Processes and analyzes data.

Reports and Dashboards

Display business insights visually.

Better Decisions

Managers make informed decisions.

9 Key Points / Features

Important Features

- Dashboard creation
 - Real-time monitoring
 - Data visualization
 - Cloud integration
 - Business reporting
-

10 Memory Trick

“V D R K”

- **V** → Visualization
- **D** → Dashboards
- **R** → Reports
- **K** → KPI Tracking

Sentence:

“Visual Dashboards Report Knowledge”

11 Exam Writing Tips

✓ Draw Power BI diagram

✓ Underline keywords:

- BI Tool
- Dashboard
- KPI
- Real-Time Analytics

✓ Write one real-life example

12 Short Conclusion

Power BI is an effective Business Intelligence tool used for data visualization, reporting, and dashboard creation, helping organizations make faster and smarter business decisions.

Last Minute Revision Points

Uses

- Dashboard creation
- Data visualization
- Reporting
- KPI tracking

Advantages

- Easy to use
 - Real-time analysis
 - Better business decisions
-

★ Viva Questions with Answers

Q1) What is Power BI?

Power BI is a Business Intelligence tool developed by Microsoft.

Q2) What is the main purpose of Power BI?

To analyze and visualize business data.

Q3) What are dashboards in Power BI?

Dashboards visually display important business information.

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7) Business Reports

Asked:

Nov 2024

Apr 2024

Important:

Types of reports

Components of Business Reporting System

5/13/2026, 10:04:55 PM



Q. List and briefly explain different types of Business Reports. [5 Marks]

1 Simple Introduction

A **Business Report** is a document that presents business information in an organized format for analysis and decision making.

Business reports help managers:

- Monitor performance
- Analyze problems
- Make business decisions

2 Types of Business Reports

1) Operational Report

Meaning

Shows daily business activities and operations.

Example

Daily sales report of a retail store.

Use

Helps managers monitor routine activities.

2) Analytical Report

Meaning

Used for detailed analysis of business data.

Example

Customer buying behavior analysis report.

Use

Supports strategic decision making.

3) Financial Report

Meaning

Shows financial performance of business.

Example

Profit and loss report.

Use

Helps monitor revenue and expenses.

4) Performance Report

Meaning

Measures employee or organizational performance using KPIs.

Example

Monthly employee productivity report.

Use

Helps improve business efficiency.

5) Dashboard Report

Meaning

Visual report showing charts, graphs, and KPIs together.

Example

Hospital dashboard showing patient statistics.

Use

Supports quick understanding and decision making.

Simple Real-Life Example

Example: E-Commerce Company

An online shopping company uses:

- Operational reports → Daily orders
- Financial reports → Revenue analysis
- Analytical reports → Customer trends
- Dashboard reports → Overall business performance

4 Explain the Example

- Operational reports monitor daily work.
- Financial reports track company profit.
- Analytical reports identify customer behavior.
- Dashboard reports help managers quickly understand performance.

Thus, different reports support different business activities.

5 Simple Diagram

Business Reports



| Operational | Financial | Analytical |



Better Decision Making

6 Diagram Explanation

Operational Report

Shows daily operations.

Financial Report

Shows business financial status.

Analytical Report

Provides data analysis and insights.

All reports help managers make better decisions.

7 Key Points / Features

Features of Business Reports

- Organized presentation of data
 - Supports business analysis
 - Helps decision making
 - Monitors business performance
 - Improves communication
-

8 Memory Trick

“O F A P D”

- **O** → Operational
- **F** → Financial
- **A** → Analytical
- **P** → Performance
- **D** → Dashboard

Sentence:

“Organizations Find Analysis Perfectly Detailed”

9 Exam Writing Tips

✓ Draw report types diagram

✓ Underline keywords:

- Operational Report
- Financial Report
- Analytical Report
- Dashboard Report

✓ Write one example for each report

10 Short Conclusion

Business reports help organizations organize, analyze, and present business information effectively for better decision making.

★ Last Minute Revision Points

Types of Reports

- Operational
- Financial
- Analytical
- Performance
- Dashboard

Main Uses

- Monitoring
 - Analysis
 - Decision making
-

★ Viva Questions with Answers

Q1) What is a business report?

A business report is a document containing business information for analysis.

Q2) Which report shows daily operations?

Operational report.

Q3) Which report is used for financial analysis?

Financial report.

Q. Discuss Components of Business Reporting System. [5 Marks]

1 Simple Introduction

A **Business Reporting System** is a system that collects, processes, analyzes, and presents business information in the form of reports and dashboards.

It helps organizations monitor performance and make better decisions.

2 Components of Business Reporting System

1) Data Sources

Meaning

Business data is collected from different sources.

Examples

- Sales systems
 - Databases
 - ERP systems
 - Customer records
-

2) Data Processing

Meaning

Collected data is cleaned, organized, and processed.

Use

Removes errors and prepares data for analysis.

3) Data Analysis

Meaning

Business data is analyzed using BI tools.

Use

Helps identify trends and patterns.

4) Reporting Tools

Meaning

Reports and dashboards are generated using tools like:

- Tableau
 - Power BI
-

5) Visualization

Meaning

Data is displayed using:

- Charts
- Graphs
- Dashboards

Use

Improves understanding of business information.

6) Decision Support

Meaning

Managers use reports for business decisions.

Example

Improving sales strategy based on report analysis.

3 Simple Real-Life Example

Example: Retail Store

A retail company:

- Collects sales data
- Processes and analyzes it
- Creates dashboards
- Uses reports for business decisions

This improves profit and customer satisfaction.

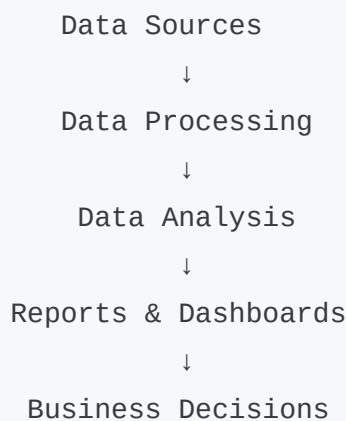
4 Explain the Example

- Data sources provide sales information.

- Processing organizes data.
- Reporting tools create dashboards.
- Managers use reports to improve sales performance.

Thus, Business Reporting System supports business intelligence.

5 Simple Diagram



6 Diagram Explanation

Data Sources

Collect raw business data.

Data Processing

Organizes and cleans data.

Data Analysis

Analyzes business performance.

Reports & Dashboards

Display information visually.

Business Decisions

Managers take corrective actions.

7 Key Points / Features

Components of Reporting System

- Data collection
 - Data processing
 - Data analysis
 - Reporting tools
 - Visualization
 - Decision support
-

8 Memory Trick

"S P A R D"

- **S** → Sources
- **P** → Processing

- **A** → Analysis
- **R** → Reporting
- **D** → Decision Making

Sentence:

“Smart People Analyze Reports Daily”

9 Exam Writing Tips

✓ Draw reporting system diagram

✓ Underline keywords:

- Data Sources
- Data Processing
- Reporting Tools
- Visualization

✓ Mention Tableau or Power BI for extra marks

10 Short Conclusion

Business Reporting Systems help organizations collect, analyze, and present business data effectively for better monitoring and decision making.

Last Minute Revision Points

Components

1. Data Sources
 2. Data Processing
 3. Data Analysis
 4. Reporting Tools
 5. Visualization
 6. Decision Support
-

Viva Questions with Answers

Q1) What is a Business Reporting System?

It is a system used to generate business reports and dashboards.

Q2) What is the role of visualization?

It helps users understand data easily.

Q3) Name two reporting tools.

- Tableau
- Power BI

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UNIT 4 → (Q3 & Q4)

🔥 MOST REPEATED QUESTIONS

1) What-if Analysis

Repeated Questions

"Write a note on : i) Multiple Goals ii) Sensitivity Analysis iii) What-If Analysis and Goal Seeking" [9]

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Q. Write a note on:

i) Multiple Goals

ii) Sensitivity Analysis

iii) What-If Analysis and Goal Seeking [9 Marks]

1 Simple Introduction

Organizations use different analytical techniques to improve planning and decision making.

Techniques like:

- Multiple Goals
- Sensitivity Analysis
- What-If Analysis
- Goal Seeking

help businesses predict outcomes and solve business problems effectively.

i) Multiple Goals

2 Meaning of Multiple Goals

In business, organizations often try to achieve more than one objective at the same time. These objectives are called **Multiple Goals**.

Sometimes goals may support each other, while sometimes they may conflict.

3 Examples of Multiple Goals

A company may want to:

- Increase profit
- Reduce cost
- Improve customer satisfaction
- Improve product quality

All these goals must be managed together.

4 Real-Life Example

Example: Hospital

A hospital wants to:

- Reduce patient waiting time
- Improve treatment quality
- Reduce operational cost

These are multiple business goals.

5 Diagram

Business Goals



| Increase Profit | Reduce Cost | Better Quality |

6 Key Points

- Multiple goals improve overall business performance.
 - Helps organizations balance different objectives.
 - Supports better planning and decision making.
-

ii) Sensitivity Analysis

7 Meaning of Sensitivity Analysis

Sensitivity Analysis studies how changes in one variable affect the final result.

It helps businesses understand:

“What will happen if input values change?”

8 Example

A company checks:

- What happens to profit if raw material cost increases?
 - What happens if product price decreases?
-

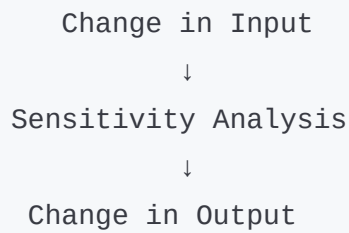
9 Real-Life Example

Example: Manufacturing Company

If production cost increases by 10%, management checks how profit changes.

This is sensitivity analysis.

10 Diagram



11 Key Points

- Helps measure risk
- Supports business forecasting
- Helps identify important variables
- Improves decision making

iii) What-If Analysis and Goal Seeking

12 Meaning of What-If Analysis

What-If Analysis checks different possible business situations by changing input values.

It answers questions like:

- What if sales increase?
- What if expenses decrease?

13 Example

A shopkeeper checks:

- What happens to profit if product price increases?
-

14 Meaning of Goal Seeking

Goal Seeking finds the required input value to achieve a desired output.

It answers:

“What should be done to reach a target?”

15 Example

A company wants profit of ₹1,00,000.

Goal seeking calculates:

- How many products should be sold?
 - What selling price is required?
-

16 Real-Life Example

Example: Online Store

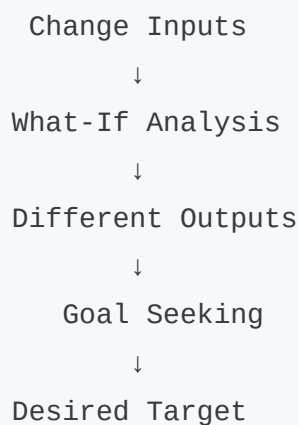
An online shopping company changes:

- Product price
- Advertising budget

to check different sales outcomes.

Managers use goal seeking to achieve target revenue.

17 Diagram



18 Diagram Explanation

What-If Analysis

Tests different business situations.

Goal Seeking

Finds required input to reach target output.

Together they improve planning and forecasting.

19 Key Points / Features

Multiple Goals

- Multiple objectives together
- Better business planning

Sensitivity Analysis

- Measures effect of variable changes
- Risk analysis

What-If Analysis

- Predicts outcomes
- Scenario testing

Goal Seeking

- Achieves target output
 - Finds required input values
-

20 Advantages

- Better forecasting
 - Improved planning
 - Supports business decisions
 - Helps in risk management
 - Improves business efficiency
-

21 Memory Trick

“M S W G”

- **M** → Multiple Goals
- **S** → Sensitivity Analysis
- **W** → What-If Analysis
- **G** → Goal Seeking

Sentence:

“Managers Study What Goals”

22 Exam Writing Tips

- ✓ Draw all diagrams neatly
- ✓ Underline keywords:

- Scenario Analysis
- Variable Change
- Goal Seeking

- Forecasting

✓ Write one example for each concept

2 3 Short Conclusion

Multiple Goals, Sensitivity Analysis, What-If Analysis, and Goal Seeking are important analytical techniques used in Business Analytics for planning, forecasting, and better decision making.

Last Minute Revision Points

Multiple Goals

- Many objectives together

Sensitivity Analysis

- Effect of variable changes

What-If Analysis

- Tests different situations

Goal Seeking

- Finds input for desired output

★ Most Important Keywords

- Multiple Goals
 - Sensitivity Analysis
 - What-If Analysis
 - Goal Seeking
 - Forecasting
 - Scenario Analysis
 - Decision Making
-

★ Viva Questions with Answers

Q1) What is What-If Analysis?

It checks outcomes by changing input values.

Q2) What is Goal Seeking?

It finds the required input to achieve target output.

Q3) Why is Sensitivity Analysis important?

It helps understand the effect of variable changes.

Q4) What are Multiple Goals?

Multiple business objectives achieved together.

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Q. Consider a scenario that a bread seller charges Rs. 30 per loaf of bread, and he sells 50 loaves a day. What would happen to his revenue if he charged more for each loaf of bread?

Evaluate the above mentioned scenario using What-If Analysis with respect to:

Case 1: The volume of bread sold doesn't depend on the price of bread.

Case 2: The amount of bread sold depends on the price charged.

(Sales increase if price decreases and sales decrease if price increases) [8 Marks]

1 Simple Introduction

What-If Analysis is a technique used to study different business situations by changing input values and observing the output.

It helps organizations predict future results and make better business decisions.

2 Given Data

- Price per loaf = ₹30
 - Number of loaves sold per day = 50
-

3 Formula Used

Revenue Formula

$$Revenue = Price \times Quantity$$

4 Current Revenue Calculation

Current Revenue:

$$Revenue = 30 \times 50 = 1500$$

👉 Current Daily Revenue = ₹1500

5 Case 1: Sales Volume Does NOT Depend on Price

Assumption

Bread seller increases price from ₹30 to ₹40.

But quantity sold remains same:

- Quantity = 50 loaves

Revenue Calculation

$$\text{Revenue} = 40 \times 50 = 2000$$

👉 New Revenue = ₹2000

Observation

- Revenue increased from ₹1500 to ₹2000
- Because sales quantity did not change.

6 Case 2: Sales Volume Depends on Price

Assumption

Bread seller increases price to ₹40.

Because price increased, demand decreases.

Assume:

- Quantity sold becomes 35 loaves

Revenue Calculation

$$\text{Revenue} = 40 \times 35 = 1400$$

👉 New Revenue = ₹1400

Observation

- Revenue decreased from ₹1500 to ₹1400
- Because higher price reduced sales volume.

7 Comparison Table

Case	Price	Quantity Sold	Revenue
Original	₹30	50	₹1500
Case 1	₹40	50	₹2000
Case 2	₹40	35	₹1400

8 Simple Real-Life Example

Example: Online Food Delivery App

If a food delivery company increases delivery charges:

- Some customers may continue ordering
- Some customers may stop ordering

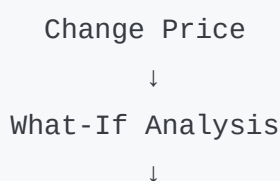
Managers use What-If Analysis to predict revenue changes before making decisions.

9 Explain the Example

- Changing delivery charges affects customer orders.
- Managers analyze different outcomes.
- This helps avoid losses and improve profit.

Thus, What-If Analysis supports smart business planning.

10 Simple Diagram



↓
Revenue Changes

1 1 Diagram Explanation

Change Price

Business changes product price.

What-If Analysis

Different business situations are tested.

Sales Same / Decrease

Demand may remain same or reduce.

Revenue Changes

Business checks impact on revenue.

1 2 Key Points / Features

Features of What-If Analysis

- Tests different business scenarios
- Predicts future outcomes
- Helps in decision making

- Supports business forecasting
 - Reduces business risk
-

13 Advantages

- Better planning
 - Smarter pricing decisions
 - Improved forecasting
 - Better profit analysis
-

14 Memory Trick

“C T P D”

- **C** → Change Inputs
- **T** → Test Scenarios
- **P** → Predict Results
- **D** → Decision Making

Sentence:

“Companies Test Prices Daily”

15 Exam Writing Tips

- ✓ Write formula clearly
- ✓ Show all calculations step-by-step
- ✓ Draw comparison table
- ✓ Underline:

- Revenue
- Quantity
- What-If Analysis

16 Short Conclusion

What-If Analysis helps businesses study the effect of changing business variables like price and quantity. It supports better planning, forecasting, and decision making.

★ Last Minute Revision Points

Formula

Revenue = Price × Quantity

Case 1

Price increases → Revenue increases if quantity remains same.

Case 2

Price increases → Revenue may decrease if quantity sold decreases.

★ Most Important Keywords

- What-If Analysis
 - Revenue
 - Scenario Analysis
 - Forecasting
 - Decision Making
 - Demand
 - Quantity Sold
-

★ Viva Questions with Answers

Q1) What is What-If Analysis?

It studies different outcomes by changing input values.

Q2) What formula is used for revenue?

Revenue = Price × Quantity

Q3) Why is What-If Analysis important?

It helps businesses predict future outcomes and make better decisions.

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Bookstore What-if Analysis Question [9]

Nov 2023

Income & Expense What-if Analysis Question [9]

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Q. Consider a scenario that you own a bookstore and have 100 books in storage. You sell 60% books for the highest price of Rs. 500 and 40% books for the lower price of Rs. 200.

Evaluate the above mentioned scenario using What-If Analysis with respect to:

Case 1: What if you sell 70% for the highest price & 30% for lowest price

Case 2: What if you sell 80% for the highest price & 20% for lowest price [9 Marks]

1 Simple Introduction

What-If Analysis is a business analysis technique used to study different business situations by changing input values and observing the results.

It helps organizations predict future outcomes and improve decision making.

2 Given Data

- Total books = 100
 - Highest price = ₹500
 - Lowest price = ₹200
-

3 Current Scenario Calculation

Books sold at ₹500

60% of 100 books:

$$60\% \times 100 = 60$$

Revenue:

$$60 \times 500 = 30000$$

Books sold at ₹200

40% of 100 books:

$$40\% \times 100 = 40$$

Revenue:

$$40 \times 200 = 8000$$

Total Revenue

$$30000 + 8000 = 38000$$

👉 Current Total Revenue = ₹38,000

4 Case 1: 70% Books Sold at Highest Price

High Price Books

70% of 100 books:

$$70\% \times 100 = 70$$

Revenue:

$$70 \times 500 = 35000$$

Low Price Books

30% of 100 books:

$$30\% \times 100 = 30$$

Revenue:

$$30 \times 200 = 6000$$

Total Revenue

$$35000 + 6000 = 41000$$

👉 New Revenue = ₹41,000

5 Case 2: 80% Books Sold at Highest Price

High Price Books

80% of 100 books:

$$80\% \times 100 = 80$$

Revenue:

$$80 \times 500 = 40000$$

Low Price Books

20% of 100 books:

$$20\% \times 100 = 20$$

Revenue:

$$20 \times 200 = 4000$$

Total Revenue

$$40000 + 4000 = 44000$$

👉 New Revenue = ₹44,000

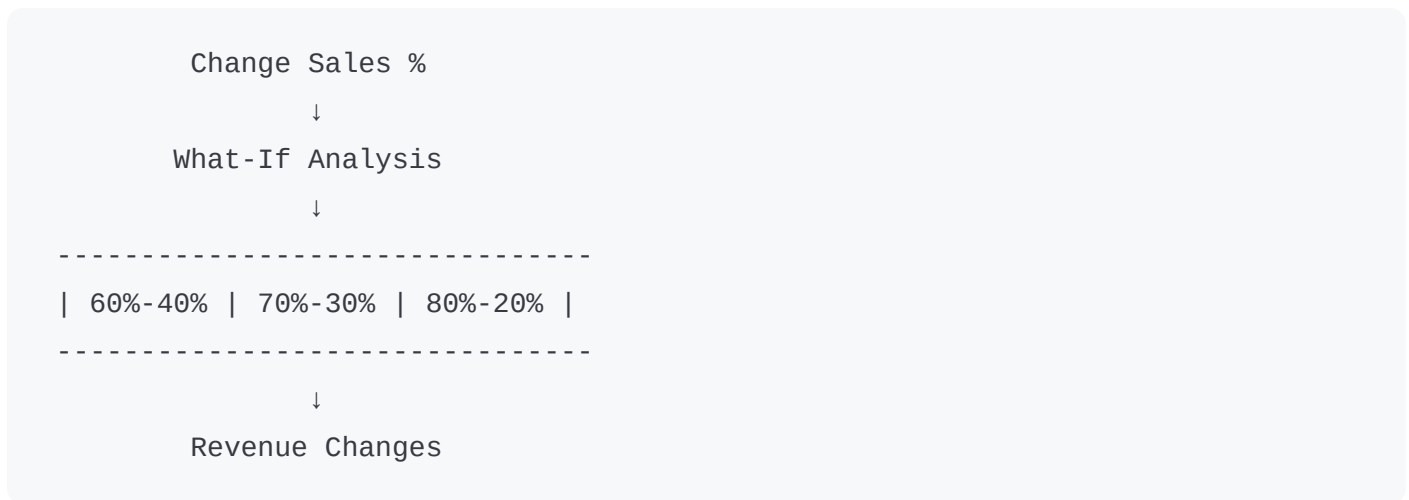
6 Comparison Table

Scenario	Revenue
Original Scenario	₹38,000
Case 1	₹41,000
Case 2	₹44,000

7 Observation

- Increasing sales of high-priced books increases total revenue.
- What-If Analysis helps predict business profit in different situations.

8 Simple Diagram



9 Diagram Explanation

Change Sales %

Business changes product sales percentage.

What-If Analysis

Different situations are tested.

Revenue Changes

Business studies impact on revenue.

10 Key Points

- Helps in forecasting
- Improves pricing strategy

- Supports profit analysis
 - Helps in business planning
-

1 1 Memory Trick

“C A R”

- **C** → Change Inputs
- **A** → Analyze Scenarios
- **R** → Revenue Prediction

Sentence:

“Companies Analyze Revenue”

1 2 Short Conclusion

What-If Analysis helps businesses predict the impact of changing sales conditions and supports better decision making and profit planning.

Viva Questions with Answers

Q1) What is What-If Analysis?

It studies business outcomes by changing input values.

Q2) What happens when high-price sales increase?

Revenue increases.

Q3) Why is What-If Analysis important?

It helps in forecasting and planning.

Q. Imagine you're managing the finances of a small business, and you want to plan your budget for the upcoming year. You have a basic income and expense sheet.

You'd like to know how different changes will affect your end-of-year balance.

Given:

- **Initial balance = \$10,000**
- **Monthly income = \$5,000**

- **Monthly expenses = \$3,000**

Perform What-If Analysis for:

i) Increase monthly income by 10%

ii) Decrease monthly expenses by 20%

iii) Increase income by 10% and decrease expenses by 20% [9 Marks]

1 Simple Introduction

What-If Analysis helps businesses test different financial situations by changing values like income and expenses.

It helps organizations predict future financial outcomes.

2 Given Data

- Initial Balance = \$10,000
 - Monthly Income = \$5,000
 - Monthly Expenses = \$3,000
-

3 Current Yearly Balance

Monthly Savings

$$5000 - 3000 = 2000$$

Yearly Savings

$$2000 \times 12 = 24000$$

Final Balance

$$10000 + 24000 = 34000$$

👉 Current Final Balance = **\$34,000**

4 Case 1: Increase Monthly Income by 10%

Increased Income

$$5000 + (10\% \times 5000) = 5500$$

Monthly Savings

$$5500 - 3000 = 2500$$

Yearly Savings

$$2500 \times 12 = 30000$$

Final Balance

$$10000 + 30000 = 40000$$

👉 Final Balance = **\$40,000**

5 Case 2: Decrease Monthly Expenses by 20%

Reduced Expenses

$$3000 - (20\% \times 3000) = 2400$$

Monthly Savings

$$5000 - 2400 = 2600$$

Yearly Savings

$$2600 \times 12 = 31200$$

Final Balance

$$10000 + 31200 = 41200$$

👉 Final Balance = **\$41,200**

6 Case 3: Increase Income by 10% and Decrease Expenses by 20%

Monthly Savings

Income = \$5500

Expenses = \$2400

$$5500 - 2400 = 3100$$

Yearly Savings

$$3100 \times 12 = 37200$$

Final Balance

$$10000 + 37200 = 47200$$

👉 Final Balance = **\$47,200**

7 Comparison Table

Scenario	Final Balance
Original	\$34,000
Case 1	\$40,000
Case 2	\$41,200
Case 3	\$47,200

8 Observation

- Increasing income improves balance.
- Reducing expenses also improves savings.
- Combining both gives maximum profit.

9 Simple Diagram

Change Income/Expenses



What-If Analysis



Different Balances



Better Financial Planning

10 Conclusion

What-If Analysis helps businesses predict financial outcomes under different situations and supports better budgeting and decision making.

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2) Descriptive / Predictive / Prescriptive Analytics

Repeated Questions

"Write a short note on: i) Descriptive analytics ii) Predictive analytics iii) Prescriptive analytics" [9]

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Q. Write a short note on:

i) Descriptive Analytics

ii) Predictive Analytics

iii) Prescriptive Analytics [9 Marks]

1 Simple Introduction

Analytics is the process of analyzing data to understand business performance and support decision making.

Businesses mainly use three types of analytics:

1. Descriptive Analytics
2. Predictive Analytics
3. Prescriptive Analytics

These analytics help organizations understand:

- What happened?

- What may happen?
 - What should be done?
-

i) Descriptive Analytics

2 Meaning of Descriptive Analytics

Descriptive Analytics studies past business data to understand:

- What happened in the business?
- Why it happened?

It summarizes historical data using:

- Reports
 - Charts
 - Dashboards
-

3 Features of Descriptive Analytics

- Analyzes past data
- Generates reports
- Uses dashboards and visualization
- Identifies trends and patterns

4 Real-Life Example

Example: Retail Store

A retail store analyzes:

- Last year sales
- Monthly profit
- Customer purchases

using charts and reports.

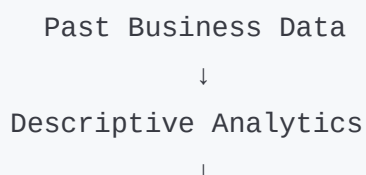
5 Explain the Example

The company studies previous sales data to understand:

- Which products sold more
- Which months had high sales

Thus, descriptive analytics explains past business performance.

6 Simple Diagram



Reports & Dashboards



Understand Past Events

7 Key Points

- Focuses on past data
- Helps understand business performance
- Uses reports and dashboards
- Supports business monitoring

ii) Predictive Analytics

8 Meaning of Predictive Analytics

Predictive Analytics uses past and present data to predict future outcomes.

It uses:

- Statistical techniques
- Machine learning
- Data mining

to forecast future trends.

9 Features of Predictive Analytics

- Predicts future events
 - Identifies future trends
 - Supports forecasting
 - Reduces business risk
-

10 Real-Life Example

Example: Online Shopping Company

An e-commerce company predicts:

- Future product demand
- Customer buying behavior
- Festival season sales

using previous customer data.

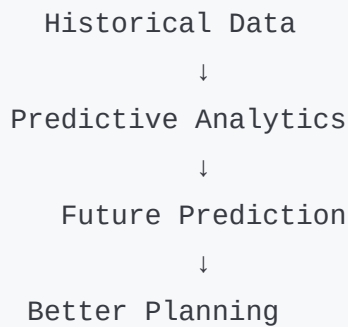
1 1 Explain the Example

The company studies past customer purchases and predicts:

- Which products customers may buy in future.

Thus, predictive analytics supports future planning.

1 2 Simple Diagram



1 3 Key Points

- Forecasts future events
- Helps in planning
- Reduces uncertainty
- Supports business growth

iii) Prescriptive Analytics

1 4 Meaning of Prescriptive Analytics

Prescriptive Analytics suggests the best action to solve business problems and achieve goals.

It answers:

“What should be done?”

15 Features of Prescriptive Analytics

- Recommends solutions
 - Supports decision making
 - Uses optimization techniques
 - Helps improve business performance
-

16 Real-Life Example

Example: Logistics Company

A delivery company uses prescriptive analytics to:

- Find shortest delivery routes
 - Reduce fuel cost
 - Improve delivery speed
-

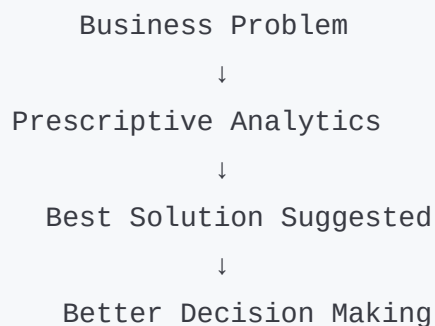
17 Explain the Example

The system analyzes traffic and delivery data and suggests:

- Best delivery routes

Thus, prescriptive analytics recommends best business actions.

18 Simple Diagram



19 Comparison Table

Analytics Type	Main Purpose	Question Answered
Descriptive	Understand past	What happened?
Predictive	Predict future	What may happen?
Prescriptive	Suggest actions	What should be done?

20 Summary Diagram

Analytics Types



| Descriptive | Predictive | Prescriptive |

21 Advantages

- Better business understanding
- Improved forecasting
- Smarter decision making
- Better planning
- Reduced business risks

22 Memory Trick

“D P P”

- **D** → Descriptive → Past
- **P** → Predictive → Future
- **P** → Prescriptive → Solution

Sentence:

“Describe Predict Provide”

23 Exam Writing Tips

✓ Draw all diagrams neatly

✓ Underline keywords:

- Forecasting
- Business Intelligence
- Future Prediction
- Decision Making

✓ Write one example for each analytics type

24 Short Conclusion

Descriptive, Predictive, and Prescriptive Analytics are important techniques used in Business Analytics for understanding past performance, predicting future outcomes, and recommending better business decisions.

Last Minute Revision Points

Descriptive Analytics

- Past data
- Reports and dashboards

Predictive Analytics

- Future prediction
- Forecasting

Prescriptive Analytics

- Suggests best actions
 - Decision support
-

★ Most Important Keywords

- Descriptive Analytics
 - Predictive Analytics
 - Prescriptive Analytics
 - Forecasting
 - Optimization
 - Business Intelligence
 - Decision Making
-

★ Viva Questions with Answers

Q1) What does descriptive analytics do?

It analyzes past business data.

Q2) What is predictive analytics used for?

Predicting future outcomes.

Q3) Which analytics suggests best solutions?

Prescriptive analytics.

Q4) Which analytics uses forecasting?

Predictive analytics.

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"What is descriptive analytics? Explain the importance of descriptive analytics with the help of an example." [5]

Nov 2023

"What is predictive analytics? Explain the importance of predictive analytics with the help of an example." [5]

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Q. What is Descriptive Analytics? Explain the importance of Descriptive Analytics with the help of an example. [5 Marks]

1 Simple Introduction

Descriptive Analytics is the process of analyzing past business data to understand:

- What happened in the business?
- Why it happened?

It converts raw data into useful information using:

- Reports
- Charts
- Dashboards
- Graphs

2 Concept Explanation

Descriptive Analytics studies historical data and summarizes it in simple form.

It helps organizations:

- Understand business performance
 - Identify trends and patterns
 - Monitor KPIs
 - Improve decision making
-

3 Importance of Descriptive Analytics

1) Understands Past Performance

Businesses can analyze:

- Sales
- Profit
- Customer behavior

Example

A company checks last year's sales performance.

2) Identifies Trends and Patterns

Helps identify:

- Seasonal sales trends
 - Customer preferences
 - Product demand
-

3) Supports Decision Making

Managers use reports and dashboards for better business decisions.

4) Improves Business Monitoring

Organizations can continuously monitor business performance.

5) Simplifies Large Data

Complex business data becomes easy to understand through charts and reports.

4 Simple Real-Life Example

Example: Retail Store

A retail company analyzes:

- Last year sales
- Monthly revenue
- Best-selling products

using charts and dashboards.

The company finds that:

- Sales increase during festival seasons.

5 Explain the Example

- Past sales data is analyzed.
- Dashboards help managers understand business performance.
- Seasonal trends are identified.
- Managers improve stock planning for future sales.

Thus, descriptive analytics helps organizations understand business activities clearly.

6 Simple Diagram



7 Diagram Explanation

Historical Data

Past business information is collected.

Descriptive Analytics

Data is analyzed and summarized.

Reports & Dashboards

Information is shown visually.

Understand Performance

Managers identify trends and patterns.

8 Key Points / Features

Features of Descriptive Analytics

- Uses historical data
- Generates reports and dashboards
- Identifies trends
- Supports monitoring
- Helps in decision making

9 Memory Trick

“P T D”

- **P** → Past Data
- **T** → Trends
- **D** → Dashboards

Sentence:

“Past Trends Displayed”

10 Exam Writing Tips

✓ Draw diagram neatly

✓ Underline keywords:

- Historical Data
- Reports
- Dashboards
- Trends

✓ Write one real-life example

11 Short Conclusion

Descriptive Analytics helps organizations analyze past business performance using reports and dashboards, improving business understanding and decision making.

★ Last Minute Revision Points

- Descriptive Analytics = Past Data Analysis
- Uses:
 - Reports
 - Charts
 - Dashboards
- Main goal:
 - Understand business performance

★ Viva Questions with Answers

Q1) What is descriptive analytics?

It analyzes past business data.

Q2) What tools are used in descriptive analytics?

Reports, charts, and dashboards.

Q3) Why is descriptive analytics important?

It helps understand business performance and trends.

Q. What is Predictive Analytics? Explain the importance of Predictive Analytics with the help of an example. [5 Marks]

1 Simple Introduction

Predictive Analytics is the process of using past and present data to predict future outcomes.

It helps businesses forecast:

- Future sales
 - Customer behavior
 - Market trends
 - Business risks
-

2 Concept Explanation

Predictive Analytics uses:

- Statistical methods
- Machine learning

- Data mining

to identify future possibilities and trends.

It answers:

“What may happen in future?”

3 Importance of Predictive Analytics

1) Future Forecasting

Businesses can predict:

- Future sales
- Product demand
- Market growth

Example

A company predicts festival season sales.

2) Better Decision Making

Managers take smarter decisions using future predictions.

3) Risk Reduction

Organizations can identify possible business risks early.

Example

Banks predict loan fraud risks.

4) Customer Behavior Analysis

Businesses predict:

- Customer buying patterns
 - Customer preferences
-

5) Improves Business Planning

Companies prepare better strategies for future growth.

4 Simple Real-Life Example

Example: Online Shopping Company

An e-commerce company studies:

- Previous customer purchases
- Product search history

- Festival sales trends

to predict:

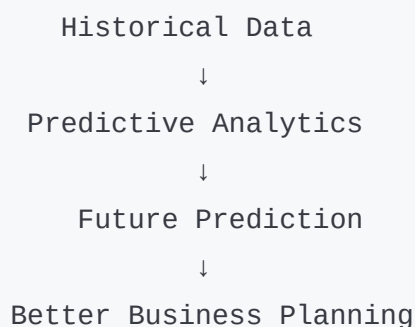
- Future customer demand
 - Best-selling products
-

5 Explain the Example

- Past customer data is analyzed.
- Predictive models forecast future sales.
- Managers plan inventory and offers accordingly.

Thus, predictive analytics improves business forecasting and planning.

6 Simple Diagram



7 Diagram Explanation

Historical Data

Past business information is collected.

Predictive Analytics

Data is analyzed using forecasting techniques.

Future Prediction

Future trends and outcomes are predicted.

Better Planning

Managers improve business strategies.

8 Key Points / Features

Features of Predictive Analytics

- Predicts future outcomes
 - Uses forecasting techniques
 - Supports business planning
 - Reduces business risks
 - Improves decision making
-

9 Memory Trick

"F P R"

- **F** → Future Prediction
- **P** → Planning
- **R** → Risk Reduction

Sentence:

"Forecast Planning Reduces Risk"

10 Exam Writing Tips

✓ Draw prediction flow diagram

✓ Underline keywords:

- Forecasting
- Future Prediction
- Customer Behavior
- Risk Analysis

✓ Write one example

1 1 Short Conclusion

Predictive Analytics helps organizations forecast future trends and improve business planning using historical and current business data.

★ Last Minute Revision Points

- Predictive Analytics = Future Forecasting
 - Uses:
 - Historical Data
 - Machine Learning
 - Statistical Analysis
 - Benefits:
 - Better planning
 - Risk reduction
 - Improved decisions
-

★ Viva Questions with Answers

Q1) What is predictive analytics?

It predicts future outcomes using past data.

Q2) Why is predictive analytics important?

It helps in forecasting and planning.

Q3) Give one application of predictive analytics.

Sales forecasting in e-commerce.

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3) Certainty, Uncertainty, Risk

"Write short note on : i) Certainty ii) Uncertainty iii) Risk" [9]

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Q. Write short note on:

i) Certainty

ii) Uncertainty

iii) Risk [9 Marks]

1 Simple Introduction

In business decision making, organizations face different situations while planning and solving problems.

These situations are mainly classified into:

1. Certainty
2. Uncertainty
3. Risk

Understanding these conditions helps managers make better business decisions.

i) Certainty

2 Meaning of Certainty

Certainty is a condition where the outcome of a decision is completely known in advance.

In certainty:

- Complete information is available
 - Future outcome is predictable
 - No doubt or confusion exists
-

3 Features of Certainty

- Known outcomes
 - Complete information available
 - Easy decision making
 - No risk involved
-

4 Real-Life Example

Example: Fixed Salary

An employee knows he will receive ₹30,000 salary every month.

The outcome is fixed and certain.

5 Explain the Example

- Salary amount is already known.
- No uncertainty or risk exists.
- Decision making becomes simple.

Thus, certainty means guaranteed outcomes.

6 Simple Diagram

Complete Information



Certainty



Known Outcome

7 Key Points

- Future outcome is known
- No risk involved
- Easy business planning
- Stable decision environment

ii) Uncertainty

8 Meaning of Uncertainty

Uncertainty is a situation where future outcomes are unknown and unpredictable.

In uncertainty:

- Complete information is not available
 - Future events cannot be predicted accurately
-

9 Features of Uncertainty

- Unknown outcomes
 - Lack of information
 - Difficult decision making
 - High confusion
-

10 Real-Life Example

Example: New Product Launch

A company launches a completely new product and does not know:

- Customer response
 - Future demand
 - Market performance
-

11 Explain the Example

- No previous data is available.
- Company cannot predict future sales accurately.
- Decisions become difficult.

Thus, uncertainty means unknown future situations.

12 Simple Diagram

Incomplete Information



Uncertainty



Unknown Outcome

13 Key Points

- Future outcome unknown
- No accurate prediction possible

- High decision complexity
 - Lack of proper information
-

iii) Risk

14 Meaning of Risk

Risk is a condition where future outcomes are uncertain, but probabilities of outcomes can be estimated.

In risk:

- Some information is available
 - Possible outcomes can be predicted partially
-

15 Features of Risk

- Partial information available
 - Probability analysis possible
 - Chances of profit or loss
 - Requires careful planning
-

16 Real-Life Example

Example: Stock Market Investment

An investor buys shares in the stock market.

There is:

- Chance of profit
- Chance of loss

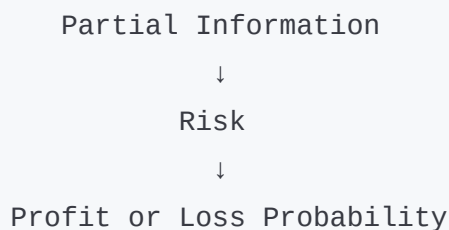
But market trends can be analyzed.

17 Explain the Example

- Future share prices are uncertain.
- Historical market data helps estimate probability.
- Investor takes decisions carefully.

Thus, risk involves uncertain outcomes with measurable probability.

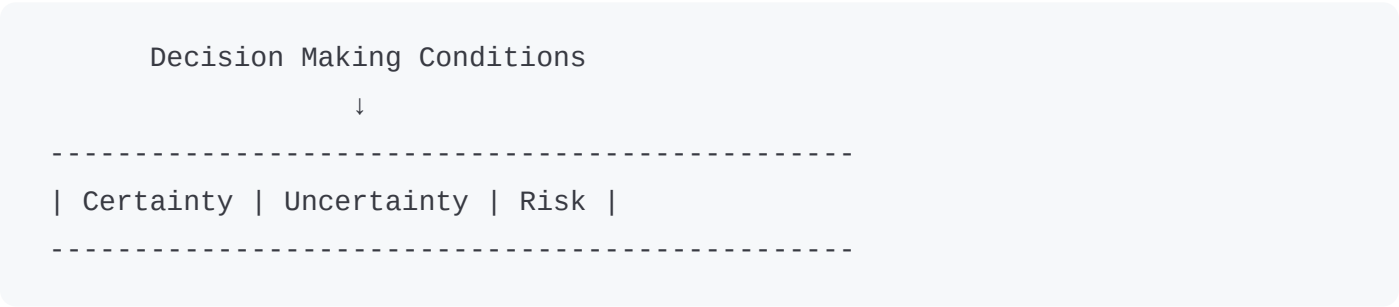
18 Simple Diagram



19 Comparison Table

Factor	Certainty	Uncertainty	Risk
Information	Complete	Incomplete	Partial
Outcome	Known	Unknown	Probable
Decision Making	Easy	Difficult	Moderate
Risk Level	No Risk	Very High	Medium

20 Summary Diagram



21 Advantages of Understanding These Concepts

- Better business planning
- Improved decision making
- Better risk management
- Helps forecasting future situations

22 Memory Trick

“C U R”

- **C** → Certainty → Known
- **U** → Uncertainty → Unknown
- **R** → Risk → Probability

Sentence:

“Certain Unknown Risks”

23 Exam Writing Tips

✓ Draw all diagrams neatly

✓ Underline keywords:

- Known Outcome
- Unknown Outcome
- Probability
- Decision Making

✓ Write one example for each concept

24 Short Conclusion

Certainty, uncertainty, and risk are important conditions in business decision making. Understanding these concepts helps organizations plan effectively and manage business situations properly.

★ Last Minute Revision Points

Certainty

- Known outcome
- No risk

Uncertainty

- Unknown outcome
- No proper information

Risk

- Probability-based outcome
- Profit or loss chances

★ Most Important Keywords

- Certainty
- Uncertainty
- Risk
- Probability
- Decision Making
- Forecasting

★ Viva Questions with Answers

Q1) What is certainty?

Certainty means the outcome is already known.

Q2) What is uncertainty?

Uncertainty means future outcomes are unknown.

Q3) What is risk?

Risk means uncertain outcomes with measurable probability.

Q4) Which condition has complete information?

Certainty.

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4) Mathematical Models

"Discuss structure of mathematical model for decision support." [8]

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Q. Discuss structure of Mathematical Model for Decision Support. [8 Marks]

1 Simple Introduction

A **Mathematical Model** is a representation of a real-world business problem using mathematical equations, variables, and formulas.

It helps organizations analyze business situations and make better decisions.

In **Decision Support Systems (DSS)**, mathematical models are used for:

- Planning
- Forecasting
- Optimization
- Problem solving

2 Concept Explanation

A mathematical model takes:

- Input data
- Business conditions
- Constraints

and produces:

- Results
- Predictions
- Solutions

using mathematical calculations.

3 Structure of Mathematical Model

The structure of a mathematical model mainly consists of the following components:

1. Decision Variables
 2. Parameters / Constants
 3. Constraints
 4. Mathematical Relationships
 5. Objective Function
 6. Output / Solution
-

4 Decision Variables

Meaning

Decision variables are values that can be changed by managers.

These variables affect the final outcome.

Example

- Number of products to manufacture
 - Selling price
 - Quantity produced
-

5 Parameters / Constants

Meaning

Parameters are fixed values used in the model.

They usually do not change frequently.

Example

- Machine capacity
 - Fixed production cost
 - Interest rate
-

6 Constraints

Meaning

Constraints are limitations or restrictions in the business problem.

Examples

- Budget limit
 - Time limit
 - Resource availability
-

7 Mathematical Relationships

Meaning

These are formulas or equations connecting variables and parameters.

Example

Revenue calculation:

$$Revenue = Price \times Quantity$$

8 Objective Function

Meaning

The objective function defines the business goal.

It may:

- Maximize profit

- Minimize cost
- Increase productivity

Example

Profit optimization:

$$Profit = Revenue - Cost$$

9 Output / Solution

Meaning

Final results generated by the model.

Example

- Best production quantity
 - Maximum profit
 - Minimum transportation cost
-

10 Simple Real-Life Example

Example: Manufacturing Company

A company wants to maximize profit.

Inputs:

- Product price
- Production cost
- Machine capacity

Constraints:

- Limited labor
- Limited raw materials

The mathematical model calculates:

- Best production quantity
- Expected profit

1 1 Explain the Example

- Variables represent production quantity.
- Constraints represent limited resources.
- Equations calculate profit.
- Model gives best business decision.

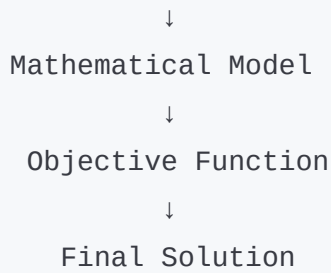
Thus, mathematical models support business planning and optimization.

1 2 Simple Diagram

Input Data



| Variables | Constraints | Parameters |



13 Diagram Explanation

Input Data

Business information is collected.

Variables

Values that can change.

Constraints

Business limitations.

Mathematical Model

Processes business data mathematically.

Objective Function

Defines business goal.

Final Solution

Best decision or output is generated.

14 Key Points / Features

Features of Mathematical Models

- Uses mathematical equations
 - Supports business decisions
 - Helps optimization
 - Improves forecasting
 - Reduces business risk
 - Supports planning and analysis
-

15 Advantages

- Better decision making
 - Accurate analysis
 - Efficient resource utilization
 - Faster problem solving
 - Improved business performance
-

16 Memory Trick

"V C O S"

- **V** → Variables
- **C** → Constraints
- **O** → Objective Function
- **S** → Solution

Sentence:

“Variables Create Optimal Solutions”

17 Exam Writing Tips

✓ Draw structure diagram neatly

✓ Underline keywords:

- Variables
- Constraints
- Objective Function
- Optimization

✓ Write one business example

18 Short Conclusion

Mathematical models play an important role in Decision Support Systems by helping organizations analyze business problems and generate better business decisions scientifically.

★ Last Minute Revision Points

Components:

1. Variables
2. Parameters
3. Constraints
4. Objective Function
5. Solution

Uses:

- Planning
 - Optimization
 - Forecasting
 - Decision Making
-

★ Most Important Keywords

- Mathematical Model
- Decision Support System
- Variables
- Constraints
- Objective Function
- Optimization
- Forecasting

★ Viva Questions with Answers

Q1) What is a mathematical model?

It is a mathematical representation of a business problem.

Q2) What is the objective function?

It defines the business goal like maximizing profit.

Q3) What are constraints?

Constraints are business limitations.

Q4) Why are mathematical models important?

They help in better business decision making.

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"Discuss the various Components of Decision Support Mathematical Models." [5]

Nov 2023

"Write a note on Mathematical Programming Optimization with an example." [7]

5/13/2026, 10:29:40 PM



Q. Discuss the various Components of Decision Support Mathematical Models. [5 Marks]

1 Simple Introduction

A **Decision Support Mathematical Model** is a mathematical representation of a business problem used in Decision Support Systems (DSS).

It helps organizations analyze business situations and make better decisions using formulas, variables, and calculations.

2 Components of Decision Support Mathematical Models

The main components are:

1. Decision Variables

2. Parameters / Constants
 3. Constraints
 4. Objective Function
 5. Mathematical Relationships
 6. Solution / Output
-

3 Decision Variables

Meaning

Variables are values that managers can change.

Example

- Product quantity
 - Selling price
 - Production units
-

4 Parameters / Constants

Meaning

These are fixed values used in the model.

Example

- Fixed cost

- Machine capacity
 - Interest rate
-

5 Constraints

Meaning

Constraints are limitations or restrictions in the business problem.

Example

- Budget limit
 - Labor limit
 - Time limit
-

6 Objective Function

Meaning

It defines the business goal.

Examples

- Maximize profit
- Minimize cost

Example formula:

$$Profit = Revenue - Cost$$

7 Mathematical Relationships

Meaning

These are equations connecting variables and parameters.

Example

$$Revenue = Price \times Quantity$$

8 Solution / Output

Meaning

Final result generated by the model.

Example

- Best production quantity
 - Maximum profit
 - Lowest transportation cost
-

9 Simple Real-Life Example

Example: Manufacturing Company

A company wants to maximize profit.

Inputs:

- Product price
- Production cost
- Available labor

Constraints:

- Limited budget
- Limited raw materials

The mathematical model calculates the best production quantity.

10 Explain the Example

- Variables represent product quantity.
- Constraints represent limited resources.
- Objective function calculates maximum profit.
- Model gives best business solution.

Thus, mathematical models support decision making.

11 Simple Diagram

Input Data



| Variables | Constraints |



Mathematical Model



Objective Function



Final Solution

1 2 Diagram Explanation

Variables

Values that can change.

Constraints

Business limitations.

Mathematical Model

Processes business data.

Objective Function

Defines business goal.

Final Solution

Best business decision.

13 Key Points / Features

- Supports decision making
 - Uses equations and formulas
 - Helps optimization
 - Improves planning
 - Reduces business risk
-

14 Memory Trick

“V C O S”

- **V** → Variables
- **C** → Constraints
- **O** → Objective Function
- **S** → Solution

Sentence:

“Variables Create Optimal Solutions”

15 Short Conclusion

Decision Support Mathematical Models help organizations solve business problems scientifically and improve business decisions effectively.

★ Last Minute Revision Points

Components:

- Variables
 - Parameters
 - Constraints
 - Objective Function
 - Solution
-

★ Viva Questions with Answers

Q1) What is a decision support model?

It is a mathematical model used for business decision making.

Q2) What is an objective function?

It defines the business goal.

Q3) What are constraints?

Constraints are business limitations.

Q. Write a note on Mathematical Programming Optimization with an example. [7 Marks]

1 Simple Introduction

Mathematical Programming Optimization is a technique used to find the best possible solution to a business problem using mathematical equations and constraints.

It helps organizations:

- Maximize profit
 - Minimize cost
 - Improve efficiency
-

2 Concept Explanation

In optimization:

- A business goal is defined.
- Constraints are identified.
- Mathematical equations are used to find the best solution.

Optimization helps organizations use resources efficiently.

3 Main Components of Optimization

1) Decision Variables

Variables whose values can change.

Example

Number of products produced.

2) Objective Function

Defines business goal.

Example

Profit maximization:

$$Profit = Revenue - Cost$$

3) Constraints

Business limitations.

Examples

- Limited labor
- Limited raw materials

- Budget limits
-

4 Simple Real-Life Example

Example: Manufacturing Company

A factory produces:

- Chairs
- Tables

Goal:

- Maximize profit

Constraints:

- Limited labor
- Limited wood
- Limited machine time

The optimization model calculates:

- Best number of chairs and tables to produce.
-

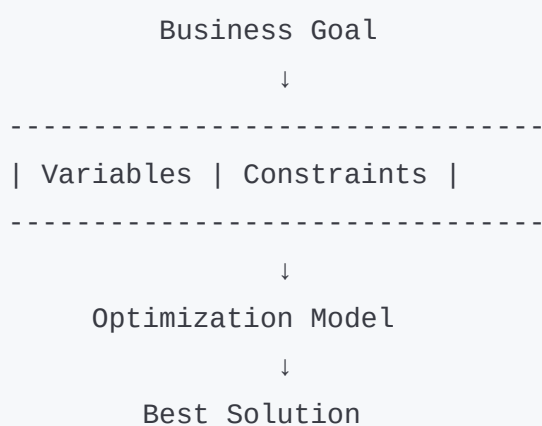
5 Explain the Example

- Variables represent products.
- Constraints represent limited resources.

- Objective function maximizes profit.
- Optimization provides best production plan.

Thus, mathematical optimization improves business performance.

6 Simple Diagram



7 Diagram Explanation

Business Goal

Organization defines target.

Variables

Values to optimize.

Constraints

Business limitations.

Optimization Model

Calculates best solution.

Best Solution

Optimal business decision.

8 Advantages of Optimization

- Maximizes profit
 - Minimizes cost
 - Improves productivity
 - Better resource utilization
 - Supports decision making
-

9 Applications of Optimization

- Production planning
 - Transportation management
 - Supply chain management
 - Inventory control
 - Financial planning
-

10 Memory Trick

"V O C"

- **V** → Variables
- **O** → Objective Function
- **C** → Constraints

Sentence:

"Variables Optimize Constraints"

1 1 Exam Writing Tips

✓ Draw optimization structure diagram

✓ Underline keywords:

- Objective Function
- Constraints
- Optimization
- Profit Maximization

✓ Write one business example

1 2 Short Conclusion

Mathematical Programming Optimization helps organizations achieve the best possible business solution by using mathematical techniques and resource constraints effectively.

★ Last Minute Revision Points

Main Components:

- Variables
- Objective Function
- Constraints

Main Goal:

- Maximize profit
 - Minimize cost
-

★ Viva Questions with Answers

Q1) What is optimization?

Finding the best solution to a business problem.

Q2) What is the objective function?

It defines business goals like maximizing profit.

Q3) Give one application of optimization.

Production planning.

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6) Decision Trees / Decision Tables

Asked:

Apr 2023

5/13/2026, 10:30:26 PM



Q. Explain Decision Trees and Decision Tables with suitable examples. [8 Marks]

1 Simple Introduction

Decision Trees and Decision Tables are important decision-making tools used in Business Analytics and Decision Support Systems (DSS).

They help organizations:

- Analyze business situations
- Compare alternatives
- Make better decisions

2 Decision Tree

3 Meaning of Decision Tree

A **Decision Tree** is a graphical representation of decisions and their possible outcomes shown in tree structure form.

It helps businesses analyze different choices and their results step-by-step.

4 Components of Decision Tree

1) Decision Node

Represents a decision point.

Shown using:

□ (Square)

2) Chance Node

Represents uncertain outcomes.

Shown using:

○ (Circle)

3) Branches

Represent possible choices or outcomes.

5 Simple Real-Life Example

Example: Product Launch Decision

A company wants to launch a new product.

Two options:

- Launch product
- Do not launch product

If launched:

- High demand → High profit
- Low demand → Loss

6 Simple Diagram of Decision Tree



7 Explain the Diagram

Decision Node

Company decides whether to launch product.

Chance Node

Market demand may be high or low.

Outcomes

Company may earn profit or face loss.

Thus, decision trees help businesses analyze possible outcomes.

8 Advantages of Decision Tree

- Easy to understand
 - Supports decision making
 - Helps analyze risk
 - Shows all possible outcomes visually
-

9 Decision Table

10 Meaning of Decision Table

A **Decision Table** is a tabular method used to represent business rules and decision conditions clearly.

It shows:

- Conditions
- Possible actions
- Decision outcomes

in table format.

1 1 Components of Decision Table

1) Conditions

Business situations or rules.

2) Actions

Business decisions or activities.

3) Rules

Combination of conditions and actions.

1 2 Simple Real-Life Example

Example: Loan Approval System

Bank checks:

- Salary
- Credit score

before approving loan.

13 Simple Decision Table

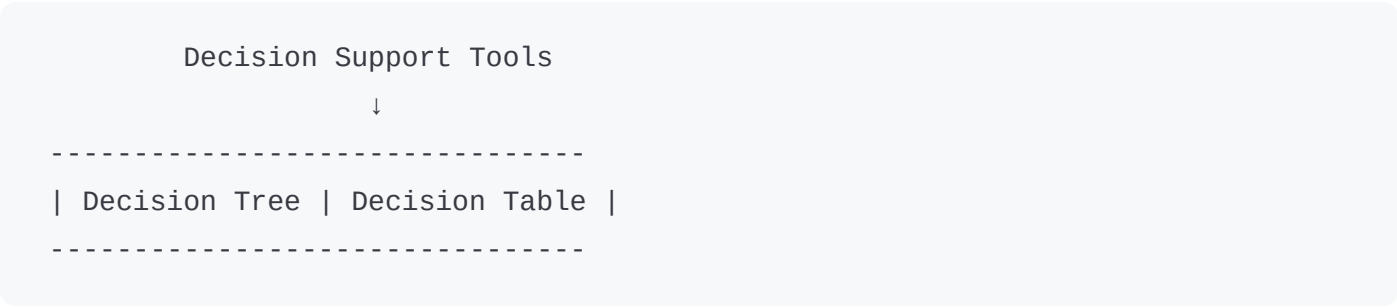
Salary High	Credit Score Good	Loan Approval
Yes	Yes	Approved
Yes	No	Rejected
No	Yes	Rejected
No	No	Rejected

14 Explain the Example

- Loan approval depends on conditions.
- Bank checks salary and credit score.
- Decision table simplifies decision making.

Thus, decision tables help manage business rules clearly.

15 Simple Summary Diagram



16 Comparison Between Decision Tree and Decision Table

Decision Tree	Decision Table
Graphical representation	Tabular representation
Shows decision flow	Shows rules and conditions
Easy for sequential decisions	Easy for complex business rules
Uses nodes and branches	Uses rows and columns

17 Key Points / Features

Decision Tree

- Visual representation
- Shows possible outcomes
- Supports risk analysis

Decision Table

- Organized rule representation
 - Easy for condition checking
 - Simplifies business decisions
-

18 Applications

- Loan approval systems
 - Product launch decisions
 - Medical diagnosis
 - Business planning
 - Risk analysis
-

19 Memory Trick

“T T”

- T → Tree → Graphical
- T → Table → Tabular

Sentence:

“Trees are Graphical, Tables are Tabular”

20 Exam Writing Tips

✓ Draw neat decision tree diagram

✓ Draw proper decision table

✓ Underline keywords:

- Decision Node
- Chance Node
- Conditions
- Rules

✓ Write one example for both

21 Short Conclusion

Decision Trees and Decision Tables are important decision-support tools that help organizations analyze business situations and make accurate decisions effectively.

★ Last Minute Revision Points

Decision Tree

- Graphical decision model
- Uses nodes and branches

Decision Table

- Tabular decision model
 - Uses conditions and actions
-

★ Most Important Keywords

- Decision Tree
 - Decision Table
 - Decision Node
 - Chance Node
 - Business Rules
 - Decision Making
 - Risk Analysis
-

★ Viva Questions with Answers

Q1) What is a decision tree?

A decision tree is a graphical representation of decisions and outcomes.

Q2) What is a decision table?

A decision table is a tabular representation of business rules.

Q3) Which tool uses nodes and branches?

Decision tree.

Q4) Which tool uses rows and columns?

Decision table.

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UNIT 5 → (Q5 & Q6)

🔥 MOST REPEATED QUESTIONS

1) BI in Finance

"Enlist the advantages of using Business Intelligence in Finance." [4]

Apr 2023

"Explain role of Business Intelligence (BI) in finance with the help of an example." [6]

5/13/2026, 10:30:49 PM



Q. Enlist the advantages of using Business Intelligence in Finance. [4 Marks]

1 Simple Introduction

Business Intelligence (BI) in finance helps organizations collect, analyze, and visualize financial data for better decision making.

It helps finance departments monitor:

- Revenue
- Expenses
- Profit
- Investments
- Risks

using dashboards, reports, and analytics tools.

2 Advantages of Using BI in Finance

1) Better Financial Decision Making

BI provides accurate financial reports and analysis.

Example

Managers can quickly decide budget allocation using dashboards.

2) Improved Financial Planning

BI helps organizations:

- Forecast future revenue
 - Plan budgets
 - Predict expenses
-

3) Risk Management

BI helps identify:

- Financial risks
- Fraud
- Unusual transactions

Example

Banks detect suspicious transactions using BI systems.

4) Real-Time Financial Monitoring

Finance managers can monitor:

- Profit
- Cash flow
- Expenses

in real time.

5) Faster Report Generation

BI tools generate:

- Financial reports
- Profit analysis
- KPI reports

quickly and accurately.

6) Better Data Visualization

Financial data is displayed using:

- Charts

- Graphs
- Dashboards

making analysis easier.

3 Simple Real-Life Example

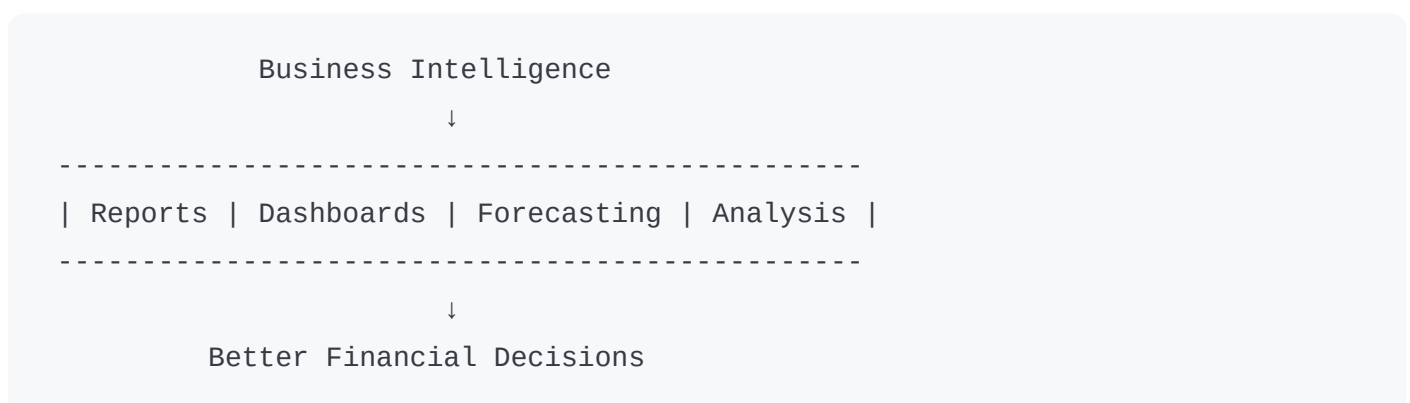
Example: Bank

A bank uses BI to:

- Monitor customer transactions
- Detect fraud
- Analyze loan performance
- Generate financial reports

This improves financial management and security.

4 Simple Diagram



5 Key Points

- Improves decision making
 - Supports financial forecasting
 - Helps risk management
 - Real-time monitoring
 - Faster report generation
-

6 Memory Trick

“F R R D”

- **F** → Forecasting
- **R** → Reporting
- **R** → Risk Management
- **D** → Decision Making

Sentence:

“Finance Reports Reduce Difficulties”

7 Short Conclusion

Business Intelligence improves financial analysis, forecasting, reporting, and decision making, helping organizations manage finances more effectively.

★ Viva Questions with Answers

Q1) What is BI in finance?

BI in finance means using analytics tools to analyze financial data.

Q2) Why is BI important in finance?

It improves financial planning and decision making.

Q3) Give one application of BI in finance.

Fraud detection in banking.

Q. Explain role of Business Intelligence (BI) in Finance with the help of an example. [6 Marks]

1 Simple Introduction

Business Intelligence (BI) plays an important role in finance by helping organizations analyze financial data and make better business decisions.

BI systems use:

- Reports
- Dashboards
- Data visualization
- Analytics

to improve financial management.

2 Role of BI in Finance

1) Financial Reporting

BI generates accurate financial reports such as:

- Profit reports
- Expense reports
- Revenue reports

Example

Monthly sales and profit reports.

2) Budget Planning and Forecasting

BI helps organizations:

- Predict future revenue
 - Plan business budgets
 - Estimate future expenses
-

3) Risk Management

BI helps identify:

- Fraud
- Financial risks
- Unusual activities

Example

Banks use BI for fraud detection.

4) Real-Time Monitoring

Managers can monitor:

- Cash flow
- Investments
- Financial performance

in real time.

5) Better Decision Making

BI dashboards help finance managers make quick and accurate decisions.

3 Simple Real-Life Example

Example: Banking Sector

A bank uses BI tools to:

- Monitor customer transactions
- Detect fraud
- Analyze loan performance
- Generate dashboards and reports

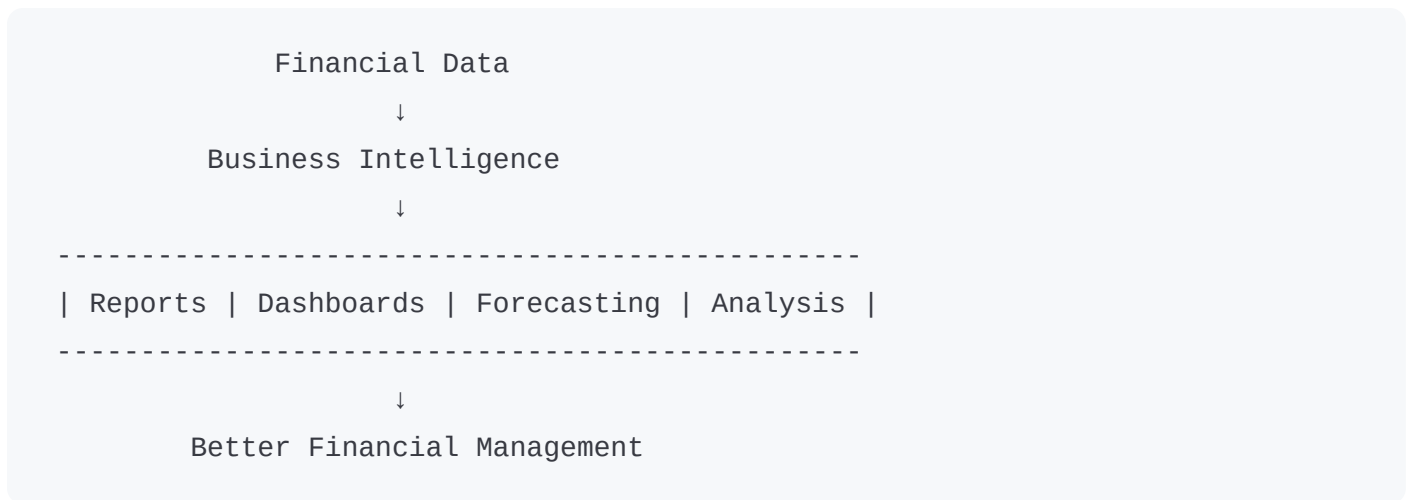
Managers use these reports to improve financial planning and reduce risks.

4 Explain the Example

- BI analyzes transaction data.
- Dashboards show financial performance visually.
- Fraud activities are identified quickly.
- Managers take faster financial decisions.

Thus, BI improves financial efficiency and security.

5 Simple Diagram



6 Diagram Explanation

Financial Data

Business financial information is collected.

BI System

Processes and analyzes financial data.

Reports & Dashboards

Shows financial insights visually.

Better Management

Managers improve financial decisions.

7 Advantages of BI in Finance

- Better financial analysis
 - Faster reporting
 - Improved forecasting
 - Fraud detection
 - Better risk management
 - Improved decision making
-

8 Memory Trick

“R F R D”

- **R** → Reports
- **F** → Forecasting
- **R** → Risk Management
- **D** → Decision Making

Sentence:

“Reports Forecast Risks Daily”

9 Exam Writing Tips

- ✓ Draw BI finance diagram
- ✓ Underline keywords:

- Forecasting
- Risk Management
- Dashboards
- Financial Reporting

✅ Write banking example for extra marks

10 Short Conclusion

Business Intelligence helps finance departments analyze financial data, reduce risks, improve forecasting, and make better business decisions effectively.

★ Last Minute Revision Points

Roles of BI in Finance

- Reporting
 - Forecasting
 - Fraud Detection
 - Risk Management
 - Decision Making
-

★ Most Important Keywords

- Business Intelligence
 - Financial Reporting
 - Forecasting
 - Risk Management
 - Fraud Detection
 - Dashboards
-

Viva Questions with Answers

Q1) What is the role of BI in finance?

BI helps analyze financial data and improve decisions.

Q2) How does BI help banks?

It helps in fraud detection and financial analysis.

Q3) Which BI tools are used in finance?

- Power BI
- Tableau

5/13/2026, 10:30:49 PM



"Explain the role of BI in finance with respect to i) Monitor Financial Performance ii) Budgeting and Forecasting iii) Financial Reporting" [7]

5/13/2026, 10:31:24 PM



Q. Explain the role of BI in Finance with respect to:

i) Monitor Financial Performance

ii) Budgeting and Forecasting

iii) Financial Reporting [7 Marks]

1 Simple Introduction

Business Intelligence (BI) in finance helps organizations collect, analyze, and visualize financial data for better decision making.

BI tools like:

- Dashboards
- Reports
- Charts
- Analytics

help finance managers monitor and improve financial activities effectively.

2 Role of BI in Finance

i) Monitor Financial Performance

Meaning

BI helps organizations continuously monitor financial activities such as:

- Revenue
- Profit
- Expenses
- Cash flow

using dashboards and reports.

Example

A company uses BI dashboards to monitor:

- Daily sales
 - Monthly profit
 - Expense trends
-

Importance

- Identifies financial problems quickly
 - Improves financial control
 - Supports faster decision making
-

ii) Budgeting and Forecasting

Meaning

BI helps organizations prepare:

- Budgets
- Future financial plans
- Revenue forecasts

using historical and current business data.

Example

A retail company predicts festival season sales and prepares budget accordingly.

Importance

- Improves financial planning
 - Reduces business risk
 - Helps resource allocation
-

iii) Financial Reporting

Meaning

BI automatically generates financial reports such as:

- Profit and loss reports
- Balance sheets
- Revenue reports

in visual and easy-to-understand format.

Example

A bank generates monthly financial reports using BI tools.

Importance

- Faster report generation
 - Accurate reporting
 - Better understanding of financial data
-

3 Simple Real-Life Example

Example: Banking Sector

A bank uses BI tools to:

- Monitor customer transactions
- Analyze profit and expenses
- Forecast future revenue
- Generate financial reports

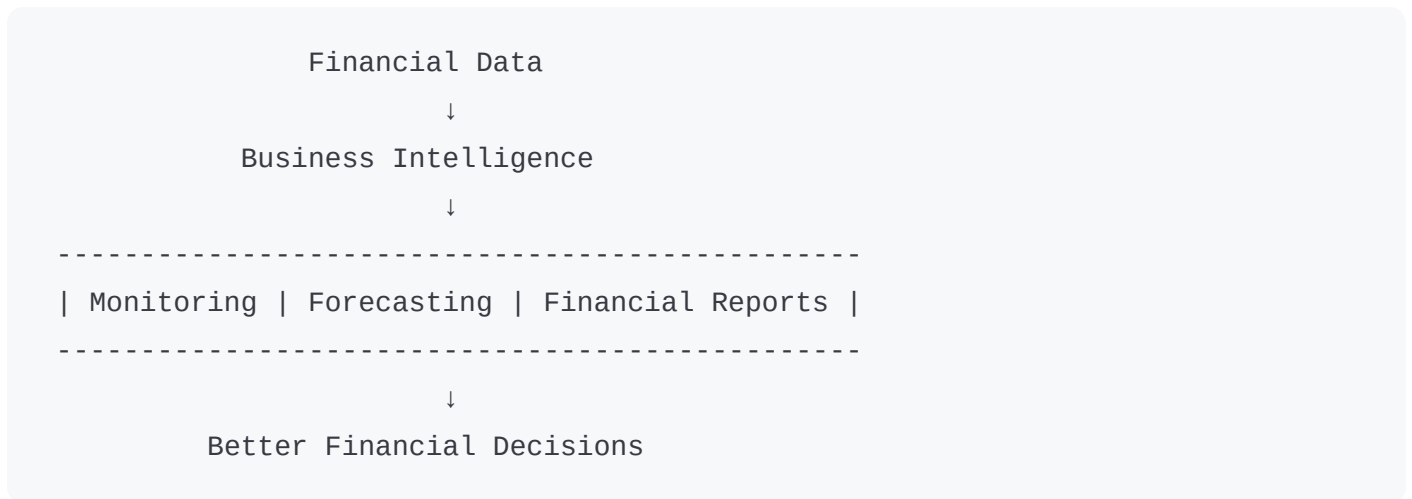
Managers use dashboards and reports for better financial decisions.

4 Explain the Example

- BI dashboards monitor financial performance in real time.
- Forecasting helps banks prepare future financial plans.
- Reports help management analyze profit and losses.

Thus, BI improves overall financial management.

5 Simple Diagram



6 Diagram Explanation

Monitoring

Tracks financial performance continuously.

Forecasting

Predicts future financial conditions.

Financial Reports

Generates visual financial reports.

Together they improve business decision making.

7 Advantages of BI in Finance

-
- Better financial analysis
 - Faster reporting
 - Improved forecasting
 - Better budgeting
 - Real-time monitoring
 - Better decision making
-

8 Key Points / Features

Monitor Financial Performance

- Tracks revenue and expenses
- Real-time dashboards

Budgeting and Forecasting

- Predicts future financial conditions
- Supports financial planning

Financial Reporting

- Generates reports automatically
 - Improves reporting accuracy
-

9 Memory Trick

“M B F”

- **M** → Monitor Performance
- **B** → Budgeting
- **F** → Financial Reporting

Sentence:

“Managers Budget Finances”

10 Exam Writing Tips

✓ Draw BI finance diagram neatly

✓ Underline keywords:

- Monitoring
- Forecasting
- Financial Reporting
- Dashboards

✓ Write banking example for extra marks

1 1 Short Conclusion

Business Intelligence plays an important role in finance by helping organizations monitor financial performance, improve budgeting and forecasting, and generate accurate financial reports for better decision making.

★ Last Minute Revision Points

Roles of BI in Finance

1. Monitor Financial Performance
2. Budgeting and Forecasting
3. Financial Reporting

Benefits

- Better analysis
 - Faster reports
 - Improved planning
-

★ Most Important Keywords

- Business Intelligence
 - Financial Performance
 - Forecasting
 - Budgeting
 - Financial Reporting
 - Dashboards
 - Decision Making
-

★ Viva Questions with Answers

Q1) What is BI in finance?

BI in finance means analyzing financial data for better decisions.

Q2) How does BI help in forecasting?

It predicts future financial trends using historical data.

Q3) Why is financial reporting important?

It helps organizations monitor profit and expenses accurately.

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2) Fraud Detection

"Explain application of Business Intelligence in Fraud Detection with a suitable example." [6]

5/13/2026, 10:31:52 PM



Q. Explain application of Business Intelligence in Fraud Detection with a suitable example. [6 Marks]

1 Simple Introduction

Fraud Detection means identifying illegal or suspicious activities in business transactions.

Business Intelligence (BI) helps organizations detect fraud by analyzing large amounts of data using:

- Reports
- Dashboards
- Data mining
- Analytics tools

BI helps organizations identify unusual patterns and prevent financial losses.

2 Application of BI in Fraud Detection

1) Monitoring Transactions

BI systems continuously monitor business transactions.

Example

Banks monitor online money transfers and credit card transactions.

2) Identifying Unusual Patterns

BI tools analyze customer behavior and detect suspicious activities.

Example

Sudden large withdrawals from bank accounts.

3) Real-Time Alerts

BI systems generate alerts when suspicious transactions occur.

Example

Credit card used in different countries within a short time.

4) Risk Analysis

BI helps organizations identify high-risk customers or transactions.

Example

Insurance companies identify fake insurance claims.

5) Data Visualization

BI dashboards display fraud-related information visually using:

- Charts
- Graphs
- Reports

This helps managers understand fraud patterns easily.

Simple Real-Life Example

Example: Banking Fraud Detection

A bank uses BI tools to monitor customer transactions.

The system checks:

- Transaction amount
- Location
- Frequency of transactions

If a customer suddenly transfers a very large amount from another country, the BI system generates a fraud alert.

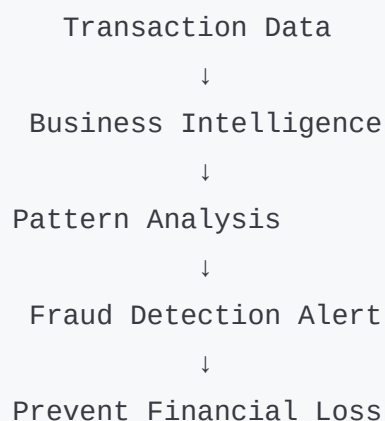
The bank temporarily blocks the transaction and verifies the customer.

4 Explain the Example

- BI analyzes transaction data continuously.
- Unusual transaction patterns are detected.
- Real-time alerts help prevent fraud quickly.
- Bank reduces financial losses and improves security.

Thus, BI plays an important role in fraud detection.

5 Simple Diagram



6 Diagram Explanation

Transaction Data

Customer transaction information is collected.

BI System

Analyzes business data.

Pattern Analysis

Detects unusual activities.

Fraud Alert

Suspicious transactions are identified.

Prevention

Organization takes action to stop fraud.

7 Advantages of BI in Fraud Detection

- Detects fraud quickly
- Reduces financial loss
- Improves security
- Real-time monitoring
- Better risk management
- Faster investigation

8 Key Points / Features

- Uses data analysis and dashboards
 - Identifies suspicious activities
 - Generates real-time alerts
 - Helps financial institutions
 - Improves business security
-

9 Memory Trick

“M A A P”

- **M** → Monitoring
- **A** → Analysis
- **A** → Alerts
- **P** → Prevention

Sentence:

“Managers Analyze Alerts Properly”

10 Exam Writing Tips

- ✓ Draw fraud detection flow diagram
- ✓ Underline keywords:

- Fraud Detection
- Pattern Analysis
- Real-Time Alerts
- Risk Management

✅ Write banking example for extra marks

1 1 Short Conclusion

Business Intelligence helps organizations detect fraud by analyzing transaction data, identifying suspicious activities, and generating real-time alerts, thereby improving business security and reducing financial losses.

★ Last Minute Revision Points

BI in Fraud Detection

- Transaction monitoring
- Pattern analysis
- Real-time alerts
- Fraud prevention

Main Benefit

- Reduces financial loss

★ Most Important Keywords

- Fraud Detection
 - Business Intelligence
 - Pattern Analysis
 - Risk Management
 - Real-Time Monitoring
 - Alerts
 - Data Analysis
-

★ Viva Questions with Answers

Q1) What is fraud detection?

Fraud detection means identifying illegal or suspicious activities.

Q2) How does BI help in fraud detection?

BI analyzes transaction data and detects suspicious patterns.

Q3) Give one example of fraud detection.

Bank credit card fraud detection.

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"With the help of a real time case study explain the applications of Business Intelligence in Fraud Detection." [7]

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Q. With the help of a real time case study explain the applications of Business Intelligence in Fraud Detection. [7 Marks]

1 Simple Introduction

Fraud Detection means identifying illegal or suspicious activities in business transactions.

Business Intelligence (BI) helps organizations detect fraud using:

- Data analysis
- Dashboards
- Reports
- Pattern analysis
- Real-time monitoring

BI systems analyze large amounts of transaction data and quickly identify unusual activities.

2 Applications of BI in Fraud Detection

1) Transaction Monitoring

BI systems continuously monitor financial transactions.

Example

Banks monitor ATM withdrawals and online transactions.

2) Pattern Analysis

BI tools identify unusual transaction patterns.

Example

Sudden large money transfers from unknown locations.

3) Real-Time Alerts

BI generates instant alerts for suspicious activities.

Example

Credit card used in two countries within one hour.

4) Risk Analysis

BI identifies high-risk customers or activities.

Example

Insurance companies detect fake insurance claims.

5) Fraud Prevention

Organizations can stop suspicious transactions before financial loss occurs.

Real-Time Case Study

Case Study: Banking Credit Card Fraud Detection

A bank uses a Business Intelligence system to monitor customer credit card transactions.

The BI system analyzes:

- Transaction amount
 - Transaction location
 - Transaction frequency
 - Customer spending behavior
-

Fraud Situation

A customer normally uses a credit card only in Pune.

Suddenly:

- A very large transaction occurs from another country.
- Multiple purchases happen within a few minutes.

The BI system identifies this as suspicious behavior.

Action Taken

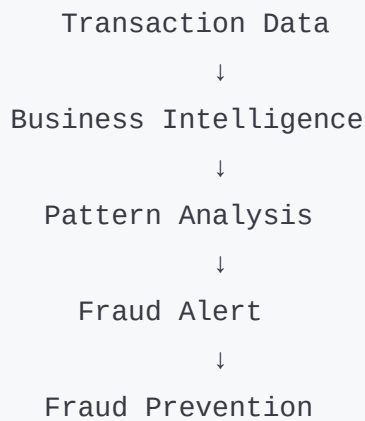
- BI dashboard generates real-time fraud alert.
 - Bank temporarily blocks the transaction.
 - Customer verification is done.
 - Fraudulent transaction is prevented.
-

4 Explain the Case Study

- BI continuously monitors transaction data.
- Pattern analysis identifies unusual activities.
- Real-time alerts help banks react quickly.
- Fraud is detected before financial loss occurs.

Thus, BI improves banking security and reduces fraud risks.

5 Simple Diagram



6 Diagram Explanation

Transaction Data

Customer transaction information is collected.

BI System

Analyzes large transaction data.

Pattern Analysis

Detects unusual transaction behavior.

Fraud Alert

Suspicious activity generates alert.

Fraud Prevention

Bank stops fraudulent transaction.

7 Advantages of BI in Fraud Detection

- Detects fraud quickly
 - Reduces financial loss
 - Real-time monitoring
 - Better risk management
 - Improves business security
 - Faster fraud investigation
-

8 Key Points / Features

- Uses dashboards and analytics
 - Monitors transactions continuously
 - Identifies suspicious patterns
 - Generates real-time alerts
 - Helps banks and financial organizations
-

9 Memory Trick

“M P A P”

- **M** → Monitoring
- **P** → Pattern Analysis
- **A** → Alerts

- **P** → Prevention

Sentence:

“Managers Prevent Alerts Properly”

10 Exam Writing Tips

- ✓ Write proper case study heading
 - ✓ Draw fraud detection flow diagram
 - ✓ Underline keywords:
 - Pattern Analysis
 - Real-Time Alerts
 - Fraud Prevention
 - Transaction Monitoring
 - ✓ Banking example gives extra marks
-

1 1 Short Conclusion

Business Intelligence plays an important role in fraud detection by analyzing transaction data, identifying suspicious activities, and preventing financial fraud through real-time monitoring and alerts.

Last Minute Revision Points

BI Applications in Fraud Detection

- Transaction monitoring
- Pattern analysis
- Fraud alerts
- Risk management
- Fraud prevention

Case Study

Bank credit card fraud detection system.

Most Important Keywords

- Business Intelligence
 - Fraud Detection
 - Pattern Analysis
 - Real-Time Alerts
 - Transaction Monitoring
 - Risk Analysis
 - Fraud Prevention
-

Viva Questions with Answers

Q1) What is fraud detection?

Fraud detection means identifying suspicious or illegal activities.

Q2) How does BI help in fraud detection?

BI analyzes transaction data and identifies unusual patterns.

Q3) Give one real-time application of BI in fraud detection.

Credit card fraud detection in banking.

Q4) What is the role of real-time alerts?

They help organizations stop fraud quickly.

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3) CRM using BI

"Discuss the real time Applications of BI in Customer Relationship Management with appropriate case study." [6]

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Q. Discuss the real time Applications of BI in Customer Relationship Management (CRM) with appropriate case study. [6 Marks]

1 Simple Introduction

Customer Relationship Management (CRM) is a system used to manage interactions with customers and improve customer satisfaction.

Business Intelligence (BI) helps CRM systems analyze customer data and improve:

- Customer service
- Sales
- Marketing
- Customer retention

using dashboards, reports, and analytics.

2 Applications of BI in CRM

1) Customer Behavior Analysis

BI analyzes customer:

- Buying patterns
- Interests
- Preferences

Example

Online shopping websites recommend products based on customer history.

2) Customer Segmentation

BI groups customers based on:

- Age
- Location
- Purchase history
- Interests

Example

E-commerce companies create separate offers for students and working professionals.

3) Personalized Marketing

BI helps companies send personalized offers and advertisements.

Example

Amazon suggests products based on previous purchases.

4) Customer Retention

BI identifies customers who may stop using services.

Example

Telecom companies offer discounts to inactive customers.

5) Improved Customer Service

BI dashboards help customer support teams analyze complaints and response times.

Example

Banks monitor customer complaints using dashboards.

6) Sales Analysis

BI helps organizations analyze:

- Product demand

- Sales trends
- Customer preferences

to improve sales strategies.

3 Real-Time Case Study

Case Study: Amazon CRM using BI

Amazon uses Business Intelligence in CRM to analyze customer behavior.

Amazon collects customer data such as:

- Search history
- Purchase history
- Product ratings
- Customer feedback

BI systems analyze this data and:

- Recommend products
 - Send personalized offers
 - Predict customer interests
-

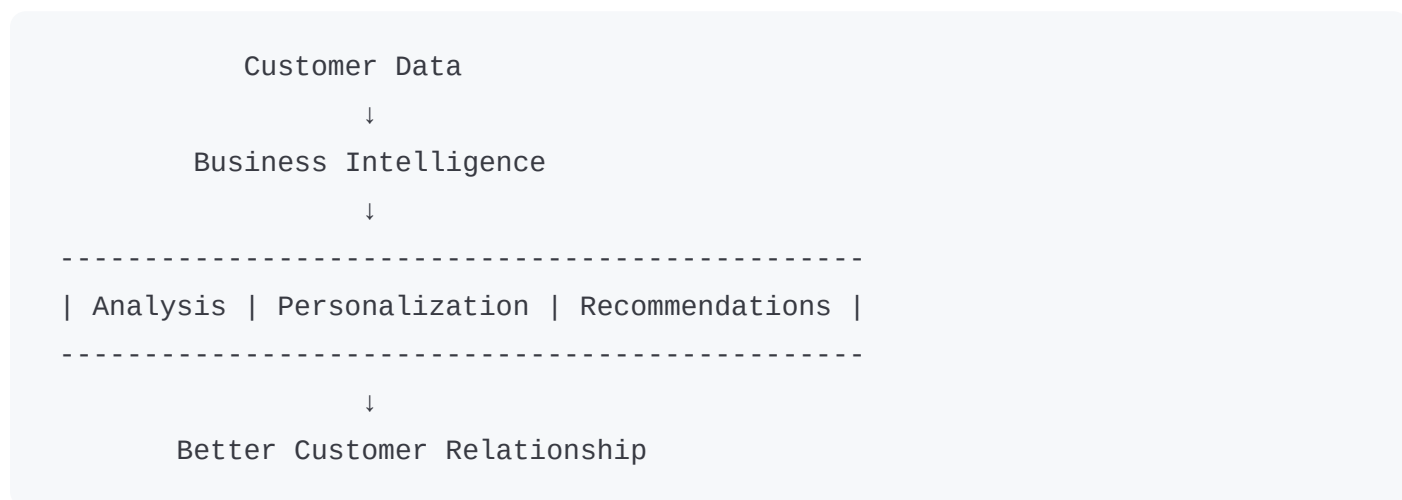
4 Explain the Case Study

- BI studies customer buying behavior.
- Amazon recommends similar products automatically.

- Personalized suggestions improve customer satisfaction.
- Better customer experience increases sales and customer loyalty.

Thus, BI improves CRM effectiveness.

5 Simple Diagram



6 Diagram Explanation

Customer Data

Customer information is collected.

BI System

Analyzes customer behavior and preferences.

Personalization

Customized offers and recommendations are generated.

Better Relationship

Improves customer satisfaction and loyalty.

7 Advantages of BI in CRM

- Better customer understanding
 - Improved customer satisfaction
 - Increased sales
 - Personalized marketing
 - Better customer retention
 - Improved decision making
-

8 Key Points / Features

- Uses customer data analysis
 - Supports personalized services
 - Improves customer loyalty
 - Helps marketing and sales teams
 - Uses dashboards and analytics tools
-

9 Memory Trick

"C P R S"

- **C** → Customer Analysis
- **P** → Personalization
- **R** → Retention
- **S** → Sales Improvement

Sentence:

"Customers Prefer Reliable Service"

10 Exam Writing Tips

- ✓ Write proper case study heading
 - ✓ Draw CRM-BI diagram neatly
 - ✓ Underline keywords:
 - Customer Retention
 - Personalized Marketing
 - Customer Behavior
 - Recommendations
 - ✓ Amazon example gives extra marks
-

1 1 Short Conclusion

Business Intelligence improves Customer Relationship Management by analyzing customer data, improving personalized services, increasing customer satisfaction, and supporting better business decisions.

★ Last Minute Revision Points

BI Applications in CRM

- Customer analysis
- Personalized marketing
- Customer retention
- Sales analysis

Case Study

Amazon recommendation system.

★ Most Important Keywords

- CRM
 - Business Intelligence
 - Customer Analysis
 - Personalized Marketing
 - Customer Retention
 - Recommendations
 - Customer Satisfaction
-

★ Viva Questions with Answers

Q1) What is CRM?

CRM is a system used to manage customer relationships.

Q2) How does BI help CRM?

BI analyzes customer data and improves customer service.

Q3) Give one example of BI in CRM.

Amazon product recommendation system.

Q4) Why is customer retention important?

Retaining customers increases long-term business profit.

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"Enlist the advantages of using Business Intelligence in Customer Relationship Management." [6]

5/13/2026, 10:37:07 PM



Q. Enlist the advantages of using Business Intelligence in Customer Relationship Management (CRM). [6 Marks]

1 Simple Introduction

Customer Relationship Management (CRM) is used to manage customer interactions and improve customer satisfaction.

Business Intelligence (BI) helps CRM systems analyze customer data using:

- Reports
- Dashboards
- Data analytics
- Visualization tools

This helps organizations improve customer service and business performance.

2 Advantages of Using BI in CRM

1) Better Understanding of Customers

BI analyzes customer:

- Buying behavior
- Preferences
- Interests

Example

Online shopping companies understand which products customers like most.

2) Improved Customer Satisfaction

Organizations can provide better services based on customer needs.

Example

Banks solve customer complaints faster using BI dashboards.

3) Personalized Marketing

BI helps companies send customized offers and advertisements.

Example

Amazon recommends products based on customer purchase history.

4) Increased Customer Retention

BI identifies customers who may stop using services.

Companies provide special offers to retain customers.

Example

Telecom companies provide discounts to inactive users.

5) Better Sales Analysis

BI helps organizations analyze:

- Product demand
- Customer purchasing trends
- Sales performance

This improves sales strategies.

6) Faster Decision Making

Managers use BI dashboards and reports for quick business decisions.

7) Improved Customer Service

Customer complaints and feedback are analyzed quickly.

Example

Hospital management systems monitor patient feedback using BI.

3 Simple Real-Life Example

Example: Amazon CRM System

Amazon uses BI in CRM to:

- Analyze customer behavior
- Recommend products
- Send personalized offers
- Improve customer satisfaction

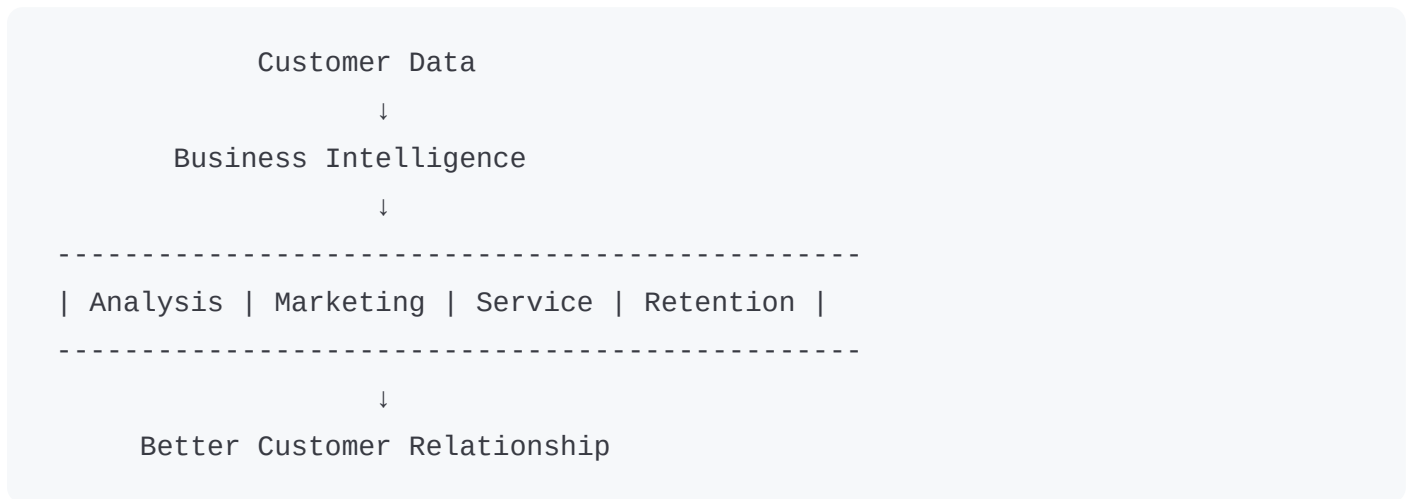
This increases sales and customer loyalty.

4 Explain the Example

- BI studies customer purchase history.
- Personalized recommendations improve user experience.
- Customers receive better service and offers.
- This increases customer retention and profit.

Thus, BI improves CRM performance effectively.

5 Simple Diagram



6 Diagram Explanation

Customer Data

Customer information is collected.

BI System

Analyzes customer behavior.

Marketing & Service

Improves customer support and offers.

Better Relationship

Increases customer satisfaction and loyalty.

7 Key Points / Features

Advantages of BI in CRM

- Better customer understanding
 - Personalized marketing
 - Improved customer service
 - Better customer retention
 - Faster decision making
 - Improved sales analysis
-

8 Memory Trick

“U S M R D”

- **U** → Understanding Customers
- **S** → Service Improvement
- **M** → Marketing Personalization
- **R** → Retention
- **D** → Decision Making

Sentence:

“Understanding Service Makes Relationships Durable”

9 Exam Writing Tips

✓ Draw CRM-BI diagram neatly

✓ Underline keywords:

- Customer Retention
- Personalized Marketing
- Customer Satisfaction
- Sales Analysis

✓ Write Amazon example for extra marks

10 Short Conclusion

Business Intelligence improves CRM by analyzing customer data, enhancing customer service, increasing customer retention, and supporting better marketing and business decisions.

★ Last Minute Revision Points

Advantages

- Better customer analysis
 - Personalized offers
 - Improved service
 - Customer retention
 - Better sales analysis
-

★ Most Important Keywords

- CRM
 - Business Intelligence
 - Customer Satisfaction
 - Personalized Marketing
 - Customer Retention
 - Sales Analysis
-

★ Viva Questions with Answers

Q1) What is CRM?

CRM is a system used to manage customer relationships.

Q2) How does BI help CRM?

BI analyzes customer data and improves customer service.

Q3) Give one application of BI in CRM.

Amazon recommendation system.

Q4) Why is customer retention important?

It helps increase long-term business profit.

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4) ERP and BI

"Write a short note on: ERP and Business Intelligence." [6]

5/13/2026, 10:39:47 PM



Q. Write a short note on: ERP and Business Intelligence. [6 Marks]

1 Simple Introduction

ERP (Enterprise Resource Planning) and **Business Intelligence (BI)** are important technologies used in organizations to improve business operations and decision making.

- ERP helps manage daily business activities.
- BI helps analyze business data and generate insights.

Together, ERP and BI improve organizational performance and efficiency.

2 ERP (Enterprise Resource Planning)

Meaning of ERP

ERP is a software system used to integrate and manage different business processes in one centralized system.

It manages departments like:

- Finance
 - HR
 - Inventory
 - Sales
 - Production
-

Features of ERP

- Centralized database
 - Real-time data sharing
 - Integrated business processes
 - Improved communication
 - Automation of tasks
-

Example of ERP

A manufacturing company uses ERP to manage:

- Employee records
 - Inventory
 - Production
 - Sales data
-

Advantages of ERP

- Improves efficiency
 - Reduces duplicate data
 - Better resource management
 - Faster operations
 - Improved coordination between departments
-

3 Business Intelligence (BI)

Meaning of BI

Business Intelligence is a technology used to collect, analyze, and visualize business data for better decision making.

BI uses:

- Dashboards
 - Reports
 - Charts
 - Analytics tools
-

Features of BI

- Data analysis
 - Data visualization
 - Real-time reporting
 - Trend analysis
 - Decision support
-

Example of BI

A retail company uses BI dashboards to analyze:

- Sales trends
 - Customer behavior
 - Profit reports
-

Advantages of BI

- Better decision making
 - Faster reporting
 - Improved forecasting
 - Trend identification
 - Better business analysis
-

4 Relationship between ERP and BI

ERP collects and stores business data.

BI analyzes ERP data and converts it into useful business insights.

Example

ERP stores sales data → BI analyzes sales performance using dashboards.

5 Simple Real-Life Example

Example: Retail Company

A retail company uses:

- ERP to manage inventory and sales records
- BI to analyze customer trends and sales performance

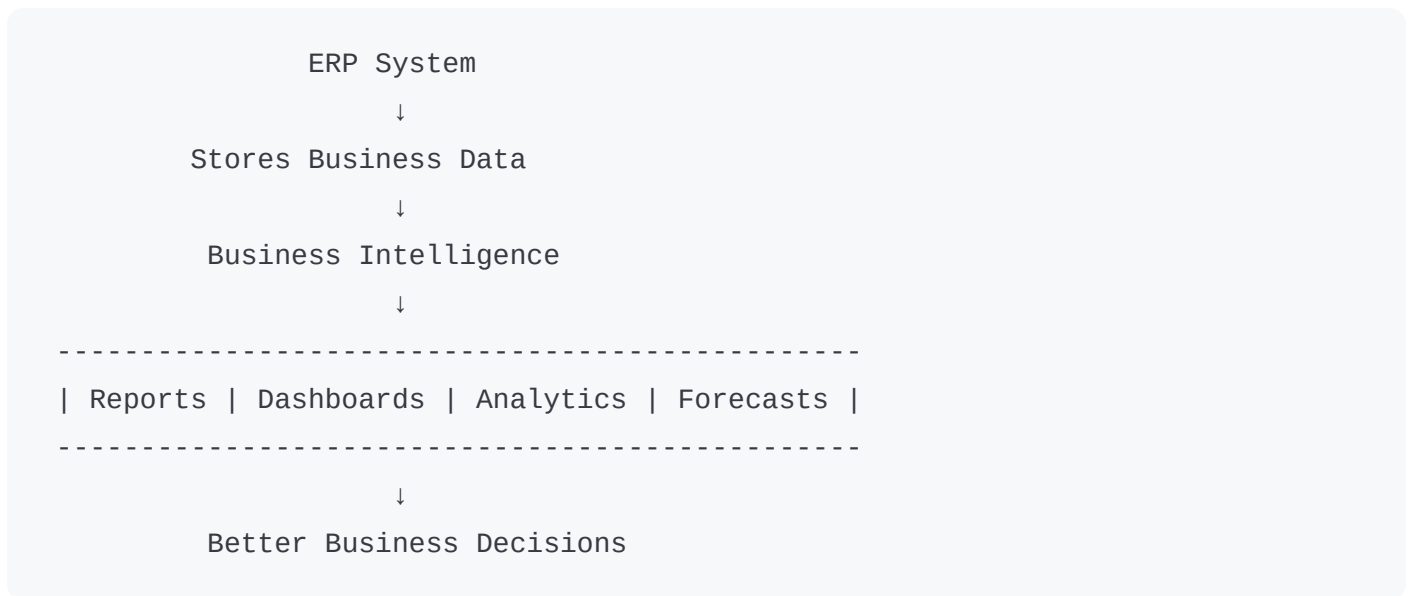
Managers use BI dashboards generated from ERP data for better decisions.

6 Explain the Example

- ERP stores daily business transactions.
- BI analyzes ERP data using reports and dashboards.
- Managers understand business performance easily.

Thus, ERP and BI work together to improve business management.

7 Simple Diagram



8 Diagram Explanation

ERP System

Collects and stores business data.

BI System

Analyzes stored business information.

Reports & Dashboards

Display business insights visually.

Better Decisions

Managers make smarter decisions.

9 Comparison between ERP and BI

ERP	BI
Manages business operations	Analyzes business data
Stores data	Generates insights
Focus on transactions	Focus on decision making
Operational system	Analytical system

10 Key Points / Features

ERP

- Integrates departments
- Stores business data
- Automates operations

BI

- Analyzes business data
- Supports decision making
- Uses dashboards and reports

11 Memory Trick

"E S – B A"

- **E** → ERP → Stores Data
- **B** → BI → Analyzes Data

Sentence:

"ERP Stores, BI Analyzes"

1 2 Exam Writing Tips

- ✓ Draw ERP-BI relationship diagram
- ✓ Underline keywords:

- Integration
- Dashboards
- Data Analysis
- Decision Making

- ✓ Write one real-life example
-

1 3 Short Conclusion

ERP and Business Intelligence together help organizations manage operations, analyze business data, and improve decision making effectively.

★ Last Minute Revision Points

ERP

- Stores and manages business data
- Integrates departments

BI

- Analyzes data
- Generates reports and dashboards

Relationship

ERP provides data → BI analyzes data

★ Most Important Keywords

- ERP
 - Business Intelligence
 - Data Analysis
 - Dashboards
 - Integration
 - Decision Making
 - Reports
-

★ Viva Questions with Answers

Q1) What is ERP?

ERP is a software system used to manage business processes.

Q2) What is BI?

BI is used to analyze business data for decision making.

Q3) How are ERP and BI related?

ERP stores data and BI analyzes that data.

Q4) Give one advantage of ERP.

Improved business integration.

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⚡ MEDIUM REPEATED

5) Banking Applications of BI

Asked:

Apr 2023

Nov 2024

Nov 2023

5/13/2026, 10:44:19 PM



Q. Explain the applications of Business Intelligence (BI) in Banking Sector. [8 Marks]

1 Simple Introduction

Business Intelligence (BI) in banking helps banks collect, analyze, and visualize financial data for better decision making.

Banks use BI tools like:

- Dashboards
- Reports
- Data analytics
- Data mining

to improve banking services, security, and customer satisfaction.

2 Applications of BI in Banking

1) Fraud Detection

BI systems monitor customer transactions and identify suspicious activities.

Example

Detecting unusual ATM withdrawals or credit card fraud.

2) Customer Relationship Management (CRM)

BI analyzes customer behavior and preferences.

Example

Banks provide personalized loan and credit card offers.

3) Risk Management

BI helps banks identify financial risks and loan risks.

Example

Checking whether a customer may fail to repay a loan.

4) Financial Reporting

BI generates:

- Profit reports
- Revenue reports
- Expense reports

quickly and accurately.

5) Loan Analysis

BI helps banks analyze:

- Loan applications
- Credit history
- Repayment capability

before approving loans.

6) Real-Time Monitoring

Banks monitor:

- Transactions
- Cash flow
- Customer activities

in real time.

7) Customer Segmentation

BI groups customers based on:

- Income
- Spending behavior
- Account activity

This helps banks provide targeted services.

8) Decision Making

BI dashboards help managers make fast and accurate banking decisions.

3 Simple Real-Life Example

Example: SBI Bank

State Bank of India uses BI systems to:

- Detect fraud
- Monitor transactions
- Analyze customer behavior

- Generate financial reports

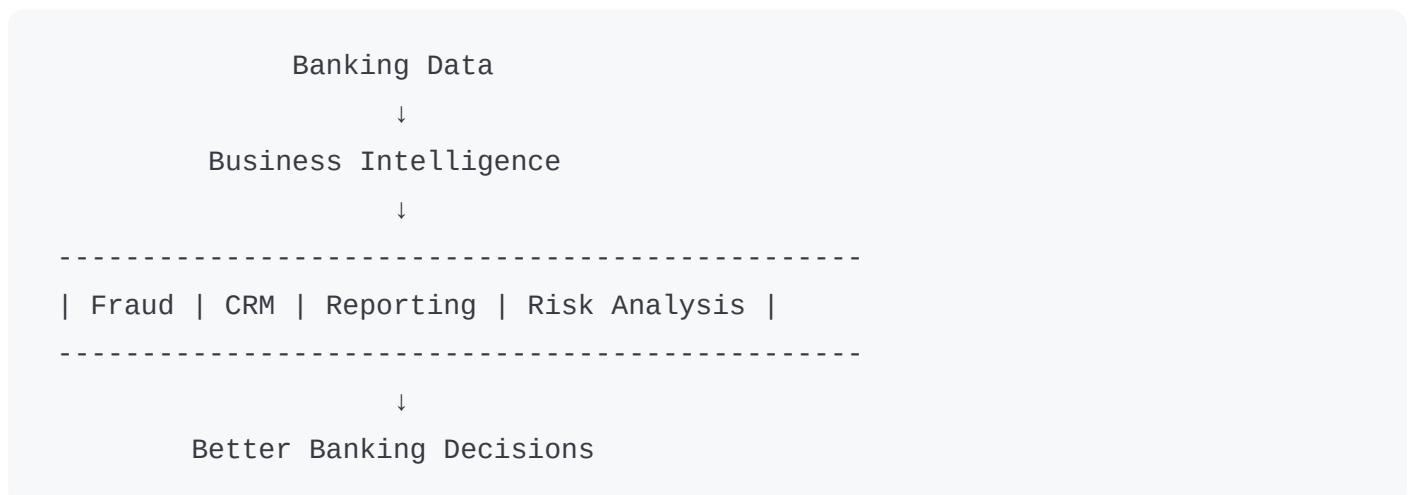
BI dashboards help bank managers improve security and customer services.

4 Explain the Example

- Transaction data is continuously analyzed.
- Fraud activities are identified quickly.
- Customer behavior analysis improves banking services.
- Managers use reports for better decisions.

Thus, BI improves efficiency and security in banking.

5 Simple Diagram



6 Diagram Explanation

Banking Data

Customer and transaction information is collected.

BI System

Analyzes banking data.

Applications

Fraud detection, CRM, reporting, and risk analysis are performed.

Better Decisions

Managers improve banking operations and services.

7 Advantages of BI in Banking

- Better fraud detection
 - Improved customer service
 - Faster reporting
 - Better risk management
 - Real-time monitoring
 - Improved decision making
-

8 Key Points / Features

- Uses dashboards and analytics
- Monitors transactions continuously

- Supports CRM and risk analysis
 - Improves banking security
 - Generates reports quickly
-

9 Memory Trick

“F C R L D”

- **F** → Fraud Detection
- **C** → CRM
- **R** → Reporting
- **L** → Loan Analysis
- **D** → Decision Making

Sentence:

“Fraud Checks Require Logical Decisions”

10 Exam Writing Tips

- ✓ Draw banking BI diagram neatly
- ✓ Underline keywords:

- Fraud Detection
- Risk Management
- CRM
- Financial Reporting

11 Short Conclusion

Business Intelligence plays an important role in banking by improving fraud detection, customer management, financial analysis, and decision making, leading to better banking services and security.

★ Last Minute Revision Points

Applications of BI in Banking

- Fraud Detection
- CRM
- Loan Analysis
- Risk Management
- Financial Reporting

Main Benefits

- Better security
 - Faster analysis
 - Improved customer service
-

★ Most Important Keywords

- Business Intelligence
 - Banking Sector
 - Fraud Detection
 - CRM
 - Risk Management
 - Financial Reporting
 - Loan Analysis
-

★ Viva Questions with Answers

Q1) What is BI in banking?

BI in banking means analyzing banking data for better decisions.

Q2) How does BI help banks?

It helps in fraud detection and customer analysis.

Q3) Give one application of BI in banking.

Credit card fraud detection.

Q4) Why is BI important in banking?

It improves security, reporting, and customer service.

5/13/2026, 10:44:20 PM



8) Types of Business Analytics / BA vs BI

Asked:

Apr 2024

Nov 2023

5/13/2026, 10:44:41 PM



Q. Explain the Types of Business Analytics. [8 Marks]

1 Simple Introduction

Business Analytics (BA) is the process of analyzing business data to improve decision making and business performance.

Business Analytics helps organizations:

- Understand past performance
- Predict future trends
- Make better business decisions

The main types of Business Analytics are:

1. Descriptive Analytics
2. Predictive Analytics
3. Prescriptive Analytics

(Some organizations also include Diagnostic Analytics.)

2 Types of Business Analytics

1) Descriptive Analytics

Meaning

Descriptive Analytics studies past business data to understand:

“What happened?”

It uses:

- Reports
- Dashboards
- Charts
- Graphs

Example

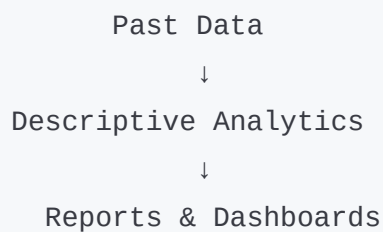
A retail company analyzes last year's sales reports.

Importance

- Understands past performance

- Identifies trends and patterns
 - Supports business monitoring
-

Diagram



2) Predictive Analytics

Meaning

Predictive Analytics uses historical and current data to predict:

“What may happen in future?”

It uses:

- Statistical analysis
 - Machine learning
 - Forecasting techniques
-

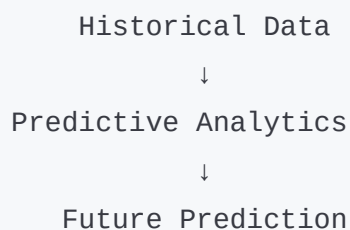
Example

An e-commerce company predicts festival season sales.

Importance

- Forecasts future trends
 - Reduces business risk
 - Improves planning
-

Diagram



3) Prescriptive Analytics

Meaning

Prescriptive Analytics suggests:

“What should be done?”

It recommends best actions using:

- Optimization
 - AI techniques
 - Decision models
-

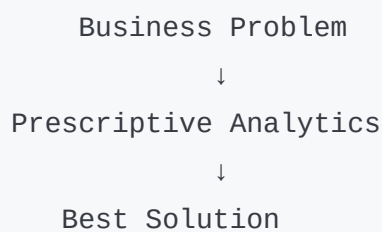
Example

A logistics company identifies shortest delivery routes.

Importance

- Supports decision making
 - Improves efficiency
 - Optimizes resources
-

Diagram



3 Comparison Table

Analytics Type	Main Purpose	Question Answered
Descriptive	Analyze past	What happened?
Predictive	Forecast future	What may happen?
Prescriptive	Suggest actions	What should be done?

4 Simple Real-Life Example

Example: Online Shopping Company

- Descriptive Analytics → Analyze previous sales
- Predictive Analytics → Predict future demand
- Prescriptive Analytics → Suggest best pricing strategy

5 Explain the Example

- Past data helps understand performance.
- Future predictions improve planning.
- Suggested actions improve profit.

Thus, all analytics types support better business decisions.

6 Summary Diagram

Business Analytics



| Descriptive | Predictive | Prescriptive |

7 Advantages of Business Analytics

- Better decision making
- Improved forecasting
- Better planning
- Increased efficiency
- Reduced business risk

8 Memory Trick

“D P P”

- **D** → Descriptive → Past
- **P** → Predictive → Future
- **P** → Prescriptive → Solution

Sentence:

“Describe Predict Provide”

9 Exam Writing Tips

✓ Draw all diagrams neatly

✓ Underline keywords:

- Forecasting
- Optimization
- Decision Making
- Historical Data

✓ Write one example for each type

10 Short Conclusion

Business Analytics helps organizations analyze past data, predict future outcomes, and recommend best actions for better business performance and decision making.

★ Last Minute Revision Points

Types of Business Analytics

1. Descriptive
2. Predictive
3. Prescriptive

Main Questions

- What happened?
 - What may happen?
 - What should be done?
-

★ Most Important Keywords

- Business Analytics
 - Descriptive Analytics
 - Predictive Analytics
 - Prescriptive Analytics
 - Forecasting
 - Optimization
 - Decision Making
-

★ Viva Questions with Answers

Q1) What is Business Analytics?

Business Analytics means analyzing business data for decision making.

Q2) Which analytics predicts future outcomes?

Predictive Analytics.

Q3) Which analytics suggests best actions?

Prescriptive Analytics.

Q. Differentiate between Business Analytics (BA) and Business Intelligence (BI). [8 Marks]

1 Simple Introduction

Business Intelligence (BI) and Business Analytics (BA) are technologies used for analyzing business data and improving decision making.

Although both use data analysis, their purpose and approach are different.

2 Difference between BA and BI

Business Intelligence (BI)	Business Analytics (BA)
Focuses on past and present data	Focuses on future prediction
Answers "What happened?"	Answers "What may happen?"
Uses dashboards and reports	Uses forecasting and prediction
Mainly descriptive analysis	Mainly predictive and prescriptive analysis

Business Intelligence (BI)

Helps monitor business performance

Uses reporting tools

Supports operational decisions

Example: Sales dashboard

Business Analytics (BA)

Helps improve future planning

Uses statistical and analytical tools

Supports strategic decisions

Example: Sales forecasting

3 Simple Real-Life Example

Example: Retail Company

BI

Company analyzes last month sales reports using dashboards.

BA

Company predicts next month sales using forecasting techniques.

4 Explain the Example

- BI helps understand past performance.
 - BA helps predict future trends.
 - Together they improve business decisions.
-

5 Simple Diagram

Business Data



| BI → Past Analysis |
BA → Future Analysis

6 Diagram Explanation

BI

Analyzes historical business performance.

BA

Predicts future outcomes and solutions.

7 Advantages

BI

- Better monitoring
- Faster reporting
- Improved visualization

BA

- Better forecasting
 - Smarter planning
 - Improved strategy
-

8 Memory Trick

“BI = Backward Insight”

“BA = Beyond Analysis”

- BI → Past
 - BA → Future
-

9 Exam Writing Tips

✓ Draw comparison table properly

✓ Underline keywords:

- Historical Data
- Forecasting
- Dashboards
- Prediction

✓ Write one example for both

10 Short Conclusion

Business Intelligence helps organizations understand current and past performance, while Business Analytics helps predict future trends and improve strategic decision making.

Last Minute Revision Points

BI

- Past analysis
- Dashboards
- Reporting

BA

- Future prediction
 - Forecasting
 - Optimization
-

Most Important Keywords

- Business Intelligence
- Business Analytics
- Forecasting
- Dashboards

- Historical Data
 - Prediction
 - Decision Making
-

Viva Questions with Answers

Q1) What is BI?

BI analyzes past and present business data.

Q2) What is BA?

BA predicts future business outcomes.

Q3) Which is future-oriented?

Business Analytics.

Q4) Which uses dashboards?

Business Intelligence.

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UNIT 6 → (Q7 & Q8)

🔥 MOST REPEATED QUESTIONS

1) Issues of Legality, Privacy and Ethics

"Write a short note on: Issues of legality, Privacy and Ethics." [6]

5/13/2026, 10:45:33 PM



Q. Write a short note on: Issues of Legality, Privacy and Ethics. [6 Marks]

1 Simple Introduction

In Business Analytics and Business Intelligence, organizations collect and analyze large amounts of data.

While using data, organizations face important issues related to:

1. Legality
2. Privacy
3. Ethics

These issues are important to protect customer information and use data responsibly.

2 Issues of Legality

Meaning

Legality means following laws and government rules while collecting, storing, and using data.

Organizations must use data legally and avoid unauthorized activities.

Examples of Legal Issues

- Using customer data without permission
 - Software piracy
 - Illegal data sharing
 - Copyright violation
-

Real-Life Example

A company shares customer phone numbers with another company without permission.

This is illegal and may lead to legal action.

Importance

- Protects customer rights
- Avoids legal penalties
- Ensures lawful business operations

3 Issues of Privacy

Meaning

Privacy means protecting personal and confidential information of users.

Organizations must keep customer data secure and confidential.

Examples of Privacy Issues

- Data leakage
 - Hacking of personal information
 - Unauthorized access to bank details
 - Sharing medical records without permission
-

Real-Life Example

A hospital accidentally exposes patient medical records online.

This violates patient privacy.

Importance

- Protects confidential information
 - Builds customer trust
 - Prevents misuse of personal data
-

4 Issues of Ethics

Meaning

Ethics means using data in a fair, honest, and responsible manner.

Even if something is legal, it should also be morally correct.

Examples of Ethical Issues

- Misusing customer data
 - Manipulating reports
 - Biased decision making
 - Spreading false information
-

Real-Life Example

A company secretly tracks customer online behavior without informing users.

This is unethical behavior.

Importance

- Builds trust and transparency
 - Improves company reputation
 - Encourages responsible data usage
-

5 Simple Real-Life Example

Example: Social Media Platform

A social media company collects user data for advertisements.

Issues may arise if:

- Data is shared without permission
- User privacy is violated
- Data is misused for unethical purposes

Thus, organizations must use data carefully and responsibly.

6 Simple Diagram

Data Collection



| Legality | Privacy | Ethics |



Responsible Data Usage

7 Diagram Explanation

Legality

Follow laws and regulations.

Privacy

Protect personal information.

Ethics

Use data fairly and responsibly.

Together they ensure safe and responsible business practices.

8 Key Points / Features

Legality

- Lawful data usage
- Avoid illegal activities

Privacy

- Protect confidential data
- Prevent unauthorized access

Ethics

- Honest and fair data use
- Responsible decision making

9 Advantages of Following These Principles

- Builds customer trust
- Improves company reputation
- Avoids legal problems
- Protects sensitive information
- Encourages ethical business practices

10 Memory Trick

“L P E”

- **L** → Legality
- **P** → Privacy

- **E** → Ethics

Sentence:

“Legal Privacy Ethics”

1 1 Exam Writing Tips

✓ Draw simple LPE diagram

✓ Underline keywords:

- Confidential Information
- Responsible Usage
- Legal Compliance
- Ethical Behavior

✓ Write one example for each issue

1 2 Short Conclusion

Legality, Privacy, and Ethics are important issues in Business Intelligence and Analytics. Organizations must use data legally, securely, and ethically to protect users and maintain trust.

Last Minute Revision Points

Legality

- Follow laws

Privacy

- Protect personal data

Ethics

- Fair and responsible usage
-

Most Important Keywords

- Legality
 - Privacy
 - Ethics
 - Confidential Data
 - Data Security
 - Ethical Behavior
 - Responsible Data Usage
-

Viva Questions with Answers

Q1) What is legality?

Legality means following laws while using data.

Q2) What is privacy?

Privacy means protecting personal information.

Q3) What is ethics?

Ethics means fair and responsible use of data.

Q4) Why are these issues important?

They protect customer data and improve trust.

5/13/2026, 10:45:34 PM



2) Cloud Computing and BI

"Write a short note on: Cloud Computing and Business Intelligence." [6]

5/13/2026, 10:45:56 PM



Q. Write a short note on: Cloud Computing and Business Intelligence. [6 Marks]

1 Simple Introduction

Cloud Computing and **Business Intelligence (BI)** are important modern technologies used by organizations for data storage, analysis, and decision making.

- Cloud Computing provides online storage and computing services through the internet.
- Business Intelligence helps analyze business data using reports, dashboards, and analytics tools.

Together, Cloud Computing and BI improve business performance and efficiency.

2 Cloud Computing

Meaning of Cloud Computing

Cloud Computing is a technology that provides:

- Storage
- Software
- Servers
- Data processing

through the internet instead of local computers.

Users can access data and applications anytime from anywhere.

Features of Cloud Computing

- Online data storage
- Remote access
- Scalability
- Cost reduction
- Easy data sharing

Examples of Cloud Services

- Google Drive
- Microsoft OneDrive
- Amazon Web Services (AWS)

Advantages of Cloud Computing

- Low infrastructure cost
 - Access from anywhere
 - Easy backup and recovery
 - Better scalability
 - Faster deployment
-

3 Business Intelligence (BI)

Meaning of BI

Business Intelligence is a technology used to:

- Analyze business data
- Generate reports
- Create dashboards
- Support decision making

BI converts raw data into meaningful business information.

Features of BI

- Data analysis

- Data visualization
 - Dashboard creation
 - Reporting
 - Forecasting
-

Examples of BI Tools

- Microsoft Power BI
 - Tableau
-

Advantages of BI

- Better decision making
 - Faster reporting
 - Improved forecasting
 - Better business monitoring
-

4 Relationship between Cloud Computing and BI

Cloud Computing stores and processes large business data online.

BI tools use cloud data for:

- Analysis
- Reporting
- Dashboard generation

This combination is called:

“Cloud BI”

5 Simple Real-Life Example

Example: E-Commerce Company

An online shopping company stores customer and sales data on cloud servers.

BI tools analyze this cloud data to:

- Monitor sales
- Predict customer demand
- Generate dashboards

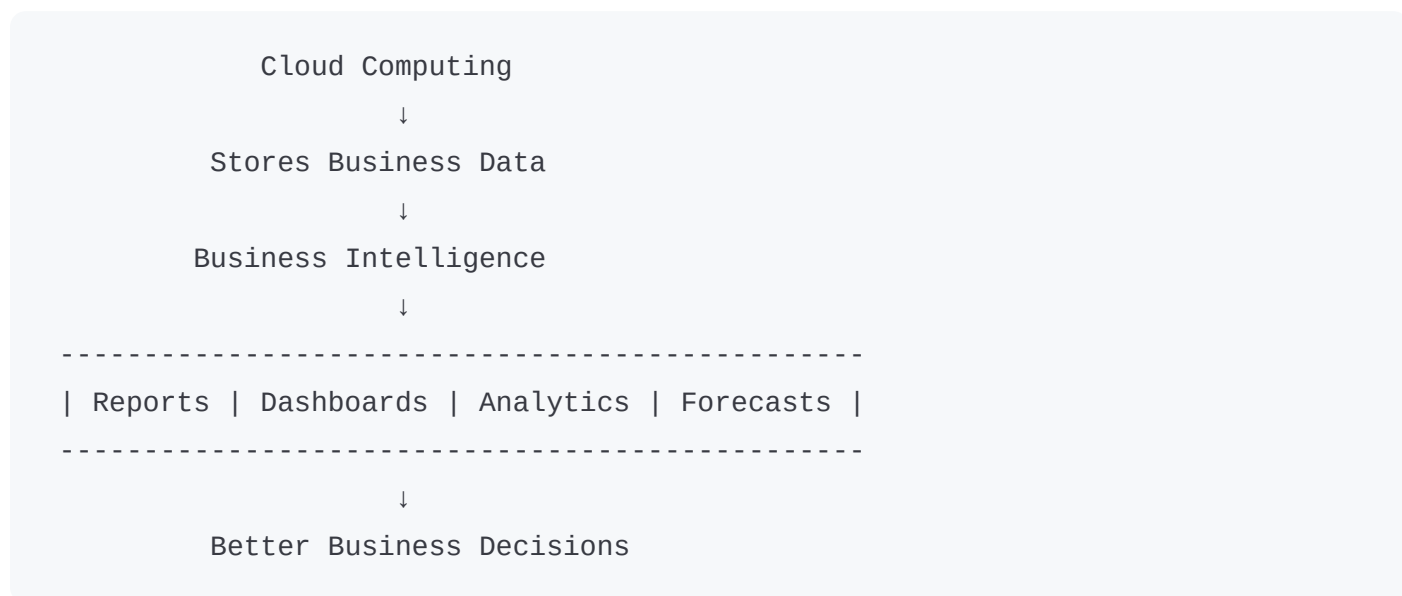
Managers can access reports from anywhere using the internet.

6 Explain the Example

- Cloud stores business data securely online.
- BI analyzes the cloud data.
- Dashboards provide real-time business insights.
- Managers make faster business decisions.

Thus, Cloud Computing and BI improve business flexibility and analysis.

7 Simple Diagram



8 Diagram Explanation

Cloud Computing

Stores and manages data online.

BI System

Analyzes cloud data.

Reports & Dashboards

Visualize business insights.

Better Decisions

Managers improve planning and decision making.

9 Key Points / Features

Cloud Computing

- Online data storage
- Remote access
- Scalability

Business Intelligence

- Data analysis
- Reporting
- Dashboard creation

Cloud BI

- Real-time analytics
 - Faster access to reports
 - Better collaboration
-

10 Advantages of Cloud BI

- Access from anywhere
 - Reduced infrastructure cost
 - Real-time reporting
 - Better scalability
 - Improved business efficiency
-

11 Memory Trick

“C S - B A”

- **C** → Cloud → Stores Data
- **B** → BI → Analyzes Data

Sentence:

“Cloud Stores, BI Analyzes”

12 Exam Writing Tips

✓ Draw Cloud BI diagram neatly

✓ Underline keywords:

- Cloud Storage
- Dashboards
- Real-Time Analytics
- Decision Making

✓ Write e-commerce example

13 Short Conclusion

Cloud Computing and Business Intelligence together help organizations store, analyze, and access business data efficiently, improving reporting, forecasting, and decision making.

Last Minute Revision Points

Cloud Computing

- Online storage
- Remote access
- Scalability

BI

- Data analysis
- Dashboards
- Reporting

Cloud BI

- Cloud stores data
 - BI analyzes data
-

★ Most Important Keywords

- Cloud Computing
 - Business Intelligence
 - Cloud BI
 - Data Analysis
 - Dashboards
 - Real-Time Reporting
 - Scalability
-

★ Viva Questions with Answers

Q1) What is Cloud Computing?

Cloud Computing provides online storage and services through the internet.

Q2) What is BI?

BI analyzes business data for decision making.

Q3) What is Cloud BI?

Cloud BI means using BI tools with cloud data.

Q4) Give one advantage of Cloud BI.

Access business reports from anywhere.

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3) BI Search and Text Analytics

"Explain BI Search and Text Analytics with the help of diagram." [6]

5/13/2026, 10:46:28 PM



Q. Explain BI Search and Text Analytics with the help of diagram. [6 Marks]

1 Simple Introduction

BI Search and **Text Analytics** are important techniques used in Business Intelligence (BI) to analyze large amounts of business information.

They help organizations:

- Search business data quickly
- Analyze text information
- Identify patterns and trends
- Improve decision making

2 BI Search

Meaning of BI Search

BI Search is a process used to search and retrieve business information from:

- Databases
- Reports
- Documents
- Emails
- Dashboards

It helps users quickly find useful business data.

Features of BI Search

- Fast information retrieval
 - Easy data access
 - Supports decision making
 - Searches structured and unstructured data
-

Example of BI Search

A manager searches:

- Monthly sales reports
- Customer transaction details
- Product performance reports

using a BI system.

Importance of BI Search

- Saves time
 - Improves data access
 - Helps managers find business insights quickly
-

3 Text Analytics

Meaning of Text Analytics

Text Analytics is the process of analyzing text data to extract useful information.

It analyzes:

- Customer reviews
- Emails
- Social media comments
- Feedback forms

using analytical techniques.

Features of Text Analytics

- Sentiment analysis
 - Keyword extraction
 - Trend identification
 - Customer opinion analysis
-

Example of Text Analytics

An online shopping company analyzes customer reviews to identify:

- Positive feedback
 - Negative feedback
 - Customer satisfaction level
-

Importance of Text Analytics

- Improves customer understanding
 - Helps improve products and services
 - Supports business decisions
-

4 Simple Real-Life Example

Example: E-Commerce Website

An online shopping company uses:

- BI Search to find sales and customer reports
- Text Analytics to analyze customer reviews and feedback

Managers identify customer problems and improve services accordingly.

5 Explain the Example

- BI Search retrieves required business data quickly.
- Text Analytics studies customer opinions.
- Negative reviews help identify service problems.
- Company improves customer satisfaction and product quality.

Thus, BI Search and Text Analytics improve business analysis and decision making.

6 Simple Diagram



7 Diagram Explanation

BI Search

Finds required business information quickly.

Text Analytics

Analyzes customer opinions and text data.

Insights & Decisions

Organizations improve products and business strategies.

8 Advantages of BI Search and Text Analytics

- Faster information retrieval
 - Better customer understanding
 - Improved decision making
 - Identifies customer trends
 - Supports business intelligence
 - Improves customer satisfaction
-

9 Key Points / Features

BI Search

- Searches business data
- Retrieves reports quickly
- Supports managers

Text Analytics

- Analyzes customer reviews
 - Identifies sentiments
 - Extracts useful information
-

10 Memory Trick

“S T I”

- **S** → Search Data
- **T** → Text Analysis
- **I** → Insights

Sentence:

“Search Text Insights”

1 1 Exam Writing Tips

- ✓ Draw BI Search & Text Analytics diagram
- ✓ Underline keywords:

- Information Retrieval
- Sentiment Analysis
- Customer Reviews
- Business Insights

✅ Write e-commerce example

1 2 Short Conclusion

BI Search and Text Analytics help organizations search business information efficiently and analyze customer opinions, improving business intelligence and decision making.

★ Last Minute Revision Points

BI Search

- Information retrieval
- Fast report access

Text Analytics

- Analyze text data
- Sentiment analysis

Main Goal

- Better business insights
-

★ Most Important Keywords

- BI Search
 - Text Analytics
 - Information Retrieval
 - Sentiment Analysis
 - Customer Reviews
 - Business Intelligence
 - Decision Making
-

★ Viva Questions with Answers

Q1) What is BI Search?

BI Search retrieves business information quickly.

Q2) What is Text Analytics?

Text Analytics analyzes text data to extract useful information.

Q3) Give one example of Text Analytics.

Customer review analysis.

Q4) Why is Text Analytics important?

It helps understand customer opinions and improve services.

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4) Analytics Applications for Consumers

"Discuss the Analytical Applications for Consumers with respect to online shopping websites." [5]

Apr 2023

"Write a short note on Analytics Applications for Consumers." [6]

5/13/2026, 10:46:58 PM



Q. Discuss the Analytical Applications for Consumers with respect to Online Shopping Websites. [5 Marks]

1 Simple Introduction

Analytics Applications for Consumers are techniques used by online shopping websites to analyze customer behavior and improve customer experience.

These applications help companies:

- Understand customer preferences
- Recommend products
- Improve sales
- Provide better services

using data analysis and Business Intelligence techniques.

2 Analytical Applications in Online Shopping Websites

1) Product Recommendation System

Websites analyze customer:

- Search history
- Purchase history
- Browsing behavior

to recommend products.

Example

Amazon suggests products like:

“Customers also bought...”

2) Personalized Advertisements

Analytics helps websites display personalized ads based on customer interests.

Example

Showing mobile advertisements to users searching for smartphones.

3) Customer Behavior Analysis

Websites study:

- Customer clicks
- Time spent on products
- Shopping patterns

to understand customer interests.

4) Sales and Demand Prediction

Analytics predicts:

- Future product demand
- Festival season sales
- Popular products

This helps companies manage inventory.

5) Customer Feedback Analysis

Customer reviews and ratings are analyzed using text analytics.

Example

Identifying whether customer reviews are positive or negative.

6) Dynamic Pricing

Online shopping websites adjust product prices based on:

- Demand
 - Competition
 - Customer behavior
-

3 Simple Real-Life Example

Example: Amazon Online Shopping Website

Amazon uses analytics to:

- Recommend products
- Analyze customer buying behavior
- Predict future demand
- Show personalized advertisements

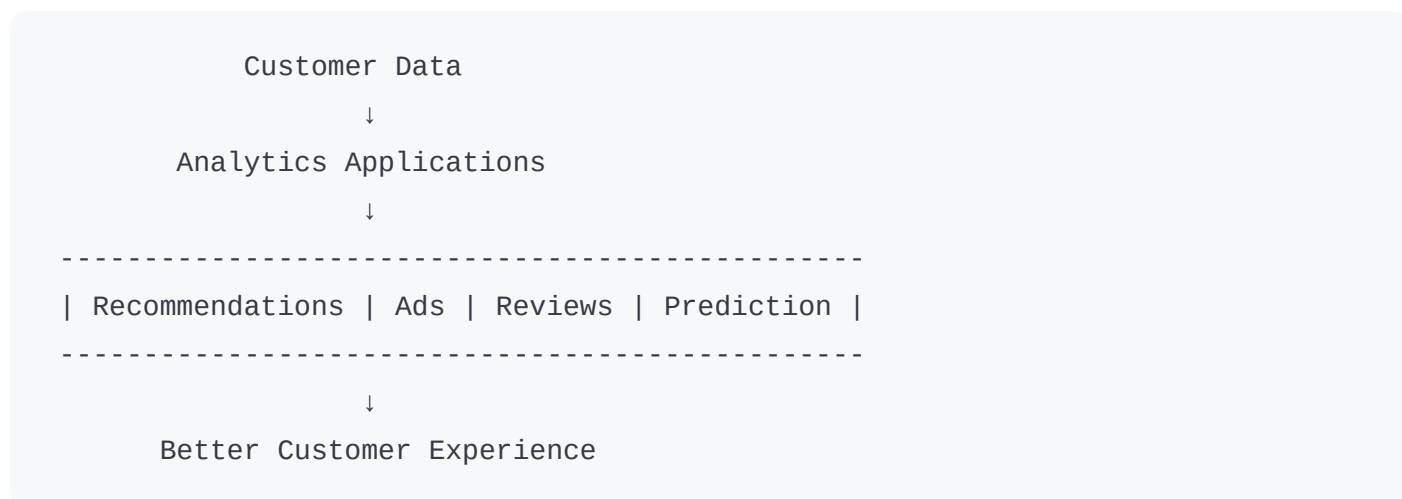
This improves customer satisfaction and increases sales.

4 Explain the Example

- Customer purchase history is analyzed.
- Personalized recommendations improve shopping experience.
- Demand prediction helps maintain stock availability.
- Customer reviews help improve product quality.

Thus, analytics applications improve online shopping services.

5 Simple Diagram



6 Diagram Explanation

Customer Data

Customer browsing and purchase information is collected.

Analytics Applications

Data is analyzed using BI and analytics techniques.

Recommendations & Prediction

Personalized services and future predictions are generated.

Better Experience

Improves customer satisfaction and sales.

7 Advantages of Analytics Applications

- Better customer understanding
 - Personalized shopping experience
 - Improved customer satisfaction
 - Increased sales
 - Better inventory management
 - Faster business decisions
-

8 Key Points / Features

- Product recommendation systems
 - Personalized advertisements
 - Customer behavior analysis
 - Demand forecasting
 - Feedback analysis
-

9 Memory Trick

“R A B F D”

- **R** → Recommendations

- **A** → Advertisements
- **B** → Behavior Analysis
- **F** → Feedback Analysis
- **D** → Demand Prediction

Sentence:

“Recommendations And Behavior Forecast Demand”

10 Exam Writing Tips

✓ Draw analytics application diagram

✓ Underline keywords:

- Product Recommendation
- Customer Behavior
- Demand Prediction
- Personalized Ads

✓ Amazon example gives extra marks

1 1 Short Conclusion

Analytics Applications help online shopping websites understand customer behavior, improve customer experience, increase sales, and provide personalized services effectively.

Last Minute Revision Points

Applications

- Product recommendation
- Personalized ads
- Customer analysis
- Demand prediction
- Feedback analysis

Main Goal

- Better customer experience
-

★ Most Important Keywords

- Analytics Applications
 - Online Shopping
 - Product Recommendation
 - Customer Behavior
 - Personalized Marketing
 - Demand Forecasting
 - Customer Feedback
-

★ Viva Questions with Answers

Q1) What are analytics applications?

Applications that analyze customer data for better services.

Q2) How do online shopping websites use analytics?

They analyze customer behavior and recommend products.

Q3) Give one example of analytics application.

Amazon product recommendation system.

Q. Write a short note on Analytics Applications for Consumers. [6 Marks]

1 Simple Introduction

Analytics Applications for Consumers are systems that analyze customer data to improve customer experience, business services, and decision making.

Organizations use analytics to understand:

- Customer behavior
- Customer preferences
- Buying patterns
- Customer feedback

2 Applications of Analytics for Consumers

1) Product Recommendation

Companies recommend products based on customer interests.

Example

Netflix recommends movies based on watch history.

2) Personalized Marketing

Analytics helps businesses send customized offers and advertisements.

Example

Online stores send discount offers based on customer purchases.

3) Customer Feedback Analysis

Organizations analyze customer reviews and ratings.

Example

Restaurants analyze customer feedback from food delivery apps.

4) Demand Prediction

Businesses predict future customer demand.

Example

E-commerce websites predict festival season sales.

5) Customer Segmentation

Customers are grouped based on:

- Age
- Interests
- Buying behavior

This improves marketing strategies.

6) Improved Customer Service

Analytics helps organizations solve customer problems quickly.

Example

Banks analyze customer complaints using dashboards.

3 Simple Real-Life Example

Example: Netflix

Netflix analyzes:

- Viewing history
- Search patterns
- Ratings

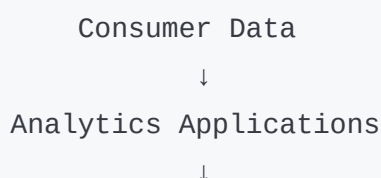
to recommend movies and shows to users.

4 Explain the Example

- Analytics studies customer interests.
- Personalized recommendations improve user experience.
- Better suggestions increase customer satisfaction.

Thus, analytics improves consumer services.

5 Simple Diagram



| Recommendations | Feedback | Prediction |



Better Customer Experience

6 Advantages

- Personalized services
- Better customer understanding
- Improved customer satisfaction
- Better marketing
- Increased sales

7 Memory Trick

“R F P S”

- **R** → Recommendations
- **F** → Feedback Analysis
- **P** → Prediction
- **S** → Segmentation

Sentence:

“Recommendations Forecast Personalized Services”

Short Conclusion

Analytics Applications for Consumers help businesses understand customer behavior and provide better personalized services, improving customer satisfaction and business growth.

Viva Questions with Answers

Q1) What are analytics applications for consumers?

They analyze consumer data to improve services.

Q2) Give one example of consumer analytics.

Netflix movie recommendation system.

Q3) Why are analytics applications important?

They improve customer satisfaction and sales.

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⚡ MEDIUM REPEATED

5) BI Strategy / Critical Success Factors

Asked:

Nov 2024

Apr 2024

5/13/2026, 10:47:36 PM



Q. Explain Business Intelligence (BI) Strategy and Critical Success Factors. [8 Marks]

1 Simple Introduction

A **Business Intelligence (BI) Strategy** is a plan used by organizations to collect, analyze, and use business data effectively for better decision making.

To successfully implement BI systems, organizations must focus on important elements called **Critical Success Factors (CSFs)**.

These factors help organizations achieve successful BI implementation and business growth.

2 Business Intelligence (BI) Strategy

Meaning of BI Strategy

BI Strategy is a structured plan for:

- Data collection
- Data analysis
- Reporting
- Decision making

using BI tools like:

- Dashboards
 - Reports
 - Analytics systems
-

Objectives of BI Strategy

- Improve decision making
 - Increase business efficiency
 - Better customer understanding
 - Improve forecasting
 - Support business growth
-

Components of BI Strategy

1. Data Collection
2. Data Integration

3. Data Analysis
 4. Reporting and Dashboards
 5. Decision Support
-

Example of BI Strategy

A retail company creates a BI strategy to:

- Analyze customer buying patterns
- Improve sales
- Predict future demand

using BI dashboards and analytics tools.

3 Critical Success Factors (CSFs)

Meaning of Critical Success Factors

Critical Success Factors are important conditions required for successful implementation of BI systems.

Without these factors, BI projects may fail.

Important Critical Success Factors

1) Strong Management Support

Top management must support BI projects financially and strategically.

2) Quality Data

BI systems require:

- Accurate
- Complete
- Reliable data

Poor data leads to wrong decisions.

3) Skilled Employees

Employees should have proper knowledge of:

- BI tools
 - Data analysis
 - Reporting systems
-

4) Clear Business Goals

Organizations must define:

- Business objectives
- Performance targets
- BI requirements

clearly before implementation.

5) Proper Technology and Infrastructure

Organizations need:

- Good hardware
- Software
- Databases
- Network systems

for successful BI implementation.

6) User Involvement

End users should participate in BI planning and implementation.

This improves system usability.

7) Continuous Monitoring and Improvement

BI systems should be updated and monitored regularly.

4 Simple Real-Life Example

Example: Retail Company

A retail company implements BI strategy to improve sales analysis.

The company:

- Collects customer data
- Uses dashboards
- Predicts future demand

Critical success factors like:

- Good management support
- Accurate sales data
- Skilled staff

help the BI project succeed.

5 Explain the Example

- BI strategy helps analyze customer behavior.

- Dashboards improve decision making.
- Accurate data improves forecasting.
- Skilled employees use BI tools effectively.

Thus, BI strategy and CSFs improve business performance.

6 Simple Diagram



7 Diagram Explanation

BI Strategy

Defines how business data will be used.

CSFs

Important factors for BI success.

Successful BI System

Improves business performance and decision making.

8 Advantages of BI Strategy

- Better decision making
 - Improved forecasting
 - Better customer understanding
 - Increased efficiency
 - Improved business growth
-

9 Key Points / Features

BI Strategy

- Data-driven planning
- Reporting and dashboards
- Business analysis

Critical Success Factors

- Management support
- Quality data

- Skilled employees
 - Proper infrastructure
-

10 Memory Trick

“M D S G T”

- **M** → Management Support
- **D** → Data Quality
- **S** → Skilled Employees
- **G** → Goals
- **T** → Technology

Sentence:

“Managers Demand Skilled Goal-oriented Teams”

1 1 Exam Writing Tips

✓ Draw BI strategy + CSF diagram

✓ Underline keywords:

- Management Support
- Quality Data
- Dashboards
- Business Goals

✓ Write one real-life example

1 2 Short Conclusion

Business Intelligence Strategy helps organizations use data effectively for decision making, while Critical Success Factors ensure successful implementation of BI systems.

Last Minute Revision Points

BI Strategy

- Data collection
- Analysis
- Dashboards
- Decision support

Critical Success Factors

- Management support
- Data quality
- Skilled staff
- Technology

Most Important Keywords

- BI Strategy

- Critical Success Factors
 - Dashboards
 - Data Quality
 - Decision Making
 - Business Goals
 - BI Implementation
-

Viva Questions with Answers

Q1) What is BI Strategy?

It is a plan for using business data effectively.

Q2) What are Critical Success Factors?

Important conditions required for successful BI implementation.

Q3) Why is data quality important in BI?

Poor data leads to wrong decisions.

Q4) Give one Critical Success Factor.

Management support.

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6) Predicting Future using Data Analysis

Asked:

Apr 2024

Nov 2023

7) Rich Reports

Asked:

Apr 2023

5/13/2026, 10:57:53 PM



Q. Explain Predicting Future using Data Analysis. [6 Marks]

1 Simple Introduction

Predicting Future using Data Analysis means using past and present business data to forecast future outcomes and trends.

Organizations analyze data using:

- Statistical techniques
- Business Analytics
- Machine Learning
- Forecasting tools

to make better future decisions.

2 Concept Explanation

Data analysis studies:

- Historical data
- Customer behavior
- Sales trends
- Market patterns

to predict:

- Future sales
- Product demand
- Customer preferences
- Business risks

This process is mainly called:

“Predictive Analytics”

3 Steps in Predicting Future using Data Analysis

1) Data Collection

Business data is collected from:

- Sales records
 - Customer databases
 - Websites
 - Social media
-

2) Data Analysis

The collected data is analyzed using:

- Charts
 - Reports
 - Statistical tools
 - BI systems
-

3) Pattern Identification

Organizations identify:

- Trends
 - Customer behavior
 - Seasonal demand
-

4) Forecasting Future Outcomes

Future predictions are generated.

Example

Predicting next month sales.

5) Decision Making

Managers use predictions for:

- Planning
 - Marketing
 - Inventory management
-

4 Simple Real-Life Example

Example: Online Shopping Website

Amazon analyzes:

- Customer purchases
- Search history
- Festival sales trends

to predict:

- Future product demand
- Best-selling products

This helps manage stock and improve sales.

5 Explain the Example

- Historical customer data is analyzed.
- Patterns and trends are identified.
- Future demand is predicted.
- Managers prepare inventory and offers accordingly.

Thus, predictive data analysis improves business planning.

6 Simple Diagram



7 Diagram Explanation

Historical Data

Past business information is collected.

Data Analysis

Data is studied using analytical tools.

Pattern Identification

Business trends are identified.

Future Prediction

Future outcomes are forecasted.

Decision Making

Organizations improve planning and strategies.

8 Advantages

- Better forecasting
 - Improved planning
 - Reduced business risk
 - Better customer understanding
 - Improved decision making
-

9 Applications

- Sales forecasting
- Demand prediction
- Fraud detection

- Customer behavior analysis
 - Stock market prediction
-

10 Memory Trick

“D P F”

- **D** → Data Analysis
- **P** → Pattern Identification
- **F** → Future Prediction

Sentence:

“Data Predicts Future”

1 1 Exam Writing Tips

✓ Draw prediction process diagram

✓ Underline keywords:

- Predictive Analytics
- Forecasting
- Pattern Analysis
- Historical Data

✓ Write Amazon example for extra marks

12 Short Conclusion

Predicting future using data analysis helps organizations forecast future trends, improve planning, reduce risks, and make smarter business decisions.

★ Last Minute Revision Points

Main Steps

1. Data Collection
2. Data Analysis
3. Pattern Identification
4. Future Prediction

Main Goal

- Better forecasting and planning
-

★ Most Important Keywords

- Predictive Analytics
- Forecasting
- Historical Data
- Pattern Analysis
- Decision Making

- Future Prediction
-

★ Viva Questions with Answers

Q1) What is predictive analytics?

It predicts future outcomes using past data.

Q2) Why is data analysis important?

It helps organizations forecast future trends.

Q3) Give one application of predictive analytics.

Sales forecasting.

Q. Write a short note on Rich Reports. [5 Marks]

1 Simple Introduction

Rich Reports are advanced business reports that present information using:

- Charts
- Graphs
- Dashboards
- Images
- Interactive elements

instead of only plain text.

Rich reports help users understand business information quickly and effectively.

2 Features of Rich Reports

- Interactive reports
 - Attractive visualization
 - Real-time updates
 - Easy understanding of data
 - Supports dashboards and multimedia
-

3 Components of Rich Reports

Rich reports may include:

- Tables
- Charts
- Graphs

- KPIs
 - Dashboards
 - Hyperlinks
 - Drill-down analysis
-

4 Advantages of Rich Reports

1) Better Data Visualization

Complex data becomes easy to understand.

2) Faster Decision Making

Managers quickly identify trends and problems.

3) Interactive Analysis

Users can:

- Filter data
- Drill down into reports
- Analyze details

4) Real-Time Monitoring

Reports can display live business data.

5) Improved Communication

Visual reports improve communication between departments.

5 Simple Real-Life Example

Example: Hospital Dashboard Report

A hospital uses rich reports showing:

- Patient count
- Bed availability
- Emergency cases
- Doctor schedules

using charts and dashboards.

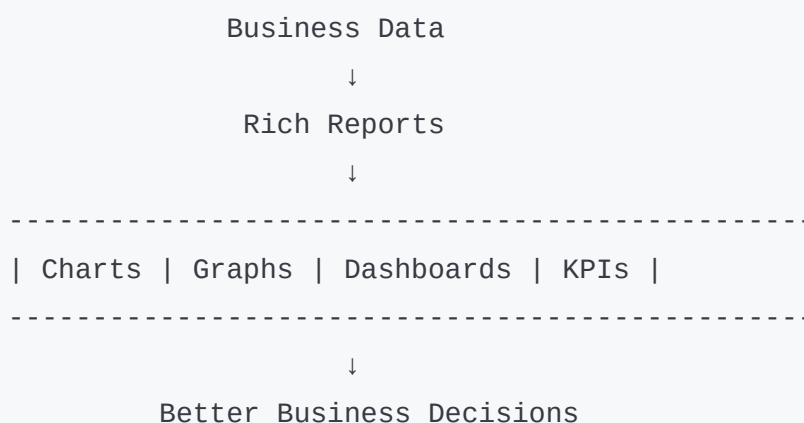
Managers understand hospital performance quickly.

6 Explain the Example

- Rich reports visually display hospital data.
- Dashboards help managers monitor operations in real time.
- Visual reports improve decision making.

Thus, rich reports improve business analysis and monitoring.

7 Simple Diagram



8 Diagram Explanation

Business Data

Raw business information is collected.

Rich Reports

Data is converted into visual reports.

Charts & Dashboards

Information becomes easier to understand.

Better Decisions

Managers make faster and smarter decisions.

9 Applications of Rich Reports

- Banking dashboards
 - Hospital management systems
 - Sales analysis
 - Financial reporting
 - CRM systems
-

10 Memory Trick

“C G D K”

- **C** → Charts
- **G** → Graphs
- **D** → Dashboards
- **K** → KPIs

Sentence:

“Charts Give Dashboard Knowledge”

1 1 Exam Writing Tips

✓ Draw Rich Report diagram

✓ Underline keywords:

- Dashboards
- Interactive Reports
- Real-Time Updates
- Visualization

✓ Write hospital dashboard example

1 2 Short Conclusion

Rich Reports improve business analysis by presenting data visually and interactively, helping organizations make faster and better decisions.

Last Minute Revision Points

Features

- Interactive
- Visual
- Real-time

- Easy analysis

Components

- Charts
 - Graphs
 - Dashboards
 - KPIs
-

★ Most Important Keywords

- Rich Reports
 - Dashboards
 - Visualization
 - KPIs
 - Interactive Reports
 - Business Intelligence
-

★ Viva Questions with Answers

Q1) What are rich reports?

Rich reports are visual and interactive business reports.

Q2) What are the components of rich reports?

Charts, graphs, dashboards, and KPIs.

Q3) Why are rich reports important?

They improve data understanding and decision making.

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